Matrices

Question1

For some $\alpha, \beta \in \mathbf{R}$, let $A = \begin{bmatrix} \alpha & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & \beta \end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = O$. Then $\left(\det\left(\operatorname{adj}\left(A^3 - B^3\right)\right)\right)^2$ is equal to _____ .

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Answer: 225

Solution:

Using characteristic equation,

$$\begin{vmatrix} \alpha - \lambda & 2 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

$(\lambda - 2)(\lambda - \alpha) - 2 = 0$

$\lambda^2 + \lambda(-2 - \alpha) + 2\alpha - 2 = 0$

$\Rightarrow A^2 + A(-2 - \alpha) + (2\alpha - 2)I = 0$

$\Rightarrow -2 - \alpha = -4 \Rightarrow \alpha = 2$

Similarly,

$$\begin{vmatrix} 1 - K & 2 \\ 1 & \beta - K \end{vmatrix} = 0$$

$\Rightarrow (K - \beta)(K - 1) - 1 = 0$

$K^2 + K(-1 - \beta) + \beta - 1 = 0$

$\Rightarrow -1 - \beta = -3 \Rightarrow \beta = 2$

$\Rightarrow A = \begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

Let $A^3 - B^3 = \begin{bmatrix} 15 & 20 \\ 6 & 7 \end{bmatrix} = C, \det(C) = -15$

$|\text{ adj } C|^2 = |C|^2 = 225$

Question2

Let A be a 3×3 matrix such that $\mathrm{A}+\mathrm{A}^{\mathrm{T}}=\mathrm{O}$. If $\mathrm{A}\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}3\\3\\2\end{bmatrix}, \mathrm{A}^2\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}-3\\19\\-24\end{bmatrix}$ and $\det(\operatorname{adj}(2\operatorname{adj}(\mathrm{A}+\mathrm{I})))=(2)^\alpha\cdot(3)^\beta\cdot(11)^\gamma, \alpha, \beta, \gamma$ are non-negative integers, then $\alpha+\beta+\gamma$ is equal to ____

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Answer: 18

Solution:

$A=-A^T$

$A=\begin{bmatrix}0 & \alpha & \beta\\-\alpha & 0 & \gamma\\-\beta & -\gamma & 0\end{bmatrix}$

$\begin{bmatrix}0 & \alpha & \beta\\-\alpha & 0 & \gamma\\-\beta & -\gamma & 0\end{bmatrix}\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}3\\3\\2\end{bmatrix}$

$\alpha=-3, -\beta+\gamma=2$

$\begin{bmatrix}0 & \alpha & \beta\\-\alpha & 0 & \gamma\\-\beta & -\gamma & 0\end{bmatrix}\begin{bmatrix}3\\3\\2\end{bmatrix}=\begin{bmatrix}-3\\19\\\beta=3\end{bmatrix}$

$|A+I|=\begin{vmatrix}1 & -3 & 3\\3 & 1 & 5\\-3 & -5 & 1\end{vmatrix}=44$

$\det(\operatorname{adj}(2\operatorname{adj}(A+I)))=2^6\det(\operatorname{adj}(\operatorname{adj}(A+I)))=2^6(44)^4=2^{14}\times 11^4$

Question3

Let $|\mathrm{A}|=6$, where A is a 3×3 matrix. If $\left|\operatorname{adj}\left(3\operatorname{adj}\left(\mathrm{A}^2\cdot\operatorname{adj}(2\,\mathrm{A})\right)\right)\right|=2^{\mathrm{m}}\cdot 3^{\mathrm{n}}, \mathrm{m}, \mathrm{n}\in\mathbf{N}$, then $\mathrm{m}+\mathrm{n}$ is equal to

____ .

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Answer: 62

Solution:

Let

$X = A^2 \cdot \text{adj}(2A)$

so the given expression is

$|\text{adj}\,(3\,\text{adj}(X))|.$

Since A is a 3×3 matrix and $|A| = 6$:

Step 1: Find $|A^2|$

$|A^2| = |A|^2 = 6^2 = 36$

Step 2: Find $|\text{adj}(2A)|$

First,

$|2A| = 2^3|A| = 8 \cdot 6 = 48$

For a 3×3 matrix M,

$|\text{adj}(M)| = |M|^{3-1} = |M|^2$

So,

$|\text{adj}(2A)| = |2A|^2 = 48^2$

Step 3: Find $|X|$

$|X| = |A^2| \cdot |\,\text{adj}(2A)| = 36 \cdot 48^2$

Now,

$36 = 2^2 \cdot 3^2, \qquad 48 = 2^4 \cdot 3$

So,

$48^2 = 2^8 \cdot 3^2$

Hence,

$|X| = (2^2 \cdot 3^2)(2^8 \cdot 3^2) = 2^{10} \cdot 3^4$

Step 4: Find $|3\,\text{adj}(X)|$

First,

$|\text{adj}(X)| = |X|^2 = (2^{10} \cdot 3^4)^2 = 2^{20} \cdot 3^8$

Since $3\,\text{adj}(X)$ is a 3×3 matrix,

$|3\,\text{adj}(X)| = 3^3 \cdot |\,\text{adj}(X)|$

Therefore,

$|3\,\text{adj}(X)| = 27 \cdot 2^{20} \cdot 3^8 = 2^{20} \cdot 3^{11}$

Step 5: Find the required determinant

Again using

$|\text{adj}(M)| = |M|^2$

we get

$|\text{adj}\,(3\,\text{adj}(X))| = \left(2^{20} \cdot 3^{11}\right)^2 = 2^{40} \cdot 3^{22}$

So,

$m = 40, \qquad n = 22$

and

$m + n = 40 + 22 = 62$

$\boxed{62}$

Question4

Let $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I - A) = I + A$. Then the sum of the diagonal elements of $\mathrm{B}^{\mathrm{T}}\mathrm{B}$ is equal to _____

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Answer: 3

Solution:

Observing properties of Matrix A

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$$
$$A^T = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -1 \\ -3 & 1 & 0 \end{bmatrix} = -A$$

Rearranging the given equation

$$B(I - A) = I + A$$
$$B = (I + A)(I - A)^{-1}$$

Finding B^T

$$B^T = \left[(I + A)(I - A)^{-1}\right]^T$$
$$B^T = \left((I - A)^{-1}\right)^T (I + A)^T$$
$$B^T = \left(I - A^T\right)^{-1} \left(I + A^T\right)$$

Since $A^T = -A$:

$$B^T = (I + A)^{-1}(I - A)$$
$$B^T B = \left[(I + A)^{-1}(I - A)\right] \cdot \left[(I + A)(I - A)^{-1}\right]$$
$$B^T B = (I + A)^{-1}(I + A)(I - A)(I - A)^{-1}$$
$$B^T B = I \cdot I = I$$
$$B^T B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the sum of diagonal elements $= 1 + 1 + 1 = 3$

The sum of the diagonal elements of $B^T B$ is 3 .

Question5

The number of 3×2 matrices A , which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of $\mathrm{A}^{\mathrm{T}}\mathrm{A}$ is 5 , is

______

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Answer: 312

Solution:

Let matrix A of order 3×2 is

$$A = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}, a, b, c, d, e, f \in \{-2, -1, 0, 1, 2\}$$

$$A^T = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}$$

$$A^T A = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix} \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} a^2 + c^2 + e^2 & ab + cd + ef \\ ab + cd + ef & b^2 + d^2 + f^2 \end{bmatrix}$$

sum of diagonal elements of $A^T A = 5$

$a^2 + c^2 + e^2 + b^2 + d^2 + f^2 = 5$

possible values of squares is $\{0, 1, 4\}$

Case 1 : one element is 4,1 element is 1 and four elements are 0 .

one element is $4 = 2$ choices $\{-2, 2\}$

one element is $1 = 2$ choices $\{-1, 1\}$

4 element is zero $= 1$ choice $\{0\}$

Selecting one element 4 out of 6 choice $\left\{a^2, b^2, c^2, d^2, e^2, f^2\right\}$, and have 2 choice $\{-2, 2\}$, total $= {}^6C_1 \times 2 = 12$

Selecting one element which is 1 from remaining 5 element $= {}^5C_1$, and have two choice $\{-1, 1\}$, total $= {}^5C_1 \times 2 = 10$

Selecting 4 element which is zero from remaining 4 elements $= {}^4C_4 = 1$ total matrices for Case $1 = 12 \times 10 \times 1 = 120$

Case 2 : five elements are 1 and one element is 0 .

squares $a^2, b^2, c^2, d^2, e^2, f^2 \in \{1, 1, 1, 1, 1, 0\}$

selection of 5 one from $a^2, b^2, c^2, d^2, e^2, f^2 = {}^6C_5$

and for 1 there are two choice for each elements $a, b, c, d, e, f = \{-1, 1\}$, total $= 2^5$

for 0 only 1 choices

total matrices of case $2 = {}^6C_5 \times 2^5 = 192$

total number of such matrices = Case 1+ Case 2

$= 120 + 192 = 312$

Question6

Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is _______

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Answer: 0

Solution:

Decompose Matrix A We can write matrix A as the sum of the Identity matrix I and a residual matrix C :

$$A = I + C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}, \quad \text{Let } C = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}.$$

$$C^2 = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 4-4 & -8+8 \\ 2-2 & -4+4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Since $C^2 = 0, C$ is a nilpotent matrix. Consequently, $C^n = 0$ for all $n \geq 2$.

Find A^{100} using Binomial Expansion :

$$A^{100} = (I + C)^{100} = {}^{100}C_0 I^{100} + {}^{100}C_1 I^{99} C + {}^{100}C_2 I^{98} C^2 + \cdots + {}^{100}C_{100} C^{100}$$

Since $C^2 = C^3 = \cdots = C^{100} = 0$, the expansion simplifies to :

$$\begin{aligned} A^{100} &= {}^{100}C_0 I + {}^{100}C_1 C \\ A^{100} &= 1 \cdot I + 100 \cdot C \\ A^{100} &= I + 100C \end{aligned}$$

Find Matrix B :

The problem states that $A^{100} = 100B + I$. Comparing this to our result:

$$\begin{aligned} 100B + I &= 100C + I \\ 100B &= 100C \\ B &= C \end{aligned}$$

Calculate the Sum of Elements of B^{100} :

Since $B = C$ and we know $C^2 = 0$, then: $\quad B^2 = 0$

Consequently, $B^{100} = 0$ (the zero matrix).

$$B^{100} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The sum of all the elements of B^{100} is 0 .

Question7

Consider the matrices $A = \begin{bmatrix} 2 & -2 \\ 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$. If matrices P and Q are such that $PA = B$ and $AQ = B$, then the absolute value of the sum of the diagonal elements of $2(P+Q)$ is ________.

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Answer: 34

Solution:

$A^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 2 \\ -4 & 2 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$

$\therefore\ PA = B$

$\Rightarrow\ P = BA^{-1}$

$= \frac{1}{2}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -21 & 12 \\ -7 & 4 \end{bmatrix}$

$\therefore\ AQ = B$

$\Rightarrow\ Q = A^{-1}B$

$= \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -2 & -6 \\ -5 & -15 \end{bmatrix}$

$\therefore\ |\operatorname{tr}(2(P+Q))| = |(-21) + (4) + (-2) + (-15)|$

$= |-34| = 34$

Question8

Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy

$$A^2 + \alpha(\mathrm{adj}(\mathrm{adj}(A))) + \beta(\mathrm{adj}(A)(\mathrm{adj}(\mathrm{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$ **for some $\alpha, \beta \in \mathbb{R}$.**

Then $(\alpha - \beta)^2$ is equal to ____

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Answer: 4

Solution:

We need to find α and β from

$$A^2 + \alpha\big(\,\mathrm{adj}(\mathrm{adj}(A))\big) + \beta\big(\,\mathrm{adj}(A)\,\mathrm{adj}(\mathrm{adj}(A))\big) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

Let us solve step by step.

1. Given matrix

$$A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Find $\det(A)$

Since the third row is $(0, 0, 1)$, expand along the third row:

$$\det(A) = 1 \cdot \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (1)(1) = -1.$$

So,

$$\det(A) = -1 \neq 0.$$

Hence A is invertible.

3. Use NCERT property of adjoint of adjoint

For an $n \times n$ invertible matrix,

$$\mathrm{adj}(\mathrm{adj}(A)) = (\det A)^{\,n-2} A.$$

Here $n = 3$, so

$\mathrm{adj}(\mathrm{adj}(A)) = (\det A)^1 A = (-1)A = -A.$

Thus,

$\mathrm{adj}(\mathrm{adj}(A)) = -A.$

Also, for any invertible matrix,

$A\,\mathrm{adj}(A) = \det(A)I.$

So,

$A\,\mathrm{adj}(A) = -I.$

Multiplying both sides by A^{-1} is not necessary; instead use

$\mathrm{adj}(A) = \det(A)A^{-1} = -A^{-1}$ if needed.

Now,

$\mathrm{adj}(A)\,\mathrm{adj}(\mathrm{adj}(A)) = \mathrm{adj}(A)(-A) = -(\mathrm{adj}(A)A).$

And since

$\mathrm{adj}(A)A = \det(A)I = -I,$

we get

$\mathrm{adj}(A)\,\mathrm{adj}(\mathrm{adj}(A)) = -(-I) = I.$

So the given equation becomes

$$A^2 + \alpha(-A) + \beta I = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

That is,

$$A^2 - \alpha A + \beta I = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

4. Compute A^2

$$A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Now multiply A by itself:

$$A^2 = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Entrywise calculation:

- First row:

  $(1,1) : (-1)(-1) + (1)(1) + (-1)(0) = 2$

$$(1,2) : (-1)(1) + (1)(0) + (-1)(0) = -1$$

$$(1,3) : (-1)(-1) + (1)(1) + (-1)(1) = 1$$

- Second row:

$$(2,1) : (1)(-1) + (0)(1) + (1)(0) = -1$$

$$(2,2) : (1)(1) + (0)(0) + (1)(0) = 1$$

$$(2,3) : (1)(-1) + (0)(1) + (1)(1) = 0$$

- Third row:

$$(3,1) = 0, \quad (3,2) = 0, \quad (3,3) = 1$$

Hence

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

5. Form $A^2 - \alpha A + \beta I$

First,

$$-\alpha A = \begin{bmatrix} \alpha & -\alpha & \alpha \\ -\alpha & 0 & -\alpha \\ 0 & 0 & -\alpha \end{bmatrix}$$

and

$$\beta I = \begin{bmatrix} \beta & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \beta \end{bmatrix}.$$

So,

$$A^2 - \alpha A + \beta I = \begin{bmatrix} 2 + \alpha + \beta & -1 - \alpha & 1 + \alpha \\ -1 - \alpha & 1 + \beta & -\alpha \\ 0 & 0 & 1 - \alpha + \beta \end{bmatrix}.$$

This must equal

$$\begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

6. Compare corresponding entries

From the $(1,2)$ entry:

$$-1 - \alpha = -2 \quad \Rightarrow \quad \alpha = 1.$$

From the $(2,2)$ entry:

$$1 + \beta = 0 \quad \Rightarrow \quad \beta = -1.$$

These also satisfy the other entries:

- $(1, 1)$:

  $2 + \alpha + \beta = 2 + 1 - 1 = 2$

- $(1, 3)$:

  $1 + \alpha = 2$

- $(2, 1)$:

  $-1 - \alpha = -2$

- $(2, 3)$:

  $-\alpha = -1$

- $(3, 3)$:

  $1 - \alpha + \beta = 1 - 1 - 1 = -1$

All correct.

Thus,

$\alpha = 1, \qquad \beta = -1.$

7. Find $(\alpha - \beta)^2$

$\alpha - \beta = 1 - (-1) = 2$

Therefore,

$(\alpha - \beta)^2 = 2^2 = 4.$

So the required answer is

$\boxed{4}$.

Question9

For the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -29 & 49 \\ -13 & 18 \end{bmatrix}$, if $(A^{15} + B) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, then among the following which one is true?

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Options:

A.

$x=16, y=3$

B.

$x=5, y=7$

C.

$x=11, y=2$

D.

$x=18, y=11$

Answer: C

Solution:

$A=\begin{bmatrix}3 & -4\\1 & -1\end{bmatrix}, B=\begin{bmatrix}23 & 49\\45 & 21\end{bmatrix}$
Characteristic equation of A,
$(3-\lambda)(-1-\lambda)+4=0$
$(\lambda-3)(\lambda+1)+4=\lambda^2-2\lambda+=0$
$\Rightarrow A^2-2A+I=0$
$A^2=2A-I, A^3=2A^2-A=2(2A-I)-A=3A-2I$
$A^4=4A^2+I-4A=4(2A-I)-4A+I$
$=4(2A-I)-3A=5A-4I$
$\left(A^5\right)^3=(5A-4I)^3$
$=125A^3-3\times 25A^2(4)+3(5A)\left(4^2\right)-64I$
$=125(3A-2I)-300(2A-I)+240A-64I$
$=A(375-600+240)+I(-250+300-64)$
$=15A-14I$
$\Rightarrow A^{15}+B=15A-14I+B$
$=\begin{bmatrix}2 & -11\\1 & -11\end{bmatrix} \quad \left(A^{15}+B\right)\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix} \Rightarrow 2x=11y$

Question10

If the system of equations

$3x+y+4z=3$

$2x+\alpha y-z=-3$

$x+2y+z=4$

has no solution, then the value of α is equal to:

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Options:

A.

13

B.

4

C.

19

D.

23

Answer: C

Solution:

$$\Delta = \begin{vmatrix} 3 & 1 & 4 \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix}$$
$\det(\Delta) = 0$
$\alpha = 19$

Question11

If $\mathrm{A} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then the determinant of the matrix $\left(\mathrm{A}^{2025} - 3\,\mathrm{A}^{2024} + \mathrm{A}^{2023}\right)$ is

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Options:

A.

12

B.

24

C.

28

D.

16

Answer: D

Solution:

$A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, |A| = 1$

$\left|A^{2025} - 3A^{2024} + A^{2023}\right| = \left|A^{2023}\right| \left|A^2 - 3A + I\right| = 16$

Question12

If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ **is a solution of the system of equations** $AX = B$**, where** adj $A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ **and** $\mathrm{B} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$**, then** $|x + y + z|$ **is equal to :**

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Options:

A.

3

B.

2

C.

$\frac{3}{2}$

D.

1

Answer: B

Solution:

$$X = A^{-1}B = \left(\frac{\operatorname{adj} A}{|A|}\right)B = \pm\frac{1}{10}\begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = \pm\frac{1}{10}\begin{pmatrix} 20 \\ -10 \\ 10 \end{pmatrix} = \pm\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$\therefore |x + y + z| = 2$

Question13

Let n be the number obtained on rolling a fair die. If the probability that the system

$$\begin{aligned} &x - \mathrm{n}y + z = 6 \\ &x + (\mathrm{n} - 2)y + (\mathrm{n} + 1)z = 8 \\ &\quad (\mathrm{n} - 1)y + z = 1 \end{aligned}$$

has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is :

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Options:

A.

22

B.

20

C.

24

D.

21

Answer: A

Solution:

$x - ny + z = 6$
$x - (n-2)y + (n+1)z = 8$
$(n-1)y + z = 1$

$\begin{vmatrix} 1 & -n & 1 \\ 1 & (n-2) & n+1 \\ 0 & n-1 & 1 \end{vmatrix} = 0 \Rightarrow n = 1, 2 \text{ or } n = -1 \text{ (rejected)}$

For unique solution $\mathrm{n} = 3, 4, 5, 6$

Now P (Probability when system of equations has unique solution) $= \frac{4}{6}$

So $\mathrm{k} = 4$

Now required sum $= 4 + (3 + 4 + 5 + 6) = 22$

Question14

Among the statements :

I: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$**, then** $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$**, and**

II: If $\begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x-3 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = \mathrm{p}x + \mathrm{q}$**, then** $\mathrm{p}^2 = 196\mathrm{q}^2$**,**

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Options:

A.

both are true

B.

both are false

C.

only I is true

D.

only II is true

Answer: B

Solution:

$$\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = 1\left(1-\cos^2\gamma\right) - \cos\alpha(\cos\alpha - \cos\beta\cos\gamma) + \cos\beta(\cos\alpha\cos\gamma - \cos\beta)$$

$$= 1 - \cos^2\gamma - \cos^2\alpha - \cos^2\beta + 2\cos\beta + 2 + \cos\alpha\cos\beta\cos\gamma$$

$$\begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix} = -\cos\alpha(-\cos\beta\cos\gamma) + \cos\beta(\cos\alpha\cos\gamma) = 2\cos\alpha\cos\beta - \cos\gamma$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\because q = \begin{vmatrix} 0 & 1 & -2 \\ -1 & 0 & -3 \\ 3 & -1 & -1 \end{vmatrix} = -10 - 2 = -12$$

$$\text{And } -p + q = \begin{vmatrix} 0 & 0 & -3 \\ -2 & -3 & -6 \\ 2 & -3 & -3 \end{vmatrix} = -36$$

$$\Rightarrow p = q + 36 = 24$$

$$\Rightarrow p^2 \neq 196q^2$$

Option (4) is correct.

Question15

The system of linear equations

$$x + y + z = 6$$
$$2x + 5y + az = 36$$
$$x + 2y + 3z = b$$

has :

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Options:

A.

unique solution for $a=8$ and $b=16$

B.

infinitely many solutions for $a=8$ and $b=14$

C.

infinitely many solutions for $a=8$ and $b=16$

D.

unique solution for $a=8$ and $b=14$

Answer: B

Solution:

For the system to have infinitely many solutions, the determinant of the coefficient matrix (Δ) must be zero.

$$\Delta=\begin{vmatrix}1 & 1 & 1\\ 2 & 5 & a\\ 1 & 2 & 3\end{vmatrix}=0$$
$$1(15-2a)-1(6-a)+1(4-5)=0$$
$$15-2a-6+a-1=0$$
$$8-a=0 \Rightarrow a=8$$

Now, for infinitely many solutions, $\Delta_x=\Delta_y=\Delta_z=0$.

Check Δ_x :

$$\Delta_x=\begin{vmatrix}6 & 1 & 1\\ 36 & 5 & 8\\ b & 2 & 3\end{vmatrix}=0$$
$$6(15-16)-1(108-8b)+1(72-5b)=0$$
$$6(-1)-108+8b+72-5b=0$$
$$-6-108+72+3b=0$$
$$-42+3b=0 \Rightarrow 3b=42 \Rightarrow b=14$$
$$\Delta_y=\begin{vmatrix}1 & 6 & 1\\ 2 & 36 & 8\\ 1 & 14 & 3\end{vmatrix}=1(108-112)-6(6-8)+1(28-36)=-4+12-8=0$$
$$\Delta_z=\begin{vmatrix}1 & 1 & 6\\ 2 & 5 & 36\\ 1 & 2 & 14\end{vmatrix}=1(70-72)-1(28-36)+6(4-5)=-2+8-6=0$$

If $a = 8$ and $b \neq 14$, the system has no solution.

If $a = 8$ and $b = 14$, the system has infinitely many solutions.

If $a \neq 8$, the system has a unique solution regardless of b.

Question16

Let $f(x) = \int \frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2} dx, x > 0, \lim\limits_{x \to 0} f(x) = 0$ and $f(1) = \frac{1}{4}$.

If $\mathrm{A} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix}$ and $\mathrm{B} = \mathrm{adj}(\mathrm{adj}\ \mathrm{A})$ be such that $|\mathrm{B}| = 81$, then α^2 is equal to

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Options:

A.

2

B.

4

C.

3

D.

1

Answer: B

Solution:

$$\mathrm{f(x)} = \int \frac{7x^{10} + 9x^8}{(1 + x^2 + 2x^9)^2} dx$$

$$= \int \frac{7x^{10} + 9x^8}{x^{18}\left(\frac{1}{x^9} + \frac{1}{x^7} + 2\right)^2} = \int \frac{\frac{7}{x^8} + \frac{9}{x^{10}}}{\left(\frac{1}{x^9} + \frac{1}{x^7} + 2\right)^2}$$

substitute $\frac{1}{x^9}+\frac{1}{x^7}+2=t$

$$\left(\frac{-9}{x^{10}}-\frac{7}{x^8}\right)dx=dt \Rightarrow \left(\frac{7}{x^8}+\frac{9}{x^{10}}\right)dx=-dt$$
$$f(t)=-\int\frac{dt}{t^2}=\frac{1}{t}+C$$

put value of t

$$f(x)=\frac{1}{\frac{1}{x^9}+\frac{1}{x^7}+2}+C$$
$$f(x)=\frac{x^9}{2x^9+x^2+1}+C$$
It is given that $\lim_{x\to 0} f(x)=0$
$$\lim_{x\to 0}\frac{x^9}{2x^9+x^2+1}+C=0$$
$0+C=0 \Rightarrow C=0$

so $f(x)=\dfrac{x^9}{2x^9+x^2+1}$
$$f'(x)=\frac{9x^8\left(2x^9+x^2+1\right)-x^9\left(18x^8+2x\right)}{\left(2x^9+x^2+1\right)^2}$$
$$f'(1)=\frac{9(1)^8\left(2(1)^9+(1)^2+1\right)-(1)^9\left(18(1)^8+2(1)\right)}{\left(2(1)^9+(1)^2+1\right)^2}$$
$$=\frac{9\times 4-20}{16}=\frac{16}{16}=1$$

It is given that A $=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & f'(1) & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$

substitute $f'(1)=1 \Rightarrow A=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & 1 & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$

$$|A|=1\left(4\times\frac{1}{4}-\alpha^2\right)=1-\alpha^2$$

It is given that B $=$ adj(adj A)&$|B|=81$

$|B|=|A|^{(n-1)^2}$, n $=3$ as A is 3×3 matrix

$|B|=|A|^{(3-1)^2}=81$
$|A|^4=81=3^4$
$|A|=\pm 3$

$1-\alpha^2=\pm 3\left\{|A|=1-\alpha^2\right\}$

$1-\alpha^2=-3$ or $1-\alpha^2=3$

$\alpha^2=4$ or $\alpha^2=-2$ not possible

so value of α^2 is 4

Question17

Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{\mathrm{ij}} = 2^{(\mathrm{i+j-1})}\mathrm{p_{ij}}$ and $\det(\mathrm{Q}) = 2^{10}$. Then the value of $\det(\mathrm{adj}(\mathrm{adj}\,\mathrm{P}))$ is:

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Options:

A.

81

B.

16

C.

124

D.

32

Answer: B

Solution:

Given two 3×3 matrices $P = [p_{ij}], Q = [q_{ij}]$ are given

$q_{ij} = 2^{(i+j-1)}p_{ij}, \det(Q) = 2^{10}$.

To calculate $\det(\mathrm{adj}(\mathrm{adj}\,P))$

We know $\det(\mathrm{adj}(\mathrm{adj}\,P)) = (\det(P))^{(n-1)^2}$

where n is order of matrix which is 3

so, $\det(\mathrm{adj}(\mathrm{adj}\,P)) = (\det(P))^4 - (1)$

Let P matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}$$

and $Q = \begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \end{bmatrix}$

use $q_{ij} = 2^{i+j-1} p_{ij}$ to find Q matrix

e.g. $q_{11} = 2^{1+1-1} p_{11} = 2p_{11}, q_{23} = 2^{2+3-1} p_{23} = 2^4 p_{23}$

$$Q = \begin{bmatrix} 2p_{11} & 2^2 p_{12} & 2^3 p_{13} \\ 2^2 p_{21} & 2^3 p_{22} & 2^4 p_{23} \\ 2^3 p_{31} & 2^4 p_{32} & 2^5 p_{33} \end{bmatrix}$$

$$Q = \begin{bmatrix} 2p_{11} & 2^2 p_{12} & 2^3 p_{13} \\ 2^2 p_{21} & 2^3 p_{22} & 2^4 p_{23} \\ 2^3 p_{31} & 2^4 p_{32} & 2^5 p_{33} \end{bmatrix}$$

$$\det(Q) = \begin{vmatrix} 2p_{11} & 2^2 p_{12} & 2^3 p_{13} \\ 2^2 p_{21} & 2^3 p_{22} & 2^4 p_{23} \\ 2^3 p_{31} & 2^4 p_{32} & 2^5 p_{33} \end{vmatrix} = 2^{10}$$

taking common 2 from $R_1, 2^2$ from $R_2 \& 2^3$ from R_3

$$\det(Q) = 2 \cdot 2^2 \cdot 2^3 \begin{vmatrix} p_{11} & 2p_{12} & 2^2 p_{13} \\ p_{21} & 2p_{22} & 2^2 p_{23} \\ p_{31} & 2p_{32} & 2^2 p_{33} \end{vmatrix} = 2^{10}$$

taking common 2 from C_2 and 2^2 from C_3

$$2 \cdot 2^2 \cdot 2^3 \cdot 2 \cdot 2^2 \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{vmatrix} = 2^{10}$$

$\Rightarrow 2^9 \cdot \det(P) = 2^{10}$
$\Rightarrow \det(P) = 2$
from (1) $\det(\operatorname{adj}(\operatorname{adj} P)) = (\det(P))^4 = 2^4 = 16$

Question18

Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $\mathrm{A} + \mathrm{C} = \mathrm{I}$.

If $\mathrm{BC} = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $\mathrm{CB} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

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Options:

A.

4

B.

2

C.

0

D.

-2

Answer: C

Solution:

We are given:

- $B = (I + A)^{-1} \implies B^{-1} = I + A$
- $A + C = I \implies A = I - C$

Substitute A into the first equation:

$B^{-1} = I + (I - C)$

$B^{-1} = 2I - C........ (1)$

pre-multiply B : $B \cdot B^{-1} = B(2I - C)I = 2B - BC \implies BC = 2B - I$ .... (2)

post multiply in eqn (1): $B^{-1}B = (2I - C) \cdot BI = 2B - CB \implies CB = 2B - I$ ....(3)

from (2) & (3), $BC = CB$

3. Set up the System of Equations Since $BC = CB$, we use the given matrix for BC :

$CB = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$

Now, substitute this into the vector equation :

$\begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$

This results in two linear equations :

- $x_1 - 5x_2 = 12$
- $-x_1 + 2x_2 = -6$

1. Solve for x_1 and x_2

   Adding the two equations :

$(x_1 - x_1) + (-5x_2 + 2x_2) = 12 - 6$

$\Rightarrow x_2 = -2$

Substitute $x_2 = -2$ into the second equation :

$$-x_1 + 2(-2) = -6$$
$-x_1 - 4 = -6 \implies -x_1 = -2 \implies \mathbf{x_1 = 2}$

Final result $x_1 + x_2 = 2 + (-2) = \mathbf{0}$

Question19

Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = O$ and $B^2 - 5B - 6I = O$, then among the two statements:

(S1) : $[(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$

and

(S2) : $\det(\operatorname{adj}(A + B)) = -5$

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Options:

A.

only (S1) is correct

B.

only (S2) is correct

C.

both (S1) and (S2) are correct

D.

both (S1) and (S2) are wrong

Answer: B

Solution:

$A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$
$A^2 - 4A + I = 0$

trace of matrix

$\alpha + 1 = 4$
$\alpha = 3$
$B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$
$B^2 - 5B - 6I = 0$

$\text{det.}(B) = 6 - 3\beta = -6$
$3\beta = 12$
$\beta = 4$
$B - A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}; B + A = \begin{bmatrix} 4 & 5 \\ 5 & 5 \end{bmatrix}$
$(B - A)(B + A) = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$
$\text{adj}(A) = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}, \text{adjB} = \begin{bmatrix} 2 & -3 \\ -4 & 3 \end{bmatrix}$
$\text{adj}(A + B) = \begin{bmatrix} 5 & -5 \\ -5 & 4 \end{bmatrix}$

Question20

Let $\alpha, \beta \in \mathbb{R}$ be such that the system of linear equations

$$\begin{aligned} x + 2y + z &= 5 \\ 2x + y + \alpha z &= 5 \\ 8x + 4y + \beta z &= 18 \end{aligned}$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

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Options:

A.

-4

B.

4

C.

8

D.

-8

Answer: B

Solution:

$x+2y+z=5$
$2x+y+\alpha z=5$
$8x+4y+\beta z=18$

For a system of three linear equations to have **no solution**, it is necessary that the coefficient matrix be **singular**, i.e. its determinant is zero: $\Delta=0$.

$$\Delta=\begin{vmatrix}1 & 2 & 1\\ 2 & 1 & \alpha\\ 8 & 4 & \beta\end{vmatrix}=0$$
$$\Delta=1(\beta-4\alpha)-2(2\beta-8\alpha)+1(0)=0$$
$$=\beta-4\alpha-4\beta+16\alpha=0$$
$$\Rightarrow 4\alpha=\beta$$

Now substitute $\beta=4\alpha$ in the third equation. Then the left side of the third equation becomes $8x+4y+4\alpha z$, which is exactly 4 times the left side of the second equation $2x+y+\alpha z$.

But the constants do not match: $4\times 5=20$, whereas the third equation has 18. So we get $4\times(2x+y+\alpha z)=20$ but $8x+4y+4\alpha z=18$, which is a contradiction. Hence, the system is inconsistent and has **no solution**.

Therefore, $\frac{\beta}{\alpha}=\frac{4\alpha}{\alpha}=4$.

Question21

If the system of equations

$$x+5y+6z=4$$

$$2x+3y+4z=7$$

$$x+6y+az=b$$

has infinitely many solutions, then the point (a, b) lies on the line

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Options:

A.

$y - x = 3$

B.

$x - y = 3$

C.

$x + y = 11$

D.

$x + y = 12$

Answer: B

Solution:

The given system of equations is:

$x + 5y + 6z = 4$

$2x + 3y + 4z = 7$

$x + 6y + az = b$

Find the determinant of the coefficient matrix (Δ)

The coefficient matrix is formed by the coefficients of x, y, and z.

$$\Delta = \begin{vmatrix} 1 & 5 & 6 \\ 2 & 3 & 4 \\ 1 & 6 & a \end{vmatrix}$$

Let us expand this determinant along the first row:

$\Delta = 1(3 \cdot a - 6 \cdot 4) - 5(2 \cdot a - 1 \cdot 4) + 6(2 \cdot 6 - 1 \cdot 3)$

$\Delta = 1(3a - 24) - 5(2a - 4) + 6(12 - 3)$

$\Delta = 3a - 24 - 10a + 20 + 6(9)$

$\Delta = -7a - 4 + 54$

$\Delta = -7a + 50$

For the system to have infinitely many solutions, we equate Δ to 0:

$-7a + 50 = 0 \implies 7a = 50 \implies a = \frac{50}{7}$

Find the determinant Δ_z

To find Δ_z, we replace the z-column (third column) with the constant terms from the right-hand side of the equations.

$$\Delta_z = \begin{vmatrix} 1 & 5 & 4 \\ 2 & 3 & 7 \\ 1 & 6 & b \end{vmatrix}$$

We again expand along the first row:

$\Delta_z = 1(3 \cdot b - 6 \cdot 7) - 5(2 \cdot b - 1 \cdot 7) + 4(2 \cdot 6 - 1 \cdot 3)$

$\Delta_z = 1(3b - 42) - 5(2b - 7) + 4(12 - 3)$

$\Delta_z = 3b - 42 - 10b + 35 + 4(9)$

$\Delta_z = -7b - 7 + 36$

$\Delta_z = -7b + 29$

Equating Δ_z to zero for infinitely many solutions, we get:

$-7b + 29 = 0 \implies 7b = 29 \implies b = \frac{29}{7}$

Check the options for the point (a, b)

We have obtained the point coordinates as $(a, b) = \left(\frac{50}{7}, \frac{29}{7}\right)$.

Let us evaluate the difference $a - b$ to find out which linear profile equation matches our outcome:

$a - b = \frac{50}{7} - \frac{29}{7}$

$a - b = \frac{50-29}{7}$

$a - b = \frac{21}{7} = 3$

Writing this relation in terms of x and y (where $x = a$ and $y = b$), it perfectly satisfies the line equation:

$x - y = 3$

Thus, the point (a, b) lies on the line given in Option B.

Question22

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ -2 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix}$. Then the sum of all elements of the matrix $\operatorname{adj}\left(\operatorname{adj}\left(2(\operatorname{adj}\ \mathrm{A})^{-1}\right)\right)$ is equal to:

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Options:

A.

3

B.

4

C.

-4

D.

-3

Answer: D

Solution:

Given

$$A = \begin{bmatrix} 1 & 1 & 2 \\ -2 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix}$$

We have to find the sum of all elements of

$\text{adj}\left(\text{adj}\left(2(\text{adj } A)^{-1}\right)\right)$

$$\det A = 1\begin{vmatrix} 0 & 1 \\ 3 & 5 \end{vmatrix} - 1\begin{vmatrix} -2 & 1 \\ 1 & 5 \end{vmatrix} + 2\begin{vmatrix} -2 & 0 \\ 1 & 3 \end{vmatrix}$$

$= 1(0 \cdot 5 - 1 \cdot 3) - 1((-2) \cdot 5 - 1 \cdot 1) + 2((-2) \cdot 3 - 0 \cdot 1)$

$= -3 - (-11) + 2(-6)$

$= -3 + 11 - 12 = -4$

So,

$\det A = -4$

For an $n \times n$ matrix,

$\det(\text{adj } A) = (\det A)^{n-1}$

Since A is 3×3,

$\det(\text{adj } A) = (\det A)^2 = (-4)^2 = 16$

Now let

$$B = 2(\operatorname{adj} A)^{-1}$$

We need $\operatorname{adj}(\operatorname{adj} B)$.

For a 3×3 matrix, the property is:

$$\operatorname{adj}(\operatorname{adj} B) = (\det B)\, B$$

Since

$$B = 2(\operatorname{adj} A)^{-1}$$

and for a 3×3 matrix,

$$\det(2M) = 2^3 \det(M) = 8 \det(M)$$

therefore,

$$\det B = 8 \det\left((\operatorname{adj} A)^{-1}\right)$$

Also,

$$\det\left((\operatorname{adj} A)^{-1}\right) = \frac{1}{\det(\operatorname{adj} A)} = \frac{1}{16}$$

Hence,

$$\det B = 8 \cdot \frac{1}{16} = \frac{1}{2}$$

So,

$$\operatorname{adj}(\operatorname{adj} B) = (\det B)B = \frac{1}{2}\, B$$

But

$$B = 2(\operatorname{adj} A)^{-1}$$

Thus,

$$\operatorname{adj}(\operatorname{adj} B) = \frac{1}{2} \cdot 2(\operatorname{adj} A)^{-1} = (\operatorname{adj} A)^{-1}$$

So the required matrix is simply

$$(\operatorname{adj} A)^{-1}$$

We know,

$$\operatorname{adj} A = (\det A)A^{-1}$$

Hence,

$$(\operatorname{adj} A)^{-1} = \frac{1}{\det A} A$$

Since $\det A = -4$,

$$(\operatorname{adj} A)^{-1} = \frac{1}{-4} A = -\frac{1}{4} A$$

Therefore the required matrix is

$$-\frac{1}{4} A$$

First find the sum of elements of A:

$1+1+2-2+0+1+1+3+5=12$

Therefore sum of elements of $-\frac{1}{4}A$ is

$-\frac{1}{4}\cdot 12=-3$

Final Answer

$\boxed{-3}$

So, the correct option is **D**.

Question23

Let $\mathrm{S}=\left\{\mathrm{A}=\begin{bmatrix} a & b \\ c & d \end{bmatrix}: a, b, c, d \in\{0,1,2,3,4\}\right.$ and $\left.\mathrm{A}^2-4\,\mathrm{A}+3\mathrm{I}=0\right\}$ be a set of 2×2 matrices. Then the number of matrices in S , for which the sum of the diagonal elements is equal to 4 , is :

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Options:

A.

20

B.

17

C.

21

D.

19

Answer: D

Solution:

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad a, b, c, d \in \{0, 1, 2, 3, 4\}$$

and it satisfies

$$A^2 - 4A + 3I = 0.$$

We have to count those matrices for which the sum of diagonal elements is

$$a + d = 4.$$

So we must find the number of matrices satisfying both:

1. $A^2 - 4A + 3I = 0$
2. $\text{tr}(A) = a + d = 4$

Step 1: Factor the matrix equation

We write

$$A^2 - 4A + 3I = (A - I)(A - 3I) = 0.$$

So the matrix A satisfies the polynomial

$$(x - 1)(x - 3) = 0.$$

Hence the only possible eigenvalues are 1 and 3.

Now the trace is given to be 4. Since trace is sum of eigenvalues, the eigenvalues must be

1 and 3.

So every such matrix has distinct eigenvalues 1 and 3, hence it is diagonalizable.

Step 2: Use trace and determinant

For a 2×2 matrix with eigenvalues 1 and 3,

- trace $= 1 + 3 = 4$
- determinant $= 1 \cdot 3 = 3$

So we need all matrices with entries in $\{0, 1, 2, 3, 4\}$ such that

$$a + d = 4, \qquad ad - bc = 3.$$

Let $d = 4 - a$. Then

$$a(4 - a) - bc = 3$$

that is,

$$bc = a(4 - a) - 3.$$

Now check all possible values of $a \in \{0, 1, 2, 3, 4\}$.

Step 3: Case-wise counting

Case 1: $a = 0$

Then $d = 4$, and

$bc = 0 \cdot 4 - 3 = -3$

not possible.

Case 2: $a = 1$

Then $d = 3$, and

$bc = 1 \cdot 3 - 3 = 0$

So we need ordered pairs (b, c) with $b, c \in \{0, 1, 2, 3, 4\}$ and $bc = 0$.

That means at least one of b, c is 0.

Number of such ordered pairs:

- $b = 0$: 5 choices for c
- $c = 0$: 5 choices for b
- but $(0, 0)$ counted twice

So total

$5 + 5 - 1 = 9$

matrices.

Case 3: $a = 2$

Then $d = 2$, and

$bc = 2 \cdot 2 - 3 = 1$

So

$bc = 1.$

Since $b, c \in \{0, 1, 2, 3, 4\}$, this is possible only for

$(b, c) = (1, 1).$

So total matrices in this case:

1

Case 4: $a = 3$

Then $d = 1$, and

$bc = 3 \cdot 1 - 3 = 0$

Again, number of ordered pairs (b, c) with $bc = 0$ is

9.

Case 5: $a = 4$

Then $d = 0$, and

$bc = 4 \cdot 0 - 3 = -3$

not possible.

Step 4: Total count

Total number of matrices is

$9 + 1 + 9 = 19.$

Hence the required number of matrices is

$\boxed{19}$

So, the correct option is:

$\boxed{\text{Option D}}$

Question24

Let $A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}$ and $\det(A - \alpha I) = 0$, where α is a real number. If the largest possible value of α is p, then the circle $(x - p)^2 + (y - 2p)^2 = 320$, intersects the co-ordinate axes at

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Options:

A.

1 point

B.

2 points

C.

3 points

D.

4 points

Answer: C

Solution:

We need the largest real value of α such that

$\det(A - \alpha I) = 0$

That means α is an eigenvalue of the matrix

$$A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}.$$

So first we find the eigenvalues of A.

Step 1: Form $A - \alpha I$

$$A - \alpha I = \begin{bmatrix} 1-\alpha & 2 & 7 \\ 4 & -2-\alpha & 8 \\ 3 & 8 & -7-\alpha \end{bmatrix}$$

Now,

$$\det(A - \alpha I) = \begin{vmatrix} 1-\alpha & 2 & 7 \\ 4 & -2-\alpha & 8 \\ 3 & 8 & -7-\alpha \end{vmatrix} = 0$$

Step 2: Expand the determinant

Expanding along the first row:

$$\det(A - \alpha I) = (1-\alpha)\begin{vmatrix} -2-\alpha & 8 \\ 8 & -7-\alpha \end{vmatrix} - 2\begin{vmatrix} 4 & 8 \\ 3 & -7-\alpha \end{vmatrix} + 7\begin{vmatrix} 4 & -2-\alpha \\ 3 & 8 \end{vmatrix}$$

Now calculate each minor:

$$\begin{vmatrix} -2-\alpha & 8 \\ 8 & -7-\alpha \end{vmatrix} = (-2-\alpha)(-7-\alpha) - 64$$

$$= (\alpha+2)(\alpha+7) - 64 = \alpha^2 + 9\alpha + 14 - 64 = \alpha^2 + 9\alpha - 50$$

Next,

$$\begin{vmatrix} 4 & 8 \\ 3 & -7-\alpha \end{vmatrix} = 4(-7-\alpha) - 24 = -28 - 4\alpha - 24 = -52 - 4\alpha$$

And,

$$\begin{vmatrix} 4 & -2-\alpha \\ 3 & 8 \end{vmatrix} = 32 - 3(-2-\alpha) = 32 + 6 + 3\alpha = 38 + 3\alpha$$

So,

$$\det(A - \alpha I) = (1-\alpha)(\alpha^2 + 9\alpha - 50) - 2(-52 - 4\alpha) + 7(38 + 3\alpha)$$

Now simplify:

$$(1-\alpha)(\alpha^2 + 9\alpha - 50) = \alpha^2 + 9\alpha - 50 - \alpha^3 - 9\alpha^2 + 50\alpha$$

$= -\alpha^3 - 8\alpha^2 + 59\alpha - 50$

Also,

$-2(-52 - 4\alpha) = 104 + 8\alpha$

and

$7(38 + 3\alpha) = 266 + 21\alpha$

Therefore,

$\det(A - \alpha I) = -\alpha^3 - 8\alpha^2 + 59\alpha - 50 + 104 + 8\alpha + 266 + 21\alpha$

$= -\alpha^3 - 8\alpha^2 + 88\alpha + 320$

So the equation becomes

$-\alpha^3 - 8\alpha^2 + 88\alpha + 320 = 0$

Multiplying by -1:

$\alpha^3 + 8\alpha^2 - 88\alpha - 320 = 0$

Step 3: Factorise

Try integer roots. On putting $\alpha = 8$:

$8^3 + 8(8^2) - 88(8) - 320 = 512 + 512 - 704 - 320 = 0$

So $(\alpha - 8)$ is a factor.

Divide:

$\alpha^3 + 8\alpha^2 - 88\alpha - 320 = (\alpha - 8)(\alpha^2 + 16\alpha + 40)$

Now solve

$\alpha^2 + 16\alpha + 40 = 0$

$\alpha = \frac{-16 \pm \sqrt{256-160}}{2} = \frac{-16 \pm \sqrt{96}}{2} = \frac{-16 \pm 4\sqrt{6}}{2} = -8 \pm 2\sqrt{6}$

Hence the eigenvalues are

$8, \quad -8 + 2\sqrt{6}, \quad -8 - 2\sqrt{6}$

So the largest possible value is

$p = 8$

Step 4: Put $p = 8$ in the circle

Given circle:

$(x - p)^2 + (y - 2p)^2 = 320$

Substitute $p = 8$:

$(x - 8)^2 + (y - 16)^2 = 320$

Its centre is $(8, 16)$ and radius is

$r = \sqrt{320} = 8\sqrt{5}$

Step 5: Find intersections with the coordinate axes

On the x-axis

For points on the x-axis, $y = 0$.

$(x-8)^2 + (0-16)^2 = 320$

$(x-8)^2 + 256 = 320$

$(x-8)^2 = 64$

$x - 8 = \pm 8$

So,

$x = 16 \quad \text{or} \quad x = 0$

Thus the circle meets the x-axis at

$(16, 0),\ (0, 0)$

On the y-axis

For points on the y-axis, $x = 0$.

$(0-8)^2 + (y-16)^2 = 320$

$64 + (y-16)^2 = 320$

$(y-16)^2 = 256$

$y - 16 = \pm 16$

So,

$y = 32 \quad \text{or} \quad y = 0$

Thus the circle meets the y-axis at

$(0, 32),\ (0, 0)$

Step 6: Count distinct intersection points

The distinct points are

$(16, 0),\ (0, 0),\ (0, 32)$

So the circle intersects the coordinate axes at

$\boxed{\text{3 points}}$

Hence, the correct option is:

$\boxed{\text{Option C}}$

Question25

If the system of equations :

$$x + y + z = 5$$
$$x + 2y + 3z = 9$$
$$x + 3y + \lambda z = \mu$$

has infinitely many solutions, then the value of $\lambda + \mu$ is :

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Options:

A.

16

B.

18

C.

19

D.

21

Answer: B

Solution:

$D = \lambda - 5$
$D_1 = \lambda + \mu - 18$
$D_2 = 4\lambda - 2\mu + 6$
$D_3 = \mu - 13$
Here $x = \dfrac{D_1}{D}; y = \dfrac{D_2}{D}; z = \dfrac{D_3}{D}$

For the system to have infinitely many solutions, the determinant of the coefficient matrix and all the corresponding determinants must be zero.

$D = D_1 = D_2 = D_3 = 0$

First, set $D = 0$:

$\lambda - 5 = 0 \Rightarrow \lambda = 5$

Next, set $D_3 = 0$:

$\mu - 13 = 0 \Rightarrow \mu = 13$

Now check $D_1 = 0$ using $\lambda = 5$ and $\mu = 13$:

$\lambda + \mu - 18 = 5 + 13 - 18 = 0$

Also check $D_2 = 0$:

$4\lambda - 2\mu + 6 = 4(5) - 2(13) + 6 = 20 - 26 + 6 = 0$

So the values are consistent. Therefore,

$\therefore \lambda + \mu = 5 + 13 = 18$

Question26

Consider the system of linear equations in x, y, z :

$$x + 2y + tz = 0$$
$$6x + y + 5tz = 0$$
$$3x + t^2 y + f(t)z = 0$$

where $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function. If this system has infinitely many solutions for all $t \in \mathbb{R}$, then f

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Options:

A.

is a constant function

B.

is strictly increasing on $\mathbb{R}$

C.

is strictly decreasing on R

D.

has two critical points

Answer: B

Solution:

For a homogeneous system of three linear equations in x, y, z to have infinitely many solutions, the coefficient matrix must be singular for every $t \in \mathbb{R}$.

So, we form the coefficient matrix:

$$A = \begin{pmatrix} 1 & 2 & t \\ 6 & 1 & 5t \\ 3 & t^2 & f(t) \end{pmatrix}$$

Since the system has infinitely many solutions for all t, we must have

$$\det(A) = 0 \quad \text{for all } t \in \mathbb{R}.$$

Now compute the determinant.

$$\det(A) = \begin{vmatrix} 1 & 2 & t \\ 6 & 1 & 5t \\ 3 & t^2 & f(t) \end{vmatrix}$$

Expanding along the first row,

$$\det(A) = 1 \begin{vmatrix} 1 & 5t \\ t^2 & f(t) \end{vmatrix} - 2 \begin{vmatrix} 6 & 5t \\ 3 & f(t) \end{vmatrix} + t \begin{vmatrix} 6 & 1 \\ 3 & t^2 \end{vmatrix}$$

Now evaluate each minor:

$$\begin{vmatrix} 1 & 5t \\ t^2 & f(t) \end{vmatrix} = f(t) - 5t^3$$

$$\begin{vmatrix} 6 & 5t \\ 3 & f(t) \end{vmatrix} = 6f(t) - 15t$$

$$\begin{vmatrix} 6 & 1 \\ 3 & t^2 \end{vmatrix} = 6t^2 - 3$$

So,

$$\det(A) = \left(f(t) - 5t^3\right) - 2\left(6f(t) - 15t\right) + t(6t^2 - 3)$$

$$= f(t) - 5t^3 - 12f(t) + 30t + 6t^3 - 3t$$

$$= -11f(t) + t^3 + 27t$$

Since this is zero for all t,

$$-11f(t) + t^3 + 27t = 0$$

Hence,

$$f(t) = \tfrac{t^3+27t}{11}$$

Now differentiate:

$f'(t) = \frac{3t^2+27}{11} = \frac{3(t^2+9)}{11}$

Since $t^2 + 9 > 0$ for all $t \in \mathbb{R}$, we get

$f'(t) > 0 \quad$ for all $t \in \mathbb{R}$.

Therefore, f is strictly increasing on $\mathbb{R}$.

Hence, the correct option is:

$\boxed{\text{Option B}}$

Question27

Let A be a 3×3 matrix such that

$$\mathrm{A}^{\mathrm{T}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}, \mathrm{A}^{\mathrm{T}} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \mathrm{A} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} \text{ and } \mathrm{A} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

If $\det(A) = 1$, then $\det\left(\operatorname{adj}\left(A^2 + A\right)\right)$ is equal to:

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Options:

A.

16

B.

25

C.

49

D.

64

Answer: D

Solution:

Let

$$u = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad v = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

We are given

$$A^T u = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}, \qquad A^T v = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

and

$$Au = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}, \qquad Av = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

We have to find

$$\det\left(\operatorname{adj}(A^2 + A)\right)$$

given that

$$\det(A) = 1$$

Step 1: Use the relation $A(u - v)$

First note that

$$u - v = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

So,

$$A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = A(u - v) = Au - Av$$

Now,

$$Au - Av = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Hence the first column of A is

$$A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Also, since

$$A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

the third column of A is

$$\begin{bmatrix}1\\3\\1\end{bmatrix}$$

Now use

$$A\begin{bmatrix}1\\0\\1\end{bmatrix}=A\begin{bmatrix}1\\0\\0\end{bmatrix}+A\begin{bmatrix}0\\0\\1\end{bmatrix}$$

which gives

$$\begin{bmatrix}3\\4\\4\end{bmatrix}=\begin{bmatrix}2\\1\\3\end{bmatrix}+\begin{bmatrix}1\\3\\1\end{bmatrix}$$

This is consistent.

So let

$$A=\begin{bmatrix}2 & a & 1\\1 & b & 3\\3 & c & 1\end{bmatrix}$$

Now use $\det(A)=1$.

Step 2: Find the second column using determinant

Compute determinant:

$$\det(A)=\begin{vmatrix}2 & a & 1\\1 & b & 3\\3 & c & 1\end{vmatrix}$$

Expanding along the first row,

$$\det(A)=2\begin{vmatrix}b & 3\\c & 1\end{vmatrix}-a\begin{vmatrix}1 & 3\\3 & 1\end{vmatrix}+1\begin{vmatrix}1 & b\\3 & c\end{vmatrix}$$

$$=2(b-3c)-a(1-9)+(c-3b)$$

$$=2b-6c+8a+c-3b$$

$$=8a-b-5c$$

Given $\det(A)=1$,

$$8a-b-5c=1$$

But we still need more information. For that use the conditions involving A^T.

Step 3: Use transpose conditions

Since

$$A^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

this means the third column of A^T is $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, so the third row of A is

$$\begin{bmatrix} 3 & 1 & 1 \end{bmatrix}$$

Thus,

$c = 1$

Now A becomes

$$A = \begin{bmatrix} 2 & a & 1 \\ 1 & b & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Next, use

$$A^T \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$$

But

$$A^T \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = A^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + A^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Now $A^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is the first column of A^T, i.e. the first row of A, which is

$$\begin{bmatrix} 2 \\ a \\ 1 \end{bmatrix}$$

And

$$A^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 2 \\ a \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$$

So,

$$\begin{bmatrix} 5 \\ a+1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$$

Hence,

$a = 1$

Now use $\det(A) = 1$:

$8a - b - 5c = 1$

with $a = 1,\ c = 1$,

$8 - b - 5 = 1$

$3 - b = 1$

$b = 2$

Thus,

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Step 4: Find $A^2 + A$

First compute A^2:

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Multiplying row by column:

$$A^2 = \begin{bmatrix} 8 & 5 & 6 \\ 13 & 8 & 10 \\ 10 & 6 & 7 \end{bmatrix}$$

So,

$$A^2 + A = \begin{bmatrix} 8 & 5 & 6 \\ 13 & 8 & 10 \\ 10 & 6 & 7 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 7 \\ 14 & 10 & 13 \\ 13 & 7 & 8 \end{bmatrix}$$

Step 5: Use determinant property

For a 3×3 matrix M,

$\det(\operatorname{adj} M) = \det(M)^{3-1} = \det(M)^2$

So we first find

$\det(A^2 + A)$

Now,

$A^2 + A = A(A + I)$

Hence,

$\det(A^2 + A) = \det(A)\det(A + I)$

Given $\det(A) = 1$, so

$\det(A^2 + A) = \det(A + I)$

Now

$$A + I = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

Its determinant is

$$\det(A + I) = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 3 \\ 3 & 1 & 2 \end{vmatrix}$$

Expanding along the first row:

$$= 3\begin{vmatrix} 3 & 3 \\ 1 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix}$$

$= 3(6 - 3) - (2 - 9) + (1 - 9)$

$= 3 \cdot 3 - (-7) - 8$

$= 9 + 7 - 8 = 8$

So,

$\det(A^2 + A) = 8$

Therefore,

$\det(\text{adj}(A^2 + A)) = \det(A^2 + A)^2 = 8^2 = 64$

Final Answer

$\boxed{64}$

So, the correct option is **D**.

Question28

Let M be a 3×3 matrix such that $\mathrm{M}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathrm{M}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ **and** $\mathrm{M}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$**. If** $\mathrm{M}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 11 \end{pmatrix}$**, then** $x + y + z$ **equals :**

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Options:

A.

4

B.

5

C.

7

D.

11

Answer: B

Solution:

Since the images of the standard basis vectors are given, the columns of matrix M are exactly these vectors.

So,

$$M = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{pmatrix}$$

Now we are given

$$M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 11 \end{pmatrix}$$

So,

$$\begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 11 \end{pmatrix}$$

This gives the system:

$x - z = 1$

$2x + y + z = 7$

$3x + 2y + z = 11$

Now solve step by step.

From the first equation,

$x = 1 + z$

Substitute into the second equation:

$2(1+z)+y+z=7$

$2+3z+y=7$

$y+3z=5 \quad \Rightarrow \quad y=5-3z$

Now substitute $x=1+z$ and $y=5-3z$ into the third equation:

$3(1+z)+2(5-3z)+z=11$

$3+3z+10-6z+z=11$

$13-2z=11$

$-2z=-2$

$z=1$

Then,

$x=1+z=2$

and

$y=5-3z=2$

Therefore,

$x+y+z=2+2+1=5$

So the correct answer is:

$\boxed{5}$

Option B

Question29

If f : $\mathbf{N} \rightarrow \mathbf{Z}$ is defined by

$$f(n)=\begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}, k \in N,$$

and $\sum_{n=1}^{k} f(n)=98$, then k is equal to :

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Options:

A.

3

B.

4

C.

5

D.

6

Answer: A

Solution:

$$f(n)=\begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}$$

$$\sum_{n=1}^{k} f(n)=\begin{vmatrix} \frac{k(k+1)}{2} & -1 & -5 \\ -2\frac{k(k+1)(2k+1)}{6} & 3(2k+1) & 2k+1 \\ -3\frac{k^2(k+1)^2}{4} & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}=98$$

$$\begin{aligned} &\Rightarrow \frac{\mathrm{k}(\mathrm{k}+1)(2\mathrm{k}+1)}{2}\cdot\frac{7}{3}=98 \\ &\Rightarrow \mathrm{k}(\mathrm{k}+1)(2\mathrm{k}+1)=84 \\ &\Rightarrow \mathrm{k}=3 \end{aligned}$$

Question30

Let $\mathrm{A}=\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $\mathrm{B}=[\mathrm{b}_{ij}], 1 \leq i, j \leq 3$. If $\mathrm{B}=\mathrm{A}^{99}-\mathrm{I}$, then the value of $\frac{\mathrm{b}_{31}-\mathrm{b}_{21}}{\mathrm{b}_{32}}$ is :

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Options:

A.

99

B.

199

C.

149

D.

159

Answer: C

Solution:

$$A = P + I$$

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = P + I \text{ where } P^3 = 0$$

$$A^{99} = I + 99P + {}^{99}C_2 P^2$$

$$A^{99} - I = 99 \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + {}^{99}C_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 99 \times 3 & 0 & 0 \\ 44550 & 99 \times 3 & 0 \end{bmatrix}$$

$b_{31} = 44550, b_{21} = 99 \times 3, b_{32} = 99 \times 3$

$\therefore \frac{b_{31} - b_{21}}{b_{32}} = \frac{44550 - 297}{297} = 149$

Question31

The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$$x \cos 3\theta - 8y - 12z = 0$$
$$x \cos 2\theta + 3y + 3z = 0$$
$$x + y + 3z = 0$$

has a non-trivial solution, is equal to :

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Options:

A.

π

B.

2π

C.

3π

D.

4π

Answer: D

Solution:

$$\begin{vmatrix} \cos 3\theta & -8 & -12 \\ \cos 2\theta & 3 & 3 \\ 1 & 1 & 3 \end{vmatrix} = 0$$
$$\begin{vmatrix} \cos 3\theta & -8 & -4 \\ \cos 2\theta & 3 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$(C_1 \rightarrow C_1 - C_2) \& C_2 \rightarrow C_2 - C_3$
$$\begin{vmatrix} \cos 3\theta + 8 & -4 & -3 \\ \cos 2\theta - 3 & 2 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$2\cos 3\theta + 16 + 4\cos 2\theta - 12 = 0$
$\left(4\cos^3\theta - 3\cos\theta\right) + 2\left(2\cos^2\theta - 1\right) + 2 = 0$
$4\cos^3\theta + 4\cos^2\theta - 3\cos\theta = 0$
$\cos\theta\left(4\cos^2\theta + 4\cos\theta - 3\right) = 0$
$\cos\theta(2\cos\theta + 3)(2\cos\theta - 1) = 0$
$\cos\theta = 0, \frac{1}{2}, -\frac{3}{2}$ (rejected)
$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3}$
$\text{Sum } = 4\pi$

Question32

Let $A = \begin{bmatrix} \alpha & 1 & 2 \\ 2 & 3 & 0 \\ 0 & 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -5\alpha & 0 \\ 0 & 4\alpha & -2\alpha \end{bmatrix} + \operatorname{adj}(A)$. If $\det(B) = 66$, then $\det(\operatorname{adj}(A))$ equals :

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Options:

A.

289

B.

361

C.

441

D.

529

Answer: C

Solution:

$$A=\begin{bmatrix}\alpha & 1 & 2\\ 2 & 3 & 0\\ 0 & 4 & 5\end{bmatrix}$$

$$\Rightarrow |\boldsymbol{A}|=15\boldsymbol{\alpha}-10+16=15\boldsymbol{\alpha}+6$$

$$\text{Now adj } A=\begin{bmatrix}15 & 3 & -6\\ -10 & 5\alpha & 4\\ 8 & -4\alpha & 3\alpha-2\end{bmatrix}$$

$$\therefore \mathrm{B}=\begin{bmatrix}1 & 0 & 0\\ 0 & -5\alpha & 0\\ 0 & 4\alpha & -2\alpha\end{bmatrix}+\operatorname{adj}(\mathrm{A})$$

$$B=\begin{bmatrix}16 & 3 & -6\\ -10 & 0 & 4\\ 8 & 0 & \alpha-2\end{bmatrix}$$

$\&|\mathrm{~B}|=6(5\alpha+6)=66$ (given in question)
$\Rightarrow \alpha=1$
$\Rightarrow|\mathrm{~A}|=21$

$$\boldsymbol{A}\cdot \operatorname{Adj}(\boldsymbol{A})=|\boldsymbol{A}|\boldsymbol{I} \Rightarrow |\boldsymbol{A}\cdot \operatorname{Adj}(\boldsymbol{A})|=\||\boldsymbol{A}|\boldsymbol{I}|=\left(|\boldsymbol{A}|^3\right)|\boldsymbol{I}|=|\boldsymbol{A}|^3$$

$$\Rightarrow |\boldsymbol{A}||\operatorname{Adj}(\boldsymbol{A})|=|\boldsymbol{A}|^3 \Rightarrow |\operatorname{Adj}(\boldsymbol{A})|=|\boldsymbol{A}|^2=(21)^2=441$$

Question33

If the system of linear equations :

$$x+y+z=6$$
$$x+2y+5z=10$$
$$2x+3y+\lambda z=\mu$$

has infinitely many solutions, then the value of $\lambda+\mu$ equals:

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Options:

A.

12

B.

16

C.

22

D.

28

Answer: C

Solution:

To find the condition for infinitely many solutions, we first calculate the determinant D of the coefficients of x, y, z:

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 2 & 3 & \lambda \end{vmatrix} = \lambda - 6$$

For infinitely many solutions, $D = 0$. So, $\lambda - 6 = 0 \Rightarrow \lambda = 6$.

Now, let us find D_1, the determinant obtained by replacing the first column with the constants:

$$D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 5 \\ \mu & 3 & \lambda \end{vmatrix} = 2\lambda + 3\mu - 60$$

Next, the determinant D_2 is obtained by replacing the second column with the constants:

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 10 & 5 \\ 2 & \mu & \lambda \end{vmatrix} = 4(\lambda - \mu + 10)$$

Finally, D_3 is obtained by replacing the third column with the constants:

$$D_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 10 \\ 2 & 3 & \mu \end{vmatrix} = \mu - 16$$

For the system to have infinitely many solutions, all the determinants must be zero:

$$D = D_1 = D_2 = D_3 = 0$$

Substitute $\lambda = 6$:

From $D_3 = 0$, $\mu - 16 = 0 \Rightarrow \mu = 16$.

Thus,

$$\lambda + \mu = 6 + 16 = 22$$

Question34

Let A be a square matrix of order 3 such that $\det(A) = -2$ and $\det(3\,\mathrm{adj}(-6\,\mathrm{adj}(3A))) = 2^{m+n} \cdot 3^{mn}, m > n$. Then $4m + 2n$ is equal to ________.

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Answer: 34

Solution:

As $A\,\mathrm{adj}\,A = |A|I, \det(\lambda A) = \lambda^n \det A$

$\det(3\,\mathrm{adj}(-6\,\mathrm{adj}(3A))) = 3^3 \det(\mathrm{adj}(-6\,\mathrm{adj}(3A)))$

$= 3^3(-6\,\mathrm{adj}(3A))^2$

$= 3^3(-6)^6|3A|^4$

$= 3^9 2^6 \cdot 3^{12} \cdot (-2)^4$

$= 3^{21} \cdot 2^{10}$

Now comparing with given condition

$2^{m+n} 3^{mn} = 2^{10} \cdot 3^{21}$

$m + n = 10, mn = 21$

$\Rightarrow m = 7, n = 3 (m > n)$

$\therefore 4m + 2n = 28 + 6 = 34$

Question35

Let A be a 3×3 matrix such that $\mathrm{X}^{\mathrm{T}}\mathrm{AX} = \mathrm{O}$ for all nonzero 3×1 matrices $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. If A $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$, A $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -8 \end{bmatrix}$, and $\det(\mathrm{adj}(2(\mathrm{~A} + \mathrm{I}))) = 2^\alpha 3^\beta 5^\gamma, \alpha, \beta, \gamma \in N$, then $\alpha^2 + \beta^2 + \gamma^2$ is

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Answer: 44

Solution:

$X^T AX = 0$

$(xyz)\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$(xyz)\begin{pmatrix} a_1x + a_2y + a_3z \\ b_1x + b_2y + b_3z \\ c_1x + c_2y + c_3z \end{pmatrix} = 0$

$x\left(a_1x + a_2y + a_3z\right) + y\left(b_1x + b_2y + b_3z\right)$
$+ z\left(c_1x + c_2y + c_3z\right) = 0$
$a_1 = 0, b_2 = 0c_3 = 0$
$a_2 + b_1 = 0, a_3 + c_1 = 0, b_3 = c_2 = 0$

$A =$ skew symm matrix

$A = \begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix}; \quad A = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$

$\Rightarrow A = \begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$

$x + y = 1$
$-x + z = 4$
$y + z = 5$

$\begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -8 \end{pmatrix}$

$$\begin{array}{ll} 2x + y = 0 & x = -1 \\ -x + z = 4 & y = 2 \\ -y - 2z = -8 & z = 3 \end{array}$$

$A = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix}$

$2(\mathrm{~A} + \mathrm{I}) = \begin{pmatrix} 2 & -2 & 4 \\ 2 & 2 & 6 \\ -2 & -6 & 2 \end{pmatrix}$

$2(\mathrm{~A} + \mathrm{I}) = 120 \Rightarrow \det|\operatorname{adi}(2(\mathrm{~A} + \mathrm{I}))|$
$= 120^2 = 2^6 \cdot 3^2 \cdot 5^2$
$\alpha = 6, \beta = 2, \gamma = 2$

Question36

Let M denote the set of all real matrices of order 3×3 and let $\mathrm{S} = \{-3, -2, -1, 1, 2\}$. Let

$$\mathrm{S}_1 = \left\{\mathrm{A} = [a_{\mathrm{ij}}] \in \mathrm{M} : \mathrm{A} = \mathrm{A}^{\mathrm{T}} \text{ and } a_{\mathrm{ij}} \in \ \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\right\},$$
$$\mathrm{S}_2 = \left\{\mathrm{A} = [a_{\mathrm{ij}}] \in \mathrm{M} : \mathrm{A} = -\mathrm{A}^{\mathrm{T}} \text{ and } a_{\mathrm{ij}} \in \ \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\right\},$$
$$\mathrm{S}_3 = \{\mathrm{A} = [a_{\mathrm{ij}}] \in \mathrm{M} : a_{11} + a_{22} + a_{33} = 0 \text{ and } a_{\mathrm{ij}} \in \ \mathrm{S}, \forall \mathrm{i}, \mathrm{j}\}.$$

If $n\left(S_1 \cup S_2 \cup S_3\right) = 125\alpha$, then α equls __________.

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Answer: 1613

Solution:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

No. of elements in $\mathrm{S}_1 : \mathrm{A} = \mathrm{A}^{\mathrm{T}} \Rightarrow 5^3 \times 5^3$

No. of elements in $S_2 : A = -A^T \Rightarrow 0$

since no zero in S_2

No. of elements in $\mathrm{S}_3 \Rightarrow$

$$\left.\begin{array}{c} a_{11} + a_{22} + a_{33} = 0 \Rightarrow (1, 2, -3) \Rightarrow 31 \\ \text{or} \\ (1, 1, -2) \Rightarrow 3 \\ \text{or} \\ (-1, -1, 2) \Rightarrow 3 \end{array}\right\} \Rightarrow 12 \times 5^6$$

$$\begin{aligned} & \mathrm{n}(\mathrm{S}_1 \cap \mathrm{S}_3) = 12 \times 5^3 \\ & \mathrm{n}(\mathrm{S}_1 \cup \mathrm{S}_2 \cup \mathrm{S}_3) = 5^6(1 + 12) - 12 \times 5^3 \\ & \quad \Rightarrow 5^3 \times \left[13 \times 5^3 - 12\right] = 125\alpha \\ & \quad \alpha = 1613 \end{aligned}$$

Question37

Let $S = \left\{m \in \mathbf{Z} : A^{m^2} + A^m = 3I - A^{-6}\right\}$, where $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$. Then $n(S)$ is equal to __________.

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Answer: 2

Solution:

$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}, A^3 = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}, A^4 = \begin{bmatrix} 5 & -4 \\ 4 & -3 \end{bmatrix}$

and so on

$\mathrm{A}^6 = \begin{bmatrix} 7 & -6 \\ 6 & -5 \end{bmatrix}$

$\mathrm{A}^{\mathrm{m}} = \begin{bmatrix} \mathrm{m}+1 & -\mathrm{m} \\ \mathrm{m} & -\mathrm{m}+1 \end{bmatrix},$

$\mathrm{A}^{\mathrm{m}^2} = \begin{bmatrix} \mathrm{m}^2+1 & -\mathrm{m}^2 \\ \mathrm{m}^2 & -\left(\mathrm{m}^2-1\right) \end{bmatrix}$

$\mathrm{A}^{\mathrm{m}^2} + \mathrm{A}^{\mathrm{m}} = 3\mathrm{I} - \mathrm{A}^{-6}$

$\begin{bmatrix} \mathrm{m}^2+1 & -\mathrm{m}^2 \\ \mathrm{m}^2 & -\left(\mathrm{m}^2-1\right) \end{bmatrix} + \begin{bmatrix} \mathrm{m}+1 & -\mathrm{m} \\ \mathrm{m} & -\mathrm{m}+1 \end{bmatrix}$

$= 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -5 & 6 \\ -6 & 7 \end{bmatrix}$

$= \begin{bmatrix} 8 & -6 \\ 6 & -4 \end{bmatrix}$

$= \mathrm{m}^2 + 1 + \mathrm{m} + 1 = 8$

$= \mathrm{m}^2 + \mathrm{m} - 6 = 0 \Rightarrow \mathrm{m} = -3, 2$

$\mathrm{n}(\mathrm{s}) = 2$

Question38

Let I be the identity matrix of order 3×3 and for the matrix $A = \begin{bmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{bmatrix}$, $|A| = -1$. Let B be the inverse of the matrix $\operatorname{adj}\left(\mathrm{A}\operatorname{adj}\left(A^2\right)\right)$. Then $|(\lambda \mathrm{B} + \mathrm{I})|$ is equal to______

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Answer: 38

Solution:

$B = \left[\operatorname{adj}\left(A \operatorname{adj}\left(A^2\right)\right)\right]^{-1}$
$\operatorname{Adj}\left(A^2\right) = (\operatorname{adj} A)^2 \Rightarrow A \operatorname{adj}\left(A^2\right) = A \operatorname{adj}(A) \cdot (\operatorname{adj} A)$
$= A\left(|A| A^{-1}\right)^2 = |A|^2\left(A^{-1}\right) = A^{-1}$
$\Rightarrow B = \left(\operatorname{adj}\left(A^{-1}\right)\right)^{-1} = \left(\left|\left(A^{-1}\right)\right| A\right)^{-1} = \dfrac{A^{-1}}{-1} = -A^{-1}$
$\Rightarrow B = -A^{-1}$

$$|A| = -1 = \begin{matrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{matrix}\Bigg| = -1 \Rightarrow \lambda = 3$$

$$|3B + I| = \ I - 3A^{-1}\big| = \frac{|A|\left|I - 3A^{-1}\right|}{|A|} = \frac{|A - 3I|}{|A|}$$

$$= \frac{|A - 3I|}{-1} = \frac{\begin{matrix} 0 & 2 & 3 \\ 4 & 2 & 6 \\ 7 & -1 & -1 \end{matrix}\Bigg|}{-1} = 38$$

$\Rightarrow |3B + I| = 38$

Question39

Let $A = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$. If for some $\theta \in (0, \pi)$, $A^2 = A^T$, then the sum of the diagonal elements of the matrix $(\mathrm{A} + \mathrm{I})^3 + (\mathrm{A} - \mathrm{I})^3 - 6\ \mathrm{A}$ is equal to _________ .

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Answer: 6

Solution:

Note that A is orthogonal:

$AA^T = A^T A = I$ and $A^T = A^{-1}$

Given $A^2 = A^T$, then:

$A^3 = I$
$\operatorname{Tr}(A+I)^3 + (A-l)^3 - 6A = \operatorname{Tr}\left(2A^3 + 6A - 6A\right)$
$= \operatorname{Tr}\left(2A^3\right) = \operatorname{Tr}(2I)$
$\left(\text{ Using } (A+l)^3 + (A-I)^3 = 2A^3 + 6A \text{ and } 2A^3 = 2I\right) =$
6

Question40

The number of singular matrices of order 2 , whose elements are from the set $\{2, 3, 6, 9\}$, is __________.

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Answer: 36

Solution:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

for A to be singular matrix

$ad = bc$

Case 1: exactly 1 number is used $\Rightarrow {}^4C_1$ ways

Case 2 : exactly 2 numbers is used $\Rightarrow {}^4C_2$ ways

Case 3 : exactly 3 numbers used $\Rightarrow$ none will be singular.

Case 4: exactly 4 numbers is used

$\Rightarrow ab = cd \Rightarrow 2 \times 9 = 3 \times 6$
$\Rightarrow {}^4C_1 \times 2! = 8$ matrix .

$\therefore$ Total ways $\Rightarrow 4 + 6 \times 4 + 8 = 36$ matrices.

Question41

For a 3×3 matrix M, let trace (M) denote the sum of all the diagonal elements of M. Let A be a 3×3 matrix such that $|A| = \frac{1}{2}$ and trace $(A) = 3$. If $B = \mathrm{adj}(\mathrm{adj}(2A))$, then the value of $|B| +$ trace (B) equals :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. 56

B. 132

C. 174

D. 280

Answer: D

Solution:

$B = \mathrm{adj}(\mathrm{adj}(2A)) = \det(2A) \cdot (2A)$

Since A is a 3×3 matrix with

$\det(A) = \frac{1}{2}$,

the determinant of $2A$ is computed as

$\det(2A) = 2^3 \det(A) = 8 \cdot \frac{1}{2} = 4.$

Thus,

$B = 4 \cdot (2A) = 8A.$

Now, compute the determinant and the trace of B:

Determinant of B:

$\det(B) = \det(8A) = 8^3 \det(A) = 512 \cdot \frac{1}{2} = 256.$

Trace of B:

$\text{trace}(B) = \text{trace}(8A) = 8 \cdot \text{trace}(A) = 8 \cdot 3 = 24.$

Finally, adding these results:

$\det(B) + \text{trace}(B) = 256 + 24 = 280.$

Question42

If the system of linear equations :

$$x + y + 2z = 6$$
$$2x + 3y + \mathrm{az} = \mathrm{a} + 1$$
$$-x - 3y + \mathrm{b}z = 2\ \mathrm{b}$$

where $a, b \in \mathbf{R}$, has infinitely many solutions, then $7a + 3b$ is equal to :

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Options:

A. 12

B. 9

C. 22

D. 16

Answer: D

Solution:

We begin with the system:

$$\begin{aligned} x + y + 2z &= 6, \\ 2x + 3y + az &= a + 1, \\ -x - 3y + bz &= 2b. \end{aligned}$$

Step 1. Solve the first equation for x:

$x = 6 - y - 2z.$

Step 2. Substitute $x = 6 - y - 2z$ into the second equation:

$$2(6 - y - 2z) + 3y + az = a + 1.$$

Expanding and simplifying:

$$12 - 2y - 4z + 3y + az = a + 1 \quad \Longrightarrow \quad y + (a - 4)z = a - 11.$$

Call this **Equation (I)**.

Step 3. Substitute $x = 6 - y - 2z$ into the third equation:

$$-(6 - y - 2z) - 3y + bz = 2b.$$

Expanding and simplifying:

$$-6 + y + 2z - 3y + bz = 2b \quad \Longrightarrow \quad -2y + (b + 2)z = 2b + 6.$$

Call this **Equation (II)**.

Step 4. For the system to have infinitely many solutions, the two equations in y and z must be dependent—that is, one must be a constant multiple of the other. Assume there exists a constant k such that

$$-2 = k \cdot 1 \quad \Longrightarrow \quad k = -2.$$

Apply this to the coefficient of z and the constant term.

For the z-coefficient in Equations (I) and (II):

$$b + 2 = k(a - 4) = -2(a - 4) = -2a + 8.$$

Thus,

$$b = -2a + 6.$$

For the constant term:

$$2b + 6 = k(a - 11) = -2(a - 11) = -2a + 22.$$

Substitute $b = -2a + 6$ into this equation:

$$2(-2a + 6) + 6 = -2a + 22 \quad \Longrightarrow \quad -4a + 12 + 6 = -2a + 22.$$

Simplify:

$$-4a + 18 = -2a + 22.$$

Solve for a:

$$-4a + 18 + 4a = -2a + 22 + 4a \quad \Longrightarrow \quad 18 = 2a + 22,$$

$$2a = 18 - 22 = -4 \quad \Longrightarrow \quad a = -2.$$

Substitute $a = -2$ into $b = -2a + 6$:

$$b = -2(-2) + 6 = 4 + 6 = 10.$$

Step 5. We now compute

$$7a + 3b = 7(-2) + 3(10) = -14 + 30 = 16.$$

Thus, the value of $7a + 3b$ is

$$\boxed{16}.$$

Question43

If $\mathrm{A}, \mathrm{B}, \text{and} \left(\operatorname{adj}\left(\mathrm{A}^{-1}\right)+\operatorname{adj}\left(\mathrm{B}^{-1}\right)\right)$ **are non-singular matrices of same order, then the inverse of** $A\left(\operatorname{adj}\left(A^{-1}\right)+\operatorname{adj}\left(B^{-1}\right)\right)^{-1} B$**, is equal to**

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Options:

A. $\frac{AB^{-1}}{|A|}+\frac{BA^{-1}}{|B|}$

B. $\operatorname{adj}\left(\mathrm{B}^{-1}\right)+\operatorname{adj}\left(\mathrm{A}^{-1}\right)$

C. $\mathrm{AB}^{-1}+\mathrm{A}^{-1} \mathrm{~B}$

D. $\frac{1}{|AB|}(\operatorname{adj}(B)+\operatorname{adj}(A))$

Answer: D

Solution:

$$\left[\mathrm{A}\left(\operatorname{adj}\left(\mathrm{~A}^{-1}\right)+\operatorname{adj}\left(\mathrm{B}^{-1}\right)\right)^{-1} \cdot \mathrm{~B}\right]^{-1}$$
$$\mathrm{B}^{-1} \cdot\left(\operatorname{adj}\left(\mathrm{~A}^{-1}\right)+\operatorname{adj}\left(\mathrm{B}^{-1}\right)\right) \cdot \mathrm{A}^{-1}$$
$$\mathrm{B}^{-1} \operatorname{adj}\left(\mathrm{~A}^{-1}\right) \mathrm{A}^{-1}+\mathrm{B}^{-1}\left(\operatorname{adj}\left(\mathrm{~B}^{-1}\right)\right) \cdot \mathrm{A}^{-1}$$
$$\mathrm{B}^{-1}\left|\mathrm{~A}^{-1}\right| \mathrm{I}+\left|\mathrm{B}^{-1}\right| \mathrm{IA}^{-1}$$
$$\frac{\mathrm{B}^{-1}}{|\mathrm{~A}|}+\frac{\mathrm{A}^{-1}}{|\mathrm{~B}|}$$
$$\Rightarrow \frac{\operatorname{adjB}}{|\mathrm{B}||\mathrm{A}|}+\frac{\operatorname{adj}}{|\mathrm{A}||\mathrm{B}|}$$
$$=\frac{1}{|\mathrm{~A}||\mathrm{B}|}(\operatorname{adjB}+\operatorname{adjA})$$

Question44

If the system of equations

$$(\lambda-1)x+(\lambda-4)y+\lambda z=5$$
$$\lambda x+(\lambda-1)y+(\lambda-4)z=7$$
$$(\lambda+1)x+(\lambda+2)y-(\lambda+2)z=9$$

has infinitely many solutions, then $\lambda^2+\lambda$ is equal to

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Options:

A. 20

B. 10

C. 6

D. 12

Answer: D

Solution:

$(\lambda-1)x+(\lambda-4)y+\lambda z=5$
$\lambda x+(\lambda-1)y+(\lambda-4)z=7$
$(\lambda+1)x+(\lambda+2)y-(\lambda+2)z=9$
For infinitely many solutions

$$\mathrm{D}=\begin{vmatrix}\lambda-1 & \lambda-4 & \lambda \\ \lambda & \lambda-1 & \lambda-4 \\ \lambda+1 & \lambda+2 & -(\lambda+2)\end{vmatrix}=0$$

$(\lambda-3)(2\lambda+1)=0$

$$\mathrm{D_x}=\begin{vmatrix}5 & \lambda-4 & \lambda \\ 7 & \lambda-1 & \lambda-4 \\ 9 & \lambda+2 & -(\lambda+2)\end{vmatrix}=0$$

$2(3-\lambda)(23-2\lambda)=0$
$\lambda=3$
$\therefore \lambda^2+\lambda=9+3=12$

Question45

Let $A = [a_{ij}]$ be a 3×3 matrix such that $A\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$, $A\begin{bmatrix}4\\1\\3\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix}$ and $A\begin{bmatrix}2\\1\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$, then a_{23} equals :

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Options:

A. 2

B. -1

C. 1

D. 0

Answer: B

Solution:

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$A\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow \begin{bmatrix}a_{12}\\a_{22}\\a_{32}\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix} \Rightarrow \begin{array}{l} a_{22} = 0; a_{12} = 0 \\ a_{32} = 1 \end{array}$$

$$A\begin{bmatrix}4\\1\\3\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix} \Rightarrow \begin{array}{l} 4a_{11} + a_{12} + 3a_{13} = 0 \\ 4a_{21} + a_{22} + 3a_{23} = 1 \Rightarrow 4a_{21} + 3a_{23} = 1 \\ 4a_{31} + a_{32} + 3a_{33} = 0 \end{array}$$

$$A\begin{bmatrix}2\\1\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix} \Rightarrow \begin{array}{l} 2a_{11} + a_{12} + 2a_{13} = 1 \\ 2a_{21} + a_{22} + 2a_{23} = 0 \Rightarrow a_{21} + a_{23} = 0 \\ 2a_{31} + a_{32} + 2a_{33} = 0 \end{array}$$

$$-4a_{23} + 3a_{23} = 1 \Rightarrow a_{23} = -1$$

Question46

The system of equations

$$x+y+z=6,$$
$$x+2y+5z=9,$$
$$x+5y+\lambda z=\mu,$$

has no solution if

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Options:

A. $\lambda=17, \mu=18$

B. $\lambda=17, \mu \neq 18$

C. $\lambda=15, \mu \neq 17$

D. $\lambda \neq 17, \mu \neq 18$

Answer: B

Solution:

$$\mathrm{D}=\left|\begin{array}{ccc}1 & 1 & 1 \\ 1 & 2 & 5 \\ 1 & 5 & \lambda\end{array}\right|=0$$
$$\lambda=17$$
$$D_z=\left|\begin{array}{ccc}1 & 1 & 6 \\ 1 & 2 & 9 \\ 1 & 5 & \mu\end{array}\right| \neq 0$$
$$\mu \neq 18$$

Question47

If the system of equations

$$2x-y+z=4$$
$$5x+\lambda y+3z=12$$
$$100x-47y+\mu z=212$$

has infinitely many solutions, then $\mu-2\lambda$ is equal to

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Options:

A. 56

B. 59

C. 57

D. 55

Answer: C

Solution:

$$\Delta = 0 \Rightarrow \begin{vmatrix} 2 & -1 & 1 \\ 5 & \lambda & 3 \\ 100 & -47 & \mu \end{vmatrix} = 0$$

$2(\lambda\mu + 141) + (5\mu - 300) - 235 - 100\lambda = 0 \ldots (1)$

$$\Delta_3 = 0 \Rightarrow \begin{vmatrix} 2 & -1 & 4 \\ 5 & \lambda & 12 \\ 100 & -47 & 212 \end{vmatrix} = 0$$

$6\lambda = -12 \Rightarrow \lambda = -2$

Put $\lambda = 2$ in (1)

$2(-2\mu + 141) + 5\mu - 300 - 235 + 200 = 0$

$\mu = 53$

$\therefore 57$

Question48

If the system of equations

$$x + 2y - 3z = 2$$
$$2x + \lambda y + 5z = 5$$
$$14x + 3y + \mu z = 33$$

has infinitely many solutions, then $\lambda + \mu$ is equal to :

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Options:

A. 13

B. 10

C. 12

D. 11

Answer: C

Solution:

$$\mathrm{D}=\begin{vmatrix}1 & 2 & -3\\ 2 & \lambda & 5\\ 14 & 3 & \mu\end{vmatrix}=0, \lambda\mu+42\lambda-4\mu+107=0$$

$\mathrm{D}_1=2\lambda\mu+99\lambda-10\mu+255$
$\mathrm{D}_2=13-\mu$
$D_3=5\lambda+5$
$D_2=0 \Rightarrow \mu=13 \& D_3=0 \Rightarrow \lambda=-1$
check & verify for these values $\mathrm{D} \& \mathrm{D}_2=0$

Question49

For some a, b, let

$$f(x)=\begin{vmatrix}\mathrm{a}+\frac{\sin x}{x} & 1 & \mathrm{b}\\ \mathrm{a} & 1+\frac{\sin x}{x} & \mathrm{b}\\ \mathrm{a} & 1 & \mathrm{b}+\frac{\sin x}{x}\end{vmatrix}, x \neq 0, \lim_{x \rightarrow 0} f(x)=\lambda+\mu \mathrm{a}+\nu \mathrm{b}.$$

Then $(\lambda+\mu+v)^2$ is equal to :

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Options:

A. 25

B. 16

C. 9

D. 36

Answer: B

Solution:

$$\lim_{x\to 0} f(x)=\begin{vmatrix} a+1 & 1 & b \\ a & 1+1 & b \\ a & 1 & b+1 \end{vmatrix}$$
$$=(a+1)(2(b+1)-b)+1(ab-a(b+1))+ba$$
$$=(a+1)(b+2)-a+ab$$
$$=b+a+2=\lambda+\mu a+vb$$
$$\lambda=2, \mu=1, v=1 \Rightarrow (\lambda+\mu+v)^2=16$$

Question50

Let $\mathrm{A}=\begin{bmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{bmatrix}$ **and** $\mathrm{P}=\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \theta>0$**. If** $\mathrm{B}=\mathrm{PAP}^{\top}, \mathrm{C}=\mathrm{P}^{\top}\mathrm{B}^{10}\mathrm{P}$ **and the sum of the diagonal elements of** C **is** $\frac{m}{n}$**, where** $\gcd(m,n)=1$**, then** $m+n$ **is :**

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Options:

A.

127

B.

2049

C.

258

D.

65

Answer: D

Solution:

$$P=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$$
$$\because P^TP=I$$
$$B=PAPT$$

Pre multiply by P^T (Given)

$$P^T B=P^TPAP^T=AP^T$$

Now post multiply by P

$$P^TBP=AP^TP=A$$

$$\text{So } A^2=\underbrace{P^TBP\,P^T}_{I}\ BP$$

$$A^2=P^T B^2P$$

Similarly $A^{10}=P^TB^{10}P=C$

$$A=\begin{bmatrix}\frac{1}{\sqrt{2}} & -2\\ 0 & 1\end{bmatrix}\text{ (Given)}$$
$$\Rightarrow A^2=\begin{bmatrix}\frac{1}{2} & -\sqrt{2}-2\\ 0 & 1\end{bmatrix}$$

Similarly check A^3 and so on since $C=A^{10}$

$\Rightarrow$ Sum of diagonal elements of C is $\left(\frac{1}{\sqrt{2}}\right)^{10}+1$

$$=\frac{1}{32}+1=\frac{33}{32}=\frac{m}{n}$$
$$g\,cd(m,n)=1\text{ (Given)}$$
$$\Rightarrow m+n=65$$

Question51

Let M and m respectively be the maximum and the minimum values of

$$f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & 4\sin 4x\\ \sin^2 x & 1+\cos^2 x & 4\sin 4x\\ \sin^2 x & \cos^2 x & 1+4\sin 4x\end{vmatrix}, x\in R$$

Then M^4-m^4 is equal to :

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Options:

A.

1280

B.

1040

C.

1215

D.

1295

Answer: A

Solution:

$$\begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 4x \\ \sin^2 x & 1+\cos^2 x & 4\sin 4x \\ \sin^2 x & \cos^2 x & 1+4\sin 4x \end{vmatrix}, x \in R$$

$R_2 \rightarrow R_2 - R_1 \& R_3 \rightarrow R_3 - R_1$

$$f(x)\begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 4x \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

Expand about R_1, we get

$f(x) = 2 + 4\sin 4x$

$\therefore M =$ max value of $f(x) = 6$

$\mathrm{m} =$ min value of $\mathrm{f(x)} = -2$

$\therefore \mathrm{M}^4 - \mathrm{m}^4 = 1280$

Question52

Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij}, $C_{ij} = \sum\limits_{k=1}^{2} a_{ik} A_{jk}, 1 \leq i, j \leq 2$, and $C = [C_{ij}]$, then $8|C|$ is equal to :

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Options:

A.

288

B.

262

C.

222

D.

242

Answer: D

Solution:

To solve the problem, we need to determine the determinant of matrix A:

$A = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$

The determinant of A, denoted as $|A|$, is calculated as:

$|A| = (\log_5 128)(\log_4 25) - (\log_4 5)(\log_5 8)$

Evaluating each component:

$\log_5 128$ can be simplified using change of base formula:

$\log_5 128 = \frac{\log_{10} 128}{\log_{10} 5}$

$\log_4 5$ using change of base:

$\log_4 5 = \frac{\log_{10} 5}{\log_{10} 4}$

$\log_5 8$:

$\log_5 8 = \frac{\log_{10} 8}{\log_{10} 5}$

$\log_4 25$:

$\log_4 25 = \frac{\log_{10} 25}{\log_{10} 4}$

Now, substitute these values into $|A|$:

$|A| = \left(\frac{\log_{10} 128}{\log_{10} 5}\right)\left(\frac{\log_{10} 25}{\log_{10} 4}\right) - \left(\frac{\log_{10} 5}{\log_{10} 4}\right)\left(\frac{\log_{10} 8}{\log_{10} 5}\right)$

Next, we find the cofactors of matrix A:

$A_{11} = \log_4 25$

$A_{12} = -\log_5 8$

$A_{21} = -\log_4 5$

$A_{22} = \log_5 128$

Then, calculate matrix C whose elements are given by $C_{ij} = \sum_{k=1}^{2} a_{ik} A_{jk}$:

$C_{11} = a_{11}A_{11} + a_{12}A_{12} = |A| = \frac{11}{2}$

$C_{12} = a_{11}A_{21} + a_{12}A_{22} = 0$

$C_{21} = a_{21}A_{11} + a_{22}A_{12} = 0$

$C_{22} = a_{21}A_{21} + a_{22}A_{22} = |A| = \frac{11}{2}$

Thus, matrix C is:

$C = \begin{bmatrix} \frac{11}{2} & 0 \\ 0 & \frac{11}{2} \end{bmatrix}$

To find $|C|$, we compute:

$|C| = \left(\frac{11}{2}\right)\left(\frac{11}{2}\right) = \frac{121}{4}$

Finally, calculate $8|C|$:

$8|C| = 8 \times \frac{121}{4} = 242$

Question53

Let $A = [a_{ij}]$ be a matrix of order 3×3, with $a_{ij} = (\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}, \alpha, \beta \in \mathbf{Z}$, then $\alpha + \beta$ is equal to :

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Options:

A.

210

B.

280

C.

224

D.

168

Answer: C

Solution:

$$A=\begin{bmatrix}(\sqrt{2})^2 & (\sqrt{2})^3 & (\sqrt{2})^4\\ (\sqrt{2})^3 & (\sqrt{2})^4 & (\sqrt{2})^5\\ (\sqrt{2})^4 & (\sqrt{2})^5 & (\sqrt{2})^6\end{bmatrix}$$

$$A=\begin{bmatrix}2 & 2\sqrt{2} & 4\\ 2\sqrt{2} & 4 & 4\sqrt{2}\\ 4 & 4\sqrt{2} & 8\end{bmatrix}$$

$$A^2=2^2\begin{bmatrix}1 & \sqrt{2} & 2\\ \sqrt{2} & 2 & 2\sqrt{2}\\ 2 & 2\sqrt{2} & 4\end{bmatrix}\begin{bmatrix}1 & \sqrt{2} & 2\\ \sqrt{2} & 2 & 2\sqrt{2}\\ 2 & 2\sqrt{2} & 4\end{bmatrix}$$

$$=4\begin{bmatrix}- & - & -\\ - & - & -\\ (2+4+8) & (2\sqrt{2}+4\sqrt{2}+8\sqrt{2}) & (4+8+16)\end{bmatrix}$$

$$\begin{aligned}&\text{Sum of elements of } 3^{\text{rd}} \text{ row } = 4(14+14\sqrt{2}+28)\\ &\quad = 4(42+14\sqrt{2})\\ &\quad = 168+56\sqrt{2}\\ &\quad \alpha+\beta\sqrt{2}\\ \therefore\ &\quad \alpha+\beta=168+56=224\end{aligned}$$

Question54

Let $\alpha, \beta\ (\alpha \neq \beta)$ be the values of m, for which the equations $x+y+z=1$, $x+2y+4z=m$ and $x+4y+10z=m^2$ have infinitely many solutions. Then the value of $\sum\limits_{n=1}^{10}\left(n^\alpha+n^\beta\right)$ is equal to :

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Options:

A.

3410

B.

560

C.

3080

D.

440

Answer: D

Solution:

$$\Delta=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 4 & 10\end{vmatrix}=1(20-16)-1(10-4)+1(4-2)$$

$$=4-6+2=0$$

For infinite solutions

$$\Delta_x=\Delta_y=\Delta_z=0$$

$$m^2-3x+2=0$$

$$m=1,2$$

$$\alpha=1, \beta=2$$

$$\therefore \sum_{n=1}^{10}\left(n^\alpha+n^\beta\right)=\sum_{n=1}^{10} n^1+\sum_{n=1}^{10} n^2$$

$$=\frac{10(11)}{2}+\frac{10(11)(21)}{6}$$

$$=55+385$$

$$=440$$

Question55

Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is:

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Options:

A.

$\frac{5}{8}$

B.

$\frac{1}{4}$

C.

$\frac{3}{4}$

D.

$\frac{3}{8}$

Answer: D

Solution:

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$
$$= a_{11}a_{22} - a_{21}a_{12}$$
$$= \{-1, 0, 1\}$$

x	$\mathrm{P_i}$	$\mathrm{P_iX_i}$	$\mathrm{P_1X_i}^2$
-1	$\frac{3}{16}$	$-\frac{3}{16}$	$\frac{3}{16}$
0	$\frac{10}{16}$	0	0
1	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{3}{16}$
		$\sum \mathrm{P_iX_i} = 0$	$\sum \mathrm{P_iX_i}^2 = \frac{3}{8}$

$$\therefore \mathrm{var(x)} = \sum \mathrm{P_iX_i^2} - \left(\sum \mathrm{P_i}X_i\right)^2$$
$$= \frac{3}{8} - 0 = \frac{3}{8}$$

Question56

If the system of linear equations

$$3x + y + \beta z = 3$$
$$2x + \alpha y - z = -3$$
$$x + 2y + z = 4$$

has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :

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Options:

A. 31

B. 37

C. 43

D. 49

Answer: A

Solution:

$3x + y + \beta z = 3$
$2x + \alpha y - z = -3$
$x + 2y + z = 4$
has infinite solution
$\Rightarrow \Delta = 0, \Delta_1 = \Delta_2 = \Delta_3$

$$\Delta = 0 \Rightarrow \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\Delta_2 = 0 \Rightarrow \begin{vmatrix} 3 & 3 & \beta \\ 2 & -3 & -1 \\ 1 & 4 & 1 \end{vmatrix} = 0$$

$\Rightarrow 3(-3+4) - 3(2+1) + \beta(8+3) = 0$
$\Rightarrow 3 - 9 + 11\beta = 0$
$\Rightarrow \quad \beta = \frac{6}{11}$

$$\Delta_3 = 0 \Rightarrow \begin{vmatrix} 3 & 1 & 3 \\ 2 & \alpha & -3 \\ 1 & 2 & 4 \end{vmatrix} = 0$$

$\Rightarrow \quad 3(4\alpha + 6) - 1(8+3) + 3(4-\alpha) = 0$
$12\alpha + 18 - 11 + 12 - 3\alpha = 0$
$9\alpha = -19$
$\alpha = \frac{-19}{9}$

$\therefore \quad 22\beta - 9\alpha = 31$

Question57

Let $a \in R$ and A be a matrix of order 3×3 such that $\det(A) = -4$ and $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$, where I is the identity matrix of order 3×3. If $\det((a+1)\operatorname{adj}((a-1)A))$ is $2^{\mathrm{m}}3^{\mathrm{n}}$, m, $\mathrm{n} \in \{0, 1, 2, \ldots, 20\}$, then $\mathrm{m} + \mathrm{n}$ is equal to :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. 14

B. 17

C. 15

D. 16

Answer: D

Solution:

To solve for the value of $m+n$, we first establish the matrix A and determine the value of a. The condition given is:

$$A+I=\begin{bmatrix}1 & a & 1\\ 2 & 1 & 0\\ a & 1 & 2\end{bmatrix}$$

Since I is the identity matrix of size 3×3, we have:

$$A=\begin{bmatrix}1 & a & 1\\ 2 & 1 & 0\\ a & 1 & 2\end{bmatrix}-I=\begin{bmatrix}0 & a & 1\\ 2 & 0 & 0\\ a & 1 & 1\end{bmatrix}$$

We know $\det(A)=-4$. Calculating the determinant:

$\det(A)=0(0\times 1-0\times 1)-a(2\times 1-0\times a)+1(2\times 1-0\times 1)=-2a+2$

Setting $-2a+2=-4$, we solve for a:

$-2a+2=-4$

$-2a=-6$

$a=3$

With $a=3$, the problem is to find $\det((a+1)\operatorname{adj}((a-1)A))$. With $a=3$:

Calculate $(a+1)\operatorname{adj}((a-1)A)$:

$=4\operatorname{adj}(3A)$

Calculate the determinant:

$\det(4\operatorname{adj}(3A))=4^3\times\det(\operatorname{adj}(3A))$

Using the property $\det(\operatorname{adj}(A))=\det(A)^{n-1}$ for a $n\times n$ matrix:

$\operatorname{adj}(3A)=(3A)^{2-1}=(3\times\det(A))^2$

$=(3^3\times(-4))^2$

Therefore, calculate $\det(3A)^2$:

$\det((3A)^2)=(3^3\times(-4))^2=3^6\times 4^2$

Substitute back:

$4^3\times\det((3A)^2)=4^3\times 3^6\times 4^2=2^6\times 3^6\times 2^4$

Simplifying gives:

$=2^{10}\times 3^6$

Finally, we have powers $m=10$ and $n=6$. Therefore:

$m + n = 10 + 6 = 16$

Question58

Let A $= \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}, \alpha > 0$, such that $\det(\mathrm{A}) = 0$ and $\alpha + \beta = 1$. If I denotes 2×2 identity matrix, then the matrix $(I + A)^8$ is :

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Options:

A. $\begin{bmatrix} 257 & -64 \\ 514 & -127 \end{bmatrix}$

B. $\begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$

C. $\begin{bmatrix} 1025 & -511 \\ 2024 & -1024 \end{bmatrix}$

D. $\begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$

Answer: B

Solution:

Let $|A| = 0 \Rightarrow \alpha\beta - (-6) = 0 \Rightarrow \alpha\beta = -6$ and $\alpha + \beta = 1 \Rightarrow \alpha\beta$ are roots of the equation

$x^2 - x - 6 = 0 \Rightarrow x = 3, -2.$ Since $\alpha > 0$

$$\Rightarrow \quad \alpha = 3, \beta = -2$$

$$\Rightarrow \quad A = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix} \Rightarrow I + A = \begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$$

$$(I + A)^2 = \begin{bmatrix} 10 & -3 \\ 18 & -5 \end{bmatrix} \Rightarrow (I + A)^4 = \begin{bmatrix} 46 & -15 \\ 90 & -29 \end{bmatrix}$$

$$\Rightarrow \quad (I + A)^8 = \begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$$

Question59

Let A be a 3×3 real matrix such that $A^2(A-2I)-4(A-I)=O$, where I and O are the identity and null matrices, respectively. If $A^5=\alpha A^2+\beta A+\gamma I$, where α, β, and γ are real constants, then $\alpha+\beta+\gamma$ is equal to :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. 76

B. 12

C. 4

D. 20

Answer: B

Solution:

$$A^2(A-2I)-4(A-I)=0$$
$$A^3-2A^2-4A+4I=0$$

Multiply by A

$$A^4=2A^3+4A^2-4A$$
$$A^4=2\left(2A^2+4A-4I\right)+4A^2-4A$$
$$A^4=8A^2+4A-8I$$

Multiply again by A

$$\Rightarrow A^5=8A^3+4A^2-8A$$
$$\Rightarrow A^5=8\left(2A^2+4A-4I\right)+4A^2-8A$$
$$\Rightarrow A^5=20A^2+24A-32I$$

Comparing with $A^5=\alpha A^2+\beta A+\gamma I$

$$\alpha=20, \beta=24, \gamma=-32$$
$$\therefore \alpha+\beta+\gamma=20+24-32$$
$$=44-32$$
$$=12$$

Question60

If the system of equations

$$2x + \lambda y + 3z = 5$$
$$3x + 2y - z = 7$$
$$4x + 5y + \mu z = 9$$

has infinitely many solutions, then $\left(\lambda^2 + \mu^2\right)$ is equal to :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. 30

B. 26

C. 22

D. 18

Answer: B

Solution:

$2x + \lambda y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \mu z = 9$
For infinite solutions $\Rightarrow \Delta = 0 = \Delta_1 = \Delta_2 = \Delta_3$

$$\Delta = \begin{vmatrix} 2 & \lambda & 3 \\ 3 & 2 & -1 \\ 4 & 5 & \mu \end{vmatrix} = 0$$

$\Rightarrow -4\lambda - 3\lambda\mu + 4\mu + 31 = 0$

$$\Delta_1 = \begin{vmatrix} 5 & \lambda & 3 \\ 7 & 2 & -1 \\ 9 & 5 & \mu \end{vmatrix} = 0 \Rightarrow -9\lambda - 7\lambda\mu + 10\mu + 76 = 0$$

$$\Delta_2 = \begin{vmatrix} 2 & 3 & 5 \\ 3 & -1 & 7 \\ 4 & \mu & 9 \end{vmatrix} = 0 \Rightarrow \mu + 5 = 0 \Rightarrow \mu = -5$$

$$\Delta_3 = \begin{vmatrix} 2 & \lambda & 5 \\ 3 & 2 & 7 \\ 4 & 5 & 9 \end{vmatrix} = 0 \Rightarrow \lambda + 1 = 0 \Rightarrow \lambda = -1$$

$\therefore$ For infinite solution $\mu = -5$ and $\lambda = -1$
Now $\mu^2 + \lambda^2 = 25 + 1$
$= 26$

Question61

Let A be a matrix of order 3×3 and $|A| = 5$. If $|2\,\mathrm{adj}(3A\,\mathrm{adj}(2A))| = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma}, \alpha, \beta, \gamma \in N$, then $\alpha + \beta + \gamma$ is equal to

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. 26

B. 27

C. 25

D. 28

Answer: B

Solution:

To find the expression $|2\,\mathrm{adj}(3A\,\mathrm{adj}(2A))|$, we break it down as follows:

Recognize that:

$|2\,\mathrm{adj}(3A\,\mathrm{adj}(2A))| = 2^3|3A(\mathrm{adj}(2A))|^2$

Apply properties of determinants:

$= 2^3(3^3)^2|A|^2|\mathrm{adj}(2A)|^2$

Further simplify using $|\mathrm{adj}(B)| = |B|^{n-1}$ for a 3×3 matrix:

$= 2^3 \cdot 3^6 \cdot 5^2 \cdot (|2A|^2)^2$

Simplify $|2A|$:

$= 2^3 \cdot 3^6 \cdot 5^2 \cdot (2^3)^4 \cdot |A|^4$

Continue to simplify:

$= 2^3 \cdot 3^6 \cdot 5^2 \cdot (2^3)^4 \cdot 5^4$

Expand and combine powers:

$= 2^{15} \cdot 3^6 \cdot 5^6$

Therefore, $\alpha = 15$, $\beta = 6$, and $\gamma = 6$. So, $\alpha + \beta + \gamma = 15 + 6 + 6 = 27$.

Question62

Let the matrix $A=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}$ satisfy $A^n=A^{n-2}+A^2-I$ for $n\geqslant 3$.

Then the sum of all the elements of A^{50} is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. 44

B. 39

C. 52

D. 53

Answer: D

Solution:

$$A=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}$$
$$A^2=\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}\begin{bmatrix}1&0&0\\1&0&1\\0&1&0\end{bmatrix}=\begin{bmatrix}1&0&0\\1&1&0\\1&0&1\end{bmatrix}$$
$$A^3=A+A^2-I$$
$$A^3=\begin{bmatrix}1&0&0\\2&0&1\\1&1&0\end{bmatrix}$$
$$A^4=A^2+A^2-I=2A^2-I$$
$$A^4=\begin{bmatrix}1&0&0\\2&1&0\\2&0&1\end{bmatrix}\text{ and }A^5=\begin{bmatrix}1&0&0\\3&0&1\\2&1&0\end{bmatrix}$$
$$A^{50}=\begin{bmatrix}1&0&0\\25&1&0\\25&0&1\end{bmatrix}$$
$\text{Sum of elements }=53$

Question63

Let the system of equations :

$$2x+3y+5z=9$$
$$7x+3y-2z=8$$
$$12x+3y-(4+\lambda)z=16-\mu$$

have infinitely many solutions. Then the radius of the circle centred at (λ,μ) and touching the line $4x=3y$ is :

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $\frac{7}{5}$

B. $\frac{21}{5}$

C. 7

D. $\frac{17}{5}$

Answer: A

Solution:

$$\Delta=\begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 12 & 3 & -(4+\lambda) \end{vmatrix}$$
$$=2(-12-3\lambda+6)-3(-28-7\lambda+24)+5(21-36)$$
$$=-12-6\lambda+12+21\lambda-75$$
$$=15\lambda-75$$
$$\Rightarrow 15\lambda-75=0$$
$$\Rightarrow \lambda=5$$
$$\Delta_1=\begin{vmatrix} 9 & 3 & 5 \\ 8 & 3 & -2 \\ 16-\mu & 3 & -9 \end{vmatrix}$$
$$=9(-27+6)-3(-72+32-2\mu)+5(24-48+3\mu)$$
$$=-189+120+6\mu-120+15\mu$$
$$=21\mu-189=0$$
$$\Rightarrow \mu=9$$

$$\therefore r=\left|\frac{4(5)-3(9)}{\sqrt{(4)^2+(3)^2}}\right|$$
$$r=\frac{7}{5}$$

Question64

Let A be a 3×3 matrix such that $|\mathrm{adj}(\mathrm{adj}(\mathrm{adj}\ \mathrm{A}))|=81$.

If $S=\left\{n\in\mathbb{Z}:(|\,\mathrm{adj}(\mathrm{adj}\ A)|)^{\frac{(n-1)^2}{2}}=|A|^{(3n^2-5n-4)}\right\}$, then $\sum\limits_{n\in S}\left|A^{(n^2+n)}\right|$ is equal to :

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 820

B. 866

C. 750

D. 732

Answer: D

Solution:

$$|\,\mathrm{adj}(\mathrm{adj}(\mathrm{adj}\ A))|=81$$
$$=|A|^{(n-1)^3}=(3)^4\Rightarrow|A|^8=3^4\Rightarrow|A|=3^{1/2}$$
$$|\,\mathrm{adj}(\mathrm{adj}\ A)|^{\frac{(n-1)^2}{2}}=|A|^{(3n^2-5n-4)}$$
$$\left[|A|^{(n-1)^2}\right]^{\frac{(n-1)^2}{2}}=|A|^{3n^2-5n-4}$$
$$|A|^{2(n-1)^2}=|A|^{3n^2-5n-4}$$
$$\Rightarrow 2(n-1)^2=3n^2-5n-4$$
$$n^2-n-6=0$$
$$\Rightarrow n=-2,3$$
$$\sum_{x\leftarrow5}\left|A^{n^2+n}\right|=\left|A^2\right|+\left|A^{12}\right|$$
$$=3+3^6=732$$

Question65

Let the system of equations

$x + 5y - z = 1$

$4x + 3y - 3z = 7$

$24x + y + \lambda z = \mu$

$\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system,

if x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

4

B.

5

C.

3

D.

6

Answer: C

Solution:

For infinitely many solution

$\Delta = 0$

$$\begin{vmatrix} 1 & 5 & -1 \\ 4 & 3 & -3 \\ 24 & 1 & \lambda \end{vmatrix} = 0$$

$\Rightarrow 1(3\lambda + 3) - 5(4\lambda + 72) - 1(4 - 72) = 0$

$\Rightarrow -17\lambda + 3 - 4 \times 72 - 4 = 0$

$\Rightarrow 17\lambda = -289$

$\Rightarrow \lambda = -17$

$\Delta 1 = 0$

$$\Rightarrow \begin{vmatrix} 1 & 5 & -1 \\ 7 & 3 & -3 \\ \mu & 1 & -17 \end{vmatrix} = 0$$

$\Rightarrow 1(-51 + 3) - 5(-119 + 3\mu) - 1(7 - 3\mu) = 0$

$\Rightarrow -48 + 595 - 15\mu - 7 + 3\mu = 0$

$\Rightarrow 12\mu = 540$

$\mu = 45$

$x + 5y - z = 1$

$4x + 3y - 3z = 7$

$24x + y - 17z = 45$

Let $z = 1$

$x + 5y = 1 + \lambda] \times 4$

$4x + 3y = 7 + 3\lambda$

$$\frac{\underset{-}{4\mathrm{x}}+20\mathrm{y}=4+\underset{-}{4\lambda}}{-17\mathrm{y}=3-\lambda}$$

$$\mathrm{y} = \frac{\lambda - 3}{17}, \mathrm{x} = 1 + \lambda - \frac{5\lambda - 15}{17}$$

$$= \frac{32 - 12\lambda}{17}$$

$$7 \leq \frac{\lambda - 3}{17} + \frac{32 + 12\lambda}{17} + \lambda \leq 77$$

$$7 \leq \frac{30\lambda + 29}{17} \leq 77$$

$3 \leq \lambda \leq 42$

$\lambda = 3, 20, 37$

Question66

Let $A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}$**.**

If $\det(\operatorname{adj}(\operatorname{adj}(3A))) = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, **then** $m + n$ **is equal to**

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

22

B.

20

C.

24

D.

26

Answer: C

Solution:

$$|A|=\begin{vmatrix}2 & 2+p & 2+p+q\\ 4 & 6+2p & 8+3p+2q\\ 6 & 12+3p & 20+6p+3q\end{vmatrix}$$

$\mathrm{C}_3 \rightarrow \mathrm{C}_3-\mathrm{C}_2-\mathrm{C}_1 \times \frac{\mathrm{q}}{2}$

Then $\mathrm{C}_3 \rightarrow \mathrm{C}_2-\mathrm{C}_1 \mathrm{X}\left(1+\frac{\mathrm{p}}{2}\right)$

$$\Rightarrow|\mathrm{A}|=\begin{vmatrix}2 & 0 & 0\\ 4 & 2 & 2+\mathrm{p}\\ 6 & 6 & 8+3\mathrm{p}\end{vmatrix}$$

$\Rightarrow|\mathrm{A}|=2(16+6\mathrm{p}-12-6\mathrm{p})=8=2^3$

$|\operatorname{adj}(\operatorname{adj}(3\mathrm{~A}))|=|3\mathrm{~A}|^{(3-1)^2}=|3\mathrm{~A}|^4$

$=\left(3^3|\mathrm{~A}|\right)^4=\left(3^3 \times 2^3\right)^4=2^{12} \times 3^{12}$

$\Rightarrow \mathrm{m}+\mathrm{n}=24$

Question67

Let α be a solution of $x^2+x+1=0$, and for some a and b in

$R, \begin{bmatrix} 4 & a & b \end{bmatrix} \begin{bmatrix} 1 & 16 & 13 \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$**. If** $\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3$**, then** $\boldsymbol{m + n}$ **is equal to ________**

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

11

B.

3

C.

8

D.

7

Answer: A

Solution:

Let α be a solution of the equation $x^2 + x + 1 = 0$. This implies that α is a cube root of unity, denoted as ω, where $\alpha^2 + \alpha + 1 = 0$.

We are given matrix equations with parameters a and b as follows:

$$\begin{bmatrix} 4 & a & b \end{bmatrix} \begin{bmatrix} 1 & 16 & 13 \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Solving these equations:

$[4 - a - 2b, 64 - a - 14b, 52 + 2a - 8b] = [0, 0, 0]$
$\Rightarrow a + 2b = 4$
$\Rightarrow a + 14b = 64$

From these, solve for a and b:

$12b = 60 \Rightarrow b = 5$
$a = -6$

We now substitute a and b into the given function:

$\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3$

We substitute $\alpha = \omega$ and simplify:

$$\frac{4}{\omega} + \frac{m}{1} + \frac{n}{\omega^2} = 3$$
$$\Rightarrow 4\omega^2 + m + n\omega = 3$$

Substitute $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$:

$$4\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + m + n\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 3$$
$$\Rightarrow -2 + m - \frac{n}{2} = 3$$
and $\dfrac{-4\sqrt{3}}{2} + \dfrac{n\sqrt{3}}{2} = 0$

From the above, solve for n and m:

$n = 4$
$m = 7$

Thus, $m + n = 11.$

Question68

Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Given below are two statements :

Statement I: $f(-x)$ is the inverse of the matrix $f(x)$.

Statement II: $f(x)f(y) = f(x+y)$.

In the light of the above statements, choose the correct answer from the options given below

[27-Jan-2024 Shift 1]

Options:

A.

Statement I is false but Statement II is true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Both Statement I and Statement II are true

Answer: D

Solution:

$$f(-x)=\begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$f(x)\cdot f(-x)=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}=I$$

Hence statement- I is correct

Now, checking statement II

$$f(y)=\begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$f(x)\cdot f(y)=\begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow f(x)\cdot f(y)=f(x+y)$

Hence statement-II is also correct.

Question69

Let A be a 2 × 2 real matrix and I be the identity matrix of order 2 . If the roots of the equation $|A-xI|=0$ be -1 and 3 , then the sum of the diagonal elements of the matrix A^2 is...............

[27-Jan-2024 Shift 2]

Answer: 10

Solution:

$|A-xI|=0$

Roots are -1 and 3

Sum of roots $=\operatorname{tr}(A)=2$

Product of roots $=|A|=-3$

Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

We have $a+d=2$

$ad-bc=-3$

$$A^2=\begin{bmatrix} a & b \\ c & d \end{bmatrix}\times\begin{bmatrix} a & b \\ c & d \end{bmatrix}=\begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$$

We need $a^2+bc+bc+d^2$

$=a^2+2bc+d^2$

$=(a+d)^2-2ad+2bc$

$=4-2(ad-bc)$

$=4-2(-3)$

$=4+6$

$=10$

Question70

Let $A=\begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ and $|2A|^3=2^{21}$ where $\alpha,\beta\in Z$, Then a value of α is

[29-Jan-2024 Shift 1]

Options:

A.

3

B.

5

C.

17

D.

9

Answer: B

Solution:

$|A| = \alpha^2 - \beta^2$

$|2A|^3 = 2^{21} \Rightarrow A| = 2^4$

$\alpha^2 - \beta^2 = 16$

$(\alpha+\beta)(\alpha-\beta) = 16 \Rightarrow \alpha = 4 \text{ or } 5$

Question71

Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2}A[(A+A^T)^2 + (A-A^T)^2]$ is equal to

[29-Jan-2024 Shift 1]

Options:

A.

$A^2 + I$

B.

$A^3 + I$

C.

$A^2 + A^T$

D.

$A^3 + A^T$

Answer: D

Solution:

$AA^T = I = A^TA$

On solving given expression, we get

$$\frac{1}{2}A[A^2 + (A^T)^2 + 2AA^T + A^2 + (A^T)^2 - 2AA^T]$$

$$= A[A^2 + (A^T)^2] = A^3 + A^T$$

Question72

Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $|P^{-1}AP - 2I|$ is equal to

[29-Jan-2024 Shift 2]

Options:

A.

26

B.

27

C.

66

D.

Answer: A

Solution:

$$|P^{-1}AP - 2I| = |P^{-1}AP - 2P^{-1}P|$$

$$= |P^{-1}(A - 2I)P|$$

$$= |P^{-1}|\,|A - 2I|\,|P|$$

$$= |A - 2I|$$

$$= \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69$$

So, Prime factor of 69 is 3 & 23

So, sum $= 26$

Question73

Let R = $\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ **be a non-zero 3 × 3 matrix, where** $x\sin\theta = y\sin\left(\theta + \frac{2\pi}{3}\right) = z\sin\left(\theta + \frac{4\pi}{3}\right) \neq 0, \theta \in (0, 2\pi).$ **For a square matrix M, let trace (M) denote the sum of all the diagonal entries of M. Then, among the statements:**

(I) Trace (R) = 0

(II) If trace(adj (adj(R)) = 0, then R has exactly one non-zero entry.

[30-Jan-2024 Shift 2]

Options:

A.

Both (I) and (II) are true

B.

Neither (I) nor (II) is true

C.

Only (II) is true

D.

Only (I) is true

Answer: C

Solution:

$$x\sin\theta = y\sin\left(\theta+\frac{2\pi}{3}\right) = z\sin\left(\theta+\frac{4\pi}{3}\right) \neq 0$$

$\Rightarrow x, y, z \neq 0$

Also,

$$\sin\theta + \sin\left(\theta+\frac{2\pi}{3}\right) + \sin\left(\theta+\frac{4\pi}{3}\right) = 0 \ \forall\theta \in R$$

$$\Rightarrow \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=0$$

$\Rightarrow xy+yz+zx=0$

(i) Trace (R) $= x+y+z$

If $x+y+z=0$ and $xy+yz+zx=0$

$\Rightarrow x=y=z=0$

Statement (i) is False

(ii) Let P : trace (Adj(Adj(R))) $=0$

Q : R has exactly one non zero entry

if P is false then $P \longrightarrow Q$ is always true

Statement (ii) is True

Question74

Let A be a 3×3 matrix and $\det(A) = 2$. If

$$n = \det(\underbrace{\operatorname{adj}(\operatorname{adj}(\ldots\ldots(\operatorname{adj} A)))}_{2024-\text{ times}})$$

Then the remainder when n is divided by 9 is equal to______

[31-Jan-2024 Shift 2]

Options:

Answer: 7

Solution:

$|A| = 2$

$$\underbrace{\operatorname{adj}(\operatorname{adj}(\operatorname{adj}\ldots\ldots(a)))}_{2024-\text{ times}} = |A|^{(n-1)^{2024}}$$

$= |A|^{2024}$

$= 2^{2^{2024}}$

$2^{2024} = (2^2)2^{2022} = 4(8)^{674} = 4(9-1)^{674}$

$\Rightarrow 2^{2024} \equiv 4(\text{mod } 9)$

$\Rightarrow 2^{2024} \equiv 9\text{m} + 4, \text{ m} \leftarrow \text{ even}$

$2^{9\text{m}+4} \equiv 16 \cdot (2^3)^{3\text{m}} \equiv 16(\text{mod } 9)$

$\equiv 7$

Question75

If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, C = ABA^T$ and $X = A^T C^2 A$, then $\det X$ is equal to :

[1-Feb-2024 Shift 1]

Options:

A.

243

B.

729

C.

27

D.

891

Answer: B

Solution:

$$A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix} \Rightarrow \det(A) = 3$$

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \det(B) = 1$$

Now $C = ABA^T \Rightarrow \det(C) = (\det(A))^2 x \det(B)$

$|C| = 9$

Now $|X| = |A^T C^2 A|$

$= |A^T| \, |C|^2 \, |A|$

$= |A|^2 \, |C|^2$

$= 9 \times 81$

$= 729$

Question76

Let $A = I_2 - 2MM^T$, where M is real matrix of order 2×1 such that the relation $M^TM = I_1$ holds. If λ is a real number such that the relation AX

= λX holds for some non-zero real matrix X of order 2 × 1, then the sum of squares of all possible values of λ is equal to :

[1-Feb-2024 Shift 2]

Answer: 2

Solution:

$A = I_2 - 2MM^T$

$A^2 = (I_2 - 2MM^T)(I_2 - 2MM^T)$

$= I_2 - 2MM^T - 2MM^T + 4MM^TMM^T$

$= I_2 - 4MM^T + 4MM^T$

$= I_2$

$AX = \lambda X$

$A^2X = \lambda AX$

$X = \lambda(\lambda X)$

$X = \lambda^2 X$

$X(\lambda^2 - 1) = 0$

$\lambda^2 = 1$

$\lambda = \pm 1$

Sum of square of all possible values $= 2$

Question77

If A and B are two non-zero n × n matrics such that $A^2 + B = A^2B$, then [24-Jan-2023 Shift 1]

Options:

A. $AB = I$

B. $A^2B = I$

C. $A^2 = I$ or $B = I$

D. $A^2B = BA^2$

Answer: D

Solution:

$$
\begin{aligned}
&A^2 + B = A^2B \\
&(A^2 - I)(B - I) = I \\
&A^2 + B = A^2B \\
&A^2(B - I) = B \\
&A^2 = B(B - I)^{-1} \\
&A^2 = B(A^2 - I) \\
&A^2 = BA^2 - B \\
&A^2 + B = BA^2 \\
&A^2B = BA^2
\end{aligned}
$$

Question78

The number of square matrices of order 5 with entries from the set {0, 1}, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1 , is
[24-Jan-2023 Shift 2]

Options:

A. 225

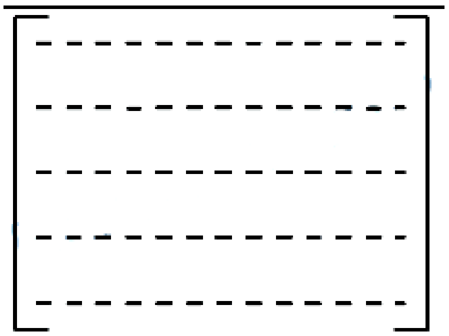
B. 120

C. 150

D. 125

Answer: B

Solution:



In each row and each column exactly one is to be placed -
∴ No. of such matrices $= 5 \times 4 \times 3 \times 2 \times 1 = 120$
Alternate :

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \rightarrow 5 \text{ ways} \\ \rightarrow 4 \text{ ways} \\ \rightarrow 3 \text{ ways} \\ \rightarrow 2 \text{ ways} \\ \rightarrow 1 \text{ ways} \end{array}$$

Step-1 : Select any 1 place for 1 's in row 1. Automatically some column will get filled with 0 's.
Step-2 : From next now select 1 place for 1 's. Automatically some column will get filled with 0 's. ⇒ Each time one less place will be available for putting 1's.
Repeat step-2 till last row.
Req. ways $= 5 \times 4 \times 3 \times 2 \times 1 = 120$

Question79

Let A be a 3 × 3 matrix such that $|\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A))| = 12^4$. Then $\left|A^{-1} \operatorname{adj} A\right|$ is equal to
[24-Jan-2023 Shift 2]

Options:

A. $2\sqrt{3}$

B. $\sqrt{6}$

C. 12

D. 1

Answer: A

Solution:

Given $|\text{adj}(\text{adj}(\text{adj} . A))| = 12^4$
$\Rightarrow |A|^{(n-1)^3} = 12^4$
We are asked
$|A^{-1} \cdot \text{adj } A|$
$= |A^{-1}| \cdot |\text{adj } A|$
$= \frac{1}{|A|} \cdot |A|^{3-1}$
$= |A| = 2\sqrt{3}$

Question80

Let x, y, z > 1 and

$$A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}.$$

Then $\left|\text{adj}(\text{adj } A^2)\right|$ is equal to

[25-Jan-2023 Shift 1]

Options:

A. 6^4

B. 2^8

C. 4^8

D. 2^4

Answer: B

Solution:

$$|A| = \frac{1}{\log x \cdot \log y \cdot \log z} \begin{vmatrix} \log x & \log y & \log z \\ \log x & 2\log y & \log z \\ \log x & \log y & 3\log z \end{vmatrix} = 2$$

$$\Rightarrow \left|\text{adj}(\text{adj } A^2)\right| = \left|A^2\right|^4 = 2^8$$

Question81

Let A = $\begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and B = $\begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where i = $\sqrt{-1}$. If

M = $A^T BA$, then the inverse of the matrix $AM^{2023}A^T$ is
[25-Jan-2023 Shift 2]

Options:

A. $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

Answer: D

Solution:

$$AA^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} & & \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} cc \frac{1}{\sqrt{10}} & \frac{-3}{\sqrt{10}} & \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} & & \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2i \\ 0 & 1 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 1 & -3i \\ 0 & 1 \end{bmatrix}$$

.

.
.
$B^{2023} = \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$
$M = A^T BA$
$M^2 = M \cdot M = A^T BA A^T BA = A^T B^2 A$
$M^3 = M^2 \cdot M = A^T B^2 AA^T BA = A^T B^3 A$
.
.
.
$M^{2023} = \ldots \cdots \cdots \cdots \cdot A^T B^{2023} A$
$AM^{2023} A^T = AA^T B^{2023} AA^T = B^{2023}$
$= \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$
Inverse of $(AM^{2023} A^T)$ is $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

Question82

Let A, B, C be 3 × 3 matrices such that A is symmetric and B and C are skew-symmetric.
Consider the statements
(S1)$A^{13}B^{26} - B^{26}A^{13}$ is symmetric
(S2) $A^{26}C^{13} - C^{13}A^{26}$ is symmetric
Then.
[25-Jan-2023 Shift 2]

Options:

A. Only S2 is true

B. Only S1 is true

C. Both S1 and S2 are false

D. Both S1 and S2 are true

Answer: A

Solution:

Given, $A^T = A, B^T = -B, C^T = -C$
Let $M = A^{13}B^{26} - B^{26}A^{13}$

Then, $M^T = (A^{13}B^{26} - B^{26}A^{13})^T$
$= (A^{13}B^{26})^T - (B^{26}A^{13})^T$
$= (B^T)^{26}(A^T)^{13} - (A^T)^{13}(B^T)^{26}$
$= B^{26}A^{13} - A^{13}B^{26} = -M$
Hence, M is skew symmetric
Let, $N = A^{26}C^{13} - C^{13}A^{26}$
then, $N^T = (A^{26}C^{13})^T - (C^{13}A^{26})^T$
$= -(C)^{13}(A)^{26} + A^{26}C^{13} = N$
Hence, N is symmetric.
∴ Only S2 is true.

Question83

Let α and β be real numbers. Consider a 3 × 3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then
[29-Jan-2023 Shift 1]

Options:

A. $\alpha = 1$

B. $\alpha = 4$

C. $\beta = 8$

D. $\beta = -8$

Answer: D

Solution:

$A^2 = 3A + \alpha I$
$A^3 = 3A^2 + \alpha A$
$A^3 = 3(3A + \alpha I) + \alpha A$
$A^3 = 9A + \alpha A + 3\alpha I$
$A^4 = (9 + \alpha)A^2 + 3\alpha A$
$= (9 + \alpha)(3A + \alpha I) + 3\alpha A$
$= A(27 + 6\alpha) + \alpha(9 + \alpha)$
$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$
$\Rightarrow \beta = \alpha(9 + \alpha) = -8$

Question84

The set of all values of t ∈ ℝ, for which the matrix

$$\begin{bmatrix} e^t & e^{-t}(\sin t-2\cos t) & e^{-t}(-2\sin t-\cos t) \\ e^t & e^{-t}(2\sin t+\cos t) & e^{-t}(\sin t-2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{bmatrix}$$

is invertible, is
[29-Jan-2023 Shift 2]

Options:

A. $\left\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\right\}$

B. $\left\{k\pi+\frac{\pi}{4}, k \in \mathbb{Z}\right\}$

C. $\{k\pi, k \in \mathbb{Z}\}$

D. $\mathbb{R}$

Answer: D

Solution:

If its invertible, then determinant value $\neq 0$
So,

$$\begin{vmatrix} e^t & e^{-t}(\sin t-2\cos t) & e^{-t}(-2\sin t-\cos t) \\ e^t & e^{-t}(2\sin t+\cos t) & e^{-t}(\sin t-2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{vmatrix} \neq 0$$

$$\Rightarrow e^t \cdot e^{-t} \cdot e^{-t}\begin{vmatrix} 1 & \sin t-2\cos t & -2\sin t-\cos t \\ 1 & 2\sin t+\cos t & \sin t-2\cos t \\ 1 & \cos t & \sin t \end{vmatrix} \neq 0$$

Applying, $R_1 \to R_1 - R_2$ then $R_2 \to R_2 - R_3$
We get

$$e^{-t}\begin{vmatrix} 0 & -\sin t-\cos t & -3\sin t+\cos t \\ 0 & 2\sin t & -2\cos t \\ 1 & \cos t & \sin t \end{vmatrix} \neq 0$$

By expanding we have,

$$e^{-t} \times 1(2\sin t\cos t+6\cos^2 t+6\sin^2 t-2\sin t\cos t) \neq 0$$

$$\Rightarrow e^{-t} \times 6 \neq 0$$

for $\forall t \in$

Question85

Let A be a symmetric matrix such that |A| = 2 and

$$\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$$ **. If the sum of the diagonal elements of A is s,**

then $\frac{\beta s}{\alpha^2}$ is equal to ________.

[29-Jan-2023 Shift 2]

Answer: 5

Solution:

$$\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$$

Now $ac - b^2 = 2$ and $2a + b = 1$ and $2b + c = 2$

solving all these above equations we get

$\frac{1-b}{2} \times \left(\frac{2-2b}{1} \right) - b^2 = 2$

$\Rightarrow (1-b)^2 - b^2 = 2$

$\Rightarrow 1 - 2b = 2$

$\Rightarrow b = -\frac{1}{2}$ and $a = \frac{3}{4}$ and $c = 3$

Hence $\alpha = 3a + \frac{3b}{2} = \frac{9}{4} - \frac{3}{4} = \frac{3}{2}$

and $\beta = 3b + \frac{3c}{2} = -\frac{3}{2} + \frac{9}{2} = 3$

also $s = a + c = \frac{15}{4}$

$\therefore \frac{\beta s}{\alpha^2} = \frac{3 \times 15}{4 \times \frac{9}{4}} = 5$

Question86

Let $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = \left|A\right| \neq 0 \left|A - d(\text{Adj } A)\right| = 0$

Then
[30-Jan-2023 Shift 1]

Options:

A. $(1+d)^2 = (m+q)^2$

B. $1+d^2 = (m+q)^2$

C. $(1+d)^2 = m^2+q^2$

D. $1+d^2 = m^2+q^2$

Answer: A

Solution:

Sol. $A = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$, $\left|A - d(\text{adj } A)\right| = 0$

$\Rightarrow |A - d(\text{adj } A)| = \left| \begin{bmatrix} m & n \\ p & q \end{bmatrix} - d \begin{bmatrix} q & -n \\ -p & m \end{bmatrix} \right|$

$= \begin{vmatrix} m-qd & n(1+d) \\ p(1+d) & q-md \end{vmatrix} = 0$

$\Rightarrow (m-qd)(q-md) - np(1+d)^2 = 0$

$\Rightarrow mq - m^2d - q^2d + mqd^2 - np(1+d)^2 = 0$

$\Rightarrow (mq-np) + d^2(mq-np) - d(m^2+q^2+2np) = 0$

$\Rightarrow d + d^3 - d((m+q)^2 - 2d) = 0$

$\Rightarrow 1 + d^2 = (m+q)^2 - 2d$

$\Rightarrow (1+d)^2 = (m+q)^2$

Question87

If P is a 3 × 3 real matrix such that $P^T = aP + (a-1)I$, where $a > 1$, then [30-Jan-2023 Shift 2]

Options:

A. P is a singular matrix

B. $|\text{Adj } P| > 1$

C. $|\text{Adj } P| = \frac{1}{2}$

D. $|\text{Adj } P| = 1$

Answer: D

Solution:

$$
\begin{aligned}
&P^T = aP + (a-1)I \\
&\Rightarrow P = aP^T + (a-1)I \\
&\Rightarrow P^T - P = a(P - P^T) \\
&\Rightarrow P = P^T, \quad \text{as} \quad a \neq -1 \\
&\text{Now,} \;\; P = aP + (a-1)I \\
&\Rightarrow P = -I \Rightarrow |P| = 1 \\
&\Rightarrow |\text{Adj } P| = 1
\end{aligned}
$$

Question88

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to:
[31-Jan-2023 Shift 1]

Options:

A. 6144

B. 4094

C. 4097

D. 2050

Answer: C

Solution:

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} = A$$

$\Rightarrow A^3 = A^4 = \ldots\ldots = A$

$(A+I)^{11} = {}^{11}C_0 A^{11} + {}^{11}C_1 A^{10} + \ldots .\ {}^{11}C_{10} A + {}^{11}C_{11} I$

$= ({}^{11}C_0 + {}^{11}C_1 + \ldots .\ {}^{11}C_{10})A + I$

$= (2^{11} - 1)A + I = 2047A + I$

$\therefore$ Sum of diagonal elements $= 2047(1 + 4 - 3) + 3$

$= 4094 + 3 = 4097$

Question89

Let $A = [a_{ij}]$, $a_{ij} \in Z \cap [0, 4]$, $1 \le i, j \le 2$. The number of matrices A such that the sum of all entries is a prime number $p \in (2, 13)$ is ________. [31-Jan-2023 Shift 2]

Answer: 204

Solution:

As given $a + b + c + d = 3$ or 5 or 7 or 11

if sum $= 3$

$(1 + x + x^2 + \ldots + x^4)^4 \rightarrow x^3$

$(1 - x^5)^4 (1 - x)^{-4} \rightarrow x^3$

$\therefore {}^{4+3-1}C_3 = {}^6C_3 = 20$

If sum = 5

$(1 - 4x^5)(1 - x)^{-4} \rightarrow x^5$

$\Rightarrow {}^{4+5-1}C_5 - 4x^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$

If sum $= 7$

$(1 - 4x^5)(1 - x)^{-4} \rightarrow x^7$

$\Rightarrow {}^{4+5-1}C_4 - {}^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$

If sum $= 11$

$(1 - 4x^5 + 6x^{10})(1 - x)^{-4} \rightarrow x^{11}$

$\Rightarrow {}^{4+11-1}C_{11} - 4 \cdot {}^{4+6-4}C_6 + 6 \cdot {}^{4+1-1}C_1$

$= {}^{14}C_{11} - 4 \cdot {}^9C_6 + 6.4 = 364 - 336 + 24 = 52$

$\therefore$ Total matrices $= 20 + 52 + 80 + 52 = 204$

Question90

Let A be a n × n matrix such that $|A| = 2$. If the determinant of the matrix $Adj(2 . Adj(2A^{-1}))$. is 2^{84}, then n is equal to ______.
[31-Jan-2023 Shift 2]

Answer: 5

Solution:

$$|\operatorname{Adj}(2\operatorname{Adj}(2A^{-1}))|$$
$$= |2\operatorname{Adj}(\operatorname{Adj}(2A^{-1}))|^{n-1}$$
$$= 2^{n(n-1)}|\operatorname{Adj}(2A^{-1})|^{n-1}$$
$$= 2^{n(n-1)}|(2A^{-1})|^{(n-1)(n-1)}$$
$$= 2^{n(n-1)}2^{n(n-1)(n-1)}|A^{-1}|^{(n-1)(n-1)}$$
$$= 2^{n(n-1)+n(n-1)(n-1)}\frac{1}{|A|^{(n-1)^2}}$$
$$= \frac{2^{n(n-1)+n(n-1)(n-1)}}{2^{(n-1)^2}}$$
$$= 2^{n(n-1)+n(n+1)^2-(n-1)^2}$$
$$= 2^{(n-1)(n^2-n+1)}$$

Now, $2^{(n-1)(n^2-n+1)}$

$2^{(n-1)(n^2-n+1)} = 2^{84}$

So, $n = 5$

Question91

If $A = \frac{1}{2}\begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then :

[1-Feb-2023 Shift 2]

Options:

A. $A^{30} - A^{25} = 2I$

B. $A^{30}+A^{25}+A=I$

C. $A^{30}+A^{25}-A=I$

D. $A^{30}=A^{25}$

Answer: C

Solution:

$$A=\frac{1}{2}\begin{bmatrix}1 & \sqrt{3}\\ -\sqrt{3} & 1\end{bmatrix}$$

$$A=\begin{bmatrix}\cos 60^{\circ} & \sin 60^{\circ}\\ -\sin 60^{\circ} & \cos 60^{\circ}\end{bmatrix}$$

If $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$ Here $\alpha=\frac{\pi}{3}$

$$=\begin{bmatrix}\cos 2\alpha & \sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{bmatrix}$$

$$A^{30}=\begin{bmatrix}\cos 30\alpha & \sin 30\alpha\\ -\sin 30\alpha & \cos 30\alpha\end{bmatrix}$$

$$A^{30}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$$

$A^{25}=A$

$A^{25}-A=0$

Question92

Let $A=[a_{ij}]_{2\times 2}$, where $a_{ij}\neq 0$ for all i, j and $A^2=I$. Let a be the sum of | all diagonal elements of A and b = | A|. Then $3a^2+4b^2$ is equal to : [6-Apr-2023 shift 1]

Options:

A. 14

B. 4

C. 3

D. 7

Answer: B

Solution:

$$A^2 = I \Rightarrow |A|^2 = 1 \Rightarrow |A| = \pm 1 = b$$

Let $A = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$

$$A^2 = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} = I$$

$$\begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta + \beta\delta \\ \alpha\gamma + \gamma\delta & \gamma\beta + \delta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \alpha^2 + \beta\gamma = 1$$

$(\alpha + \delta)\beta = 0 \Rightarrow \alpha + \delta = 0 = a$

$(\alpha + \delta)\gamma = 0$

$\beta\gamma + \delta^2 = 0$

Now $3a^2 + 4b^2 = 3(0)^2 + 4(1) = 4$

Question93

Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in N$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to : [6-Apr-2023 shift 2]

Options:

A. 40

B. 22

C. 24

D. 18

Answer: C

Solution:

$P^2 = I - P$

$P^\alpha + P^\beta = \gamma I - 29P$

$P^\alpha - P^\beta = \delta I - 13P$

$P^4 = (I - P)^2 = I + P^2 - 2P$

$P^4 = I + I - P - 2P = 2I - 3P$

$P^8 = (P^4)^2 = (2I - 3P)^2 = 4I + 9P^2 - 12P$

$= 4I + 9(I - P) - 12P$
$P^8 = 13I - 21P \ldots (1)$
$P^6 = P^4 \cdot P^2 \quad = (2I - 3P)(I - P)$
$= 2I - 5P + 3P^2$
$= 2I - 5P + 3(I - P)$
$= 5I - 8P \ldots (2)$
$(1) + (2)$
$P^8 + P^6 = 18I - 29P$
$(1) \quad - (2)$
$P^8 - P^6 = 8I - 13P$
From (A) $\quad \alpha = 8, \quad \beta = 6$
$\gamma = 18$
$\delta = 8$
$\alpha + \beta + \gamma - \delta = 32 - 8 = 24$

Question94

Let A = $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$**. If** $|adj(adj(adj\ 2A))| = (16)^n$**, then n is equal to**

[8-Apr-2023 shift 1]

Options:

A. 8

B. 9

C. 12

D. 10

Answer: D

Solution:

$|A| = 2[3] - 1[2] = 4$
$\therefore |adj(adj(adj\ 2A))|$
$= |2A|^{(n-1)^3} \Rightarrow |2A|^8 = 16^n$
$\Rightarrow (2^3 |A|)^8 = 16^n$
$\Rightarrow (2^3 \times 2^2)^8 = 16^n$
$= 2^{40} = 16^n$
$= 16^{10} = 16^n \Rightarrow n = 10$

Question95

Let P = $\left[\begin{array}{cc} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{array}\right]$ **, A =** $\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]$ **and** $Q = PAP^T$**. If**

$P^TQ^{2007}P = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$**, then** $2a + b - 3c - 4d$ **equal to**

[8-Apr-2023 shift 1]

Options:

A. 2004

B. 2007

C. 2005

D. 2006

Answer: C

Solution:

$$
\begin{aligned}
& Q = PAP^T \\
& P^T \cdot Q^{2007} \cdot P = P^T \cdot Q \cdot Q \ldots Q \cdot P \\
& = P^T(PAP^T)(P \cdot AP^T) \ldots (P^T)P. \\
& \Rightarrow (P^TP)A(P^TP)A \ldots A(P^TP)
\end{aligned}
$$

$$P^T \cdot P = \left[\begin{array}{cc} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{array}\right]\left[\begin{array}{cc} -\sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{array}\right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right] = I$$

$$\therefore P^T \cdot Q^{2007} \cdot P = A^{2007}$$

$$A^2 = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \end{array}\right]$$

$$\therefore A^{2007} = \left[\begin{array}{cc} 1 & 2007 \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} a & b \\ c & d \end{array}\right]$$

$$
\begin{aligned}
& a = 1, b = 2007, c = 0, d = 1 \\
& 2a + b - 3c - 4d = 2 + 2007 - 4 = 2005
\end{aligned}
$$

Question96

If $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $A^{-1} = \alpha A + \beta I$ and $\alpha + \beta = -2$, then $4\alpha^2 + \beta^2 + \lambda^2$ is equal to :
[8-Apr-2023 shift 2]

Options:

A. 14

B. 12

C. 19

D. 10

Answer: A

Solution:

$|A - xI| = 0 \Rightarrow \begin{vmatrix} 1-x & 5 \\ \lambda & 10-x \end{vmatrix} = 0 \Rightarrow x^2 - 11x + 10 - 5\lambda = 0$

$\Rightarrow (10 - 5\lambda)A^{-1} = -A + 11I$

$\therefore \alpha = \dfrac{-1}{10-5\lambda}$ and $\beta = \dfrac{+11}{10-5\lambda}$

$\alpha + \beta = -2 \Rightarrow \dfrac{10}{10-5\lambda} = -2 \Rightarrow 10 - 5\lambda = -5 \Rightarrow \lambda = 3$

$\therefore \alpha = \dfrac{1}{5}$ & $\beta = \dfrac{-11}{5}$

$\therefore 4a^2 + \beta^2 + \lambda^2 = \dfrac{4}{25} + \dfrac{121}{25} + 3^2 = 14$ Ans.

Question97

If A is a 3×3 matrix and $|A| = 2$, then $|3\,\mathrm{adj}(|3A|A^2)|$ is equal to :
[10-Apr-2023 shift 1]

Options:

A. $3^{12} \cdot 6^{10}$

B. $3^{11} \cdot 6^{10}$

C. $3^{12} \cdot 6^{11}$

D. $3^{10} \cdot 6^{11}$

Answer: B

Solution:

Given $|A| = 2$

Now, $|3\,\text{adj}(|3A|A^2)|$

$|3A| = 3^3 \cdot |A|$

$= 3^3 \cdot (2)$

Adj. $(|3A|A^2) = \text{adj}\{(3^3 \cdot 2)A^2\}$

$= (2.3^3)^2(\text{adj } A)^2$

$= 2^2 \cdot 3^6 \cdot (\text{adj}A)^2$

$|3\,\text{adj}(|3A|A^2)| = |2^2 \cdot 3.3^6(\text{adj } A)^2|$

$= (2^2.3^7)^3\,|\text{adj}A|^2$

$= 2^6 \cdot 3^{21}(|A|^2)^2$

$= 2^6 \cdot 3^{21}(2^2)^2$

$= 2^{10} \cdot 3^{21}$

$= 2^{10} \cdot 3^{10} \cdot 3^{11}$

$|3\,\text{adj}(|3A|A^2)| = 6^{10} \cdot 3^{11}$

Question98

If A = $\frac{1}{5!6!7!}\begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$**, then |adj(adj(2A))| is equal to**

[10-Apr-2023 shift 2]

Options:

A. 2^{16}

B. 2^8

C. 2^{12}

D. 2^{20}

Answer: A

Solution:

$$|A| = \frac{1}{5!6!7!}5!6!7!\begin{vmatrix} 1 & 6 & 42 \\ 1 & 7 & 56 \\ 1 & 8 & 72 \end{vmatrix}$$

$R_3 \rightarrow R_3 \rightarrow R_2$
$R_2 \rightarrow R_2 \rightarrow R_1$

$$|A| = \begin{vmatrix} 1 & 8 & 42 \\ 0 & 1 & 14 \\ 0 & 1 & 16 \end{vmatrix} = 2$$

$|\operatorname{adjadj}(2A)| = |2A|^{(n-1)^2}$
$= |2A|^4$
$= (2^3|A|)^4$
$= 2^{12}|A|^4 \Rightarrow 2^{16}$

Question99

Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in \mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n \in \mathbb{N}$, then n is equal to ______.
[11-Apr-2023 shift 1]

Answer: 2

Solution:

Solution:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$A^3 = A$

$$A^2 = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ac & 1 & 2+3c \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ac & a & 2+3c \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2ac+3 & a+2+3c & 2a+4+6c \\ a(a+3c)+2a & 3+2ac & 6+3a+9c \\ a+2+3c & ac+c(2+3c) & 2ac+3 \end{bmatrix}$$

Given $A^3 = A$

$2ac+3 = 0\ldots(1)$ and $a+2+3c = 1$

$a+1+3c = 0$

$a+1-\dfrac{9}{2a} = 0$

$2a^2+2a-9 = 0$

$f(1) < 0,\ f(2) > 0$

$a \in (1, 2]$

$n = 2$

Question100

Let $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to
[12-Apr-2023 shift 1]

Options:

A. 50

B. 75

C. 125

D. 100

Answer: D

Solution:

$$\text{Let } C = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, D = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$DC = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$B = CAD$$

$$B^n = \underbrace{}(CAD)(CAD)(CAD)\ldots\ldots(CAD)_{\text{n-times}}$$

$$\Rightarrow B^n = CA^nD \quad \ldots (1)$$

$$A^3 = \begin{bmatrix} 1 & \frac{3}{51} \\ 0 & 1 \end{bmatrix}$$

Similarly $A^n = A^3 = \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}$

Bn $\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & \frac{n}{51} + 2 \\ -1 & -\frac{n}{51} - 1 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{n}{51} + 1 & \frac{n}{51} \\ -\frac{n}{51} & 1 - \frac{n}{51} \end{bmatrix}$$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 25 - 50 & 25 \\ -25 & -25 - 50 \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

Sum of the elements $= 100$

Question101

The number of symmetric matrices of order 3 , with all the entries from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}, is [13-Apr-2023 shift 1]

Options:

A. 10^9

B. 10^6

C. 9^{10}

D. 6^{10}

Answer: B

Solution:

$A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$, $a, b, c, d, e, f \in \{0, 1, 2, \ldots .9\}$, Number of matrices $= 10^6$

Question102

Let B = $\begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$**, $\alpha > 2$ be the adjoint of a matrix A and $|A| = 2$.**

then $[\, \alpha \quad -2\alpha \quad \alpha \,]B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ **is equal to**

[13-Apr-2023 shift 1]

Options:

A. 16

B. 32

C. 0

D. -16

Answer: D

Solution:

Given, $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$

$|B| = 4$

$1(8-3\alpha) - 3(4-3\alpha) + \alpha(\alpha - 2\alpha) = 4$

$-\alpha^2 + 6\alpha - 8 = 0$

$\alpha = 2, 4$

Given $\alpha > 2$

So, $\alpha = 2$ is rejected

$$[\,4 \quad -8 \quad 4\,] \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix} = [-16]_{1 \times 1}$$

Question103

Let for $A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \left| A \right| = 2$. If $|2\,\text{adj}(2\,\text{adj}(2A))| = 32^n$, then $3n + \alpha$ is equal to
[13-Apr-2023 shift 2]

Options:

A. 10

B. 9

C. 12

D. 11

Answer: D

Solution:

$|A| = 2$

$\text{adj}(kA) = k^{m-1}\,\text{adj}A \quad \{m = \text{ order of matrix }\}$

$\text{adj}(2A) = 2^2\,\text{adj}\,A = 4\,\text{adj}(A)$

$\text{adj}(2\,\text{adj}(2A)) = \text{adj}(8\,\text{adj}\,A)$

$= 8^2\,\text{adj}\,\text{adj}(A)$

$|2\,\text{adj}\,2\,\text{adj}(2A)| = |2^7\,\text{adj}\,\text{adj}(A)|$

$= (2^7)^3\,|A|^{2^2}$

$= 2^{21} | A|^4$
$= 2^{21} \cdot 2^4$
$\Rightarrow 2^{25} = (32)^n$
$\Rightarrow 2^{25} = 2^{5n}$
$\Rightarrow n = 5$
$| A | = 2$
$(6 - 1) - 2(2\alpha - 1) + 3(\alpha - 3) = 2$
$\Rightarrow 5 - 4\alpha + 2 + 3\alpha - 9$
$\Rightarrow \alpha = -4$
$3n + \alpha = 11$

Question104

Let the determinant of a square matrix A of order m be m − n, where m and n satisfy 4m + n = 22 and 17m+ 4n = 93. If $\det(n\,\text{adj}(\text{adj}(mA))) = 3^a 5^b 6^c$, then a + b + c is equal to [15-Apr-2023 shift 1]

Options:

A. 101

B. 84

C. 109

D. 96

Answer: D

Solution:

$| A | = m - n$
$4m + n = 22$
$17m + 4n = 93$
$m = 5, n = 2$
$| A | = 3$
$|2\,\text{adj}(\text{adj}\,5A)| = 2^5 | 5A|^{16}$
$= 2^5 \cdot 5^{80} | A|^{16}$
$= 2^5 \cdot 5^{80} \cdot 3^{16}$
$= 3^{11} \cdot 5^{80} \cdot 6^5$
$a + b + c = 96$

Question105

Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \ldots .100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is [24-Jun-2022-Shift-2]

Answer: 100

Solution:

$$A = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -a+ab \\ 0 & b^2 \end{bmatrix}$$

$\therefore T_n = \{A \in S; A^{n(n+1)} = I\}$

$\therefore$ b must be equal to 1

$\therefore$ In this case A^2 will become identity matrix and a can take any value from 1 to 100

$\therefore$ Total number of common element will be 100 .

Question106

Let A be a 3 × 3 real matrix such that

$$A\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}; A\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ and}$$

$$A\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}1\\1\\2\end{pmatrix}$$

If $X=(x_1, x_2, x_3)^T$ and I is an identity matrix of order 3 , then the system $(A-2I)X=\begin{pmatrix}4\\1\\1\end{pmatrix}$ has :
[25-Jun-2022-Shift-1]

Options:

A. no solution

B. infinitely many solutions

C. unique solution

D. exactly two solutions

Answer: B

Solution:

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$$A = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} a+b=1 \\ d+e=1 \\ g+h=0 \end{matrix}$$

$$A = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{matrix} a+c=-1 \\ d+f=1 \\ g+i=0 \end{matrix}$$

$$A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{matrix} rc=1 \\ f=1 \\ i=2 \end{matrix}$$

Solving will get

$a=-2,\ b=3,\ c=1,\ d=-1,\ e=2,\ f=1,\ g=-1,\ h=1,\ i=2$

$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow A = 2I = \begin{bmatrix} -4 & 3 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$(A-2I)x = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow -4x_1 + 3x_2 + x_3 = 4 \ldots$ (i)

$-x_1 + x_3 = 1$

$-x_1 + x_2 = 1 \ldots$ (ii)

So $3(\text{iii}) + (\text{ii}) = (\text{i})$

$\therefore$ Infinite solution

Question107

Let A = $\left[\begin{array}{cc} 0 & -2 \\ 2 & 0 \end{array}\right]$. If M and N are two matrices given by M $= \sum\limits_{k=1}^{10} A^{2k}$ and N $= \sum\limits_{k=1}^{10} A^{2k-1}$ then MN^2 is :
[25-Jun-2022-Shift-1]

Options:

A. a non-identity symmetric matrix

B. a skew-symmetric matrix

C. neither symmetric nor skew-symmetric matrix

D. an identity matrix

Answer: A

Solution:

$$A=\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}$$

$$A^2=\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}\begin{bmatrix}0 & -2\\ 2 & 0\end{bmatrix}=\begin{bmatrix}-4 & 0\\ 0 & -4\end{bmatrix}=-4I$$

$M=A^2+A^4+A^6+......+A^{20}$

$=-4I+16I-64I+.....$ upto 10 terms

$=-I[4-16+64....+$ upto 10 terms $]$

$=-I.4\left[\dfrac{(-4)^{10}-1}{-4-1}\right]=\dfrac{4}{5}(2^{20}-1)I$

$N=A^1+A^3+A^5+.....+A^{19}$

$=A-4A+16A+.....$ upto 10 terms

$=A\left(\dfrac{(-4)^{10}-1}{-4-1}\right)=-\left(\dfrac{2^{20}-1}{5}\right)A$

$N^2=\dfrac{(2^{20}-1)^2}{2^5}A^2=\dfrac{-4}{24}(2^{20}-1)^2I$

$MN^2=\dfrac{-16}{125}(2^{20}-1)^3I=KI\quad(K\neq\pm1)$

$(MN^2)^T=(KI)^T=KI$

$\therefore$A is correct

Question108

Let A be a 3 × 3 matrix having entries from the set {−1, 0, 1}. The number of all such matrices A having sum of all the entries equal to 5 , is
[25-Jun-2022-Shift-1]

Answer: 414

Solution:

Case-I:

$1 \to 7$ times

and $-1 \to 2$ times

number of possible marrix $= \frac{9!}{7!2!} = 36$

Case-II:

$1 \to 6$ times,

$-1 \to 1$ times

and $0 \to 2$ times

number of possible marrix $= \frac{9!}{6!2!} = 252$

Case-III:

$1 \to 5$ times

and $0 \to 4$ times

number of possible marrix $= \frac{9!}{5!4!} = 126$

Hence total number of all such matrix $A = 414$

Question109

Let A = $\begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and B = $\begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set $\{(n, m) : n, m \in \{1, 2, \ldots\ldots\ldots, 10\}$. and $.nA^n + mB^m = 1\}$ is___ [25-Jun-2022-Shift-2]

Answer: 1

Solution:

$$A^2 = \begin{matrix} 2 & -2 \\ 1 & -1 \end{matrix} \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} = A$$

$\Rightarrow A^K = A, K \in I$

$$B^2 = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} = B$$

So, $B^K = B, K \in I$

$nA^n + mB^m = nA + mB$

$$= \begin{bmatrix} 2n - 2n \\ n - n \end{bmatrix} + \begin{bmatrix} -m & 2m \\ -m & 2m \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So, $2n - m = 1, -n + m = 0, 2m - n = 1$

So, $(m, n) = (1, 1)$

Question110

Let A be a 3 × 3 invertible matrix. If $|adj(24A)| = |adj(3\,adj(2A))|$, then $|A|^2$ is equal to :
[26-Jun-2022-Shift-1]

Options:

A. 6^6

B. 2^{12}

C. 2^6

D. 1

Answer: C

Solution:

We know, $|\text{adj} A| = |A|^{n-1}$

Now, $|adj 24A| = |adj 3(adj 2A)|$

$\Rightarrow |24A|^{3-1} = |3adj 2A|^{3-1}$

$\Rightarrow |24A|^2 = |3adj 2A|^2$

Also, we know, $|KA| = K^n|A|$

$\Rightarrow ((24)^2)^2 |A|^2 = ((3)^3)^2 |adj 2A|^2$

$\Rightarrow (24)^6 |A|^2 = 3^6 \cdot (|2A|^{3-1})^2$

$\Rightarrow (24)^6 |A|^2 = 3^6 \cdot |2A|^4$

$\Rightarrow (24)^6 |A|^2 = 3^6 \cdot (2^3)^4 \cdot |A|^4$

$\Rightarrow 3^6 \cdot 8^6 \cdot |A|^2 = 3^6 \cdot 8^4 \cdot |A|^4$

$\Rightarrow 8^2 = |A|^2$

$\Rightarrow |A|^2 = 64 = 2^6$

Question111

Let X = $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$**, Y = $\alpha I + \beta X + \gamma X^2$ and**

$Z = \alpha^2 I - \alpha\beta X + (\beta^2 - \alpha\gamma)X^2$, $\alpha, \beta, \gamma \in R$.

If $Y^{-1} =$ $\begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$ **, then $(\alpha - \beta + \gamma)^2$ is equal to___**

[26-Jun-2022-Shift-2]

Answer: 100

Solution:

$$\because X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore X^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore Y = \alpha I + \beta X + \gamma X^2 \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix}$$

$$\because Y \,.\, Y^{-1} = I$$

$$\therefore \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix} \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \frac{\alpha}{5} & \frac{\beta-2\alpha}{5} & \frac{\alpha-2\beta+\gamma}{5} \\ 0 & \frac{\alpha}{5} & \frac{\beta-2\alpha}{5} \\ 0 & 0 & \frac{\alpha}{5} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \alpha = 5,\ \beta = 10,\ \gamma = 15$$

$$\therefore (\alpha - \beta + \gamma)^2 = 100$$

Question112

The positive value of the determinant of the matrix A, whose

Adj(Adj(A)) = $\begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix}$**, is____**

[27-Jun-2022-Shift-1]

Answer: 14

Solution:

$|adj(adj(A))| = |A|^{2^2} = |A|^4$

$\therefore |A|^4 = \begin{vmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{vmatrix}$

$= (14)^3 \begin{vmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}$

$= (14)^3(3 - 2(-5) - 1(-1))$

$|A|^4 = (14)^4 \Rightarrow |A| = 14$

Question113

Let A and B be two 3 × 3 matrices such that AB = I and |A| = $\frac{1}{8}$**. Then |adj(B adj(2A))| is equal to**

[27-Jun-2022-Shift-2]

Options:

A. 16

B. 32

C. 64

D. 128

Answer: C

Solution:

A and B are two matrices of order 3×3.
and $AB = I$,
$|A| = \frac{1}{8}$
Now, $|A|\,|B| = 1$
$|B| = 8$
$\therefore |\operatorname{adj}\left(B(\operatorname{adj}(2A))\right| = \left|B(\operatorname{adj}(2A))\right|^{2.}$
$= |B|^2\,|\operatorname{adj}(2A)|^2$
$= 2^6\,|2A|^{2 \times 2}$
$= 2^6 \cdot 2^{12} \cdot \frac{1}{2^{12}} = 64$

Question114

Let A be a matrix of order 2 × 2, whose entries are from the set {0, 1, 2, 3, 4, 5}. If the sum of all the entries of A is a prime number p, 2 < p < 8, then the number of such matrices A is____ [27-Jun-2022-Shift-2]

Answer: 180

Solution:

$\because$ Sum of all entries of matrix A must be prime p such that $2 < p < 8$ then sum of entries may be 3,5 or 7 .
If sum is 3 then possible entries are $(0, 0, 0, 3)$, $(0, 0, 1, 2)$ or $(0, 1, 1, 1)$.
$\therefore$ Total number of matrices $= 4 + 4 + 12 = 20$
If sum of 5 then possible entries are
$(0, 0, 0, 5)$, $(0, 0, 1, 4)$, $(0, 0, 2, 3)$, $(0, 1, 1, 3)$, $(0, 1, 2, 2)$ and $(1, 1, 1, 2)$.
$\therefore$ Total number of matrices $= 4 + 12 + 12 + 12 + 12 + 4 = 56$
If sum is 7 then possible entries are
$(0, 0, 2, 5)$, $(0, 0, 3, 4)$, $(0, 1, 1, 5)$, $(0, 3, 3, 1)$, $(0, 2, 2, 3)$, $(1, 1, 1, 4)$, $(1, 2, 2, 2)$, $(1, 1, 2, 3)$ and $(0, 1, 2, 4)$.
Total number of matrices with sum $7 = 104$
$\therefore$ Total number of required matrices
$= 20 + 56 + 104 =$ 180

Question115

Let A be a matrix of order 3 × 3 and det(A) = 2. Then det(det(A) adj(5 adj(A^3))) is equal to
[28-Jun-2022-Shift-1]

Options:

A. 512×10^6

B. 256×10^6

C. 1024×10^6

D. 256×10^{11}

Answer: A

Solution:

$|A| = 2$

$|A| = \text{adj}(5\,\text{adj}\,A^3)\ \Big|$

$= |\,25\,|\,A\Big|\text{adj}(\text{adj}\,A^3)\ \Big|$

$= 25^3\,|\,A\,|^3 \cdot \left|\text{ad}\,\text{j}A^3\right|^2$

$= 25^3 \cdot 2^3 \cdot \left|A^3\right|^4$

$= 25^3 \cdot 2^3 \cdot 2^{12} = 10^6 \cdot 512$

Question116

Let A = $\begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where i = $\sqrt{-1}$. Then, the number of elements in the set {n ∈ {1, 2, ……, 100} : $A^n = A$} is____
[28-Jun-2022-Shift-2]

Answer: 25

Solution:

$$\therefore A^2 = \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1+i & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix}$$

$$A^4 = \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix} \begin{bmatrix} i & 1+i \\ 1-i & -i \end{bmatrix} = I$$

So $A^5 = A$, $A^9 = A$ and so on.
Clearly $n = 1, 5, 9, \ldots\ldots, 97$
Number of values of $n = 25$

Question117

The probability that a randomly chosen 2 × 2 matrix with all the entries from the set of first 10 primes, is singular, is equal to :
[29-Jun-2022-Shift-1]

Options:

A. $\frac{133}{10^4}$

B. $\frac{18}{10^3}$

C. $\frac{19}{10^3}$

D. $\frac{271}{10^4}$

Answer: C

Solution:

Let matrix A is singular then $|A| = 0$
Number of singular matrix $=$ All entries are same $+$ only two prime number are used in matrix
$= 10 + 10 \times 9 \times 2$
$= 190$
Required probability $= \frac{190}{10^4} = \frac{19}{10^3}$

Question118

Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all i, j = 1, 2, 3. Then, the matrix $A^2 + A^3 + \ldots\ldots + A^{10}$ is equal to : [29-Jun-2022-Shift-1]

Options:

A. $\left(\frac{3^{10} - 3}{2}\right)A$

B. $\left(\frac{3^{10} - 1}{2}\right)A$

C. $\left(\frac{3^{10} + 1}{2}\right)A$

D. $\left(\frac{3^{10} + 3}{2}\right)A$

Answer: A

Solution:

Given, $a_{ij} = 2^{j-i}$

Now, $A = \begin{bmatrix} 2^0 & 2^1 & 2^2 \\ 2^{-1} & 2^0 & 2^1 \\ 2^{-2} & 2^{-1} & 2^0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1+1 & 2+2+2 & 4+4+4 \\ \frac{1}{2}+\frac{1}{2}+\frac{1}{2} & 1+1+1 & 2+2+2 \\ \frac{1}{4}+\frac{1}{4}+\frac{1}{4} & \frac{1}{2}+\frac{1}{2}+\frac{1}{2} & 1+1+1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 12 \\ \frac{3}{2} & 3 & 6 \\ \frac{3}{4} & \frac{3}{2} & 3 \end{bmatrix}$$

$$= 3 \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & 1 \end{bmatrix}$$

$= 3A$

Similarly, $A^3 = 3^2A$

$A^4 = 3^3A$

$\therefore A^2 + A^3 + \ldots\ldots + A^{10}$

$= 3A + 3^2A + 3^3A + \ldots\ldots + 3^9A$

$= A(3 + 3^2 + 3^3 + \ldots\ldots + 3^9)$

$= A\left(\frac{3(3^9-1)}{3-1}\right) = \frac{3(3^9-1)}{2}A = \left(\frac{3^{10}-3}{2}\right)A$

Question119

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$. If

$B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \ldots.. - {}^5C_5(\text{adj } A)^5$, then the sum of all

elements of the matrix B is [29-Jun-2022-Shift-2]

Options:

A. −5

B. −6

C. −7

D. −8

Answer: C

Solution:

Given $A = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$ and

$B = I - 5_{C_1}(\text{adj } A) + 5_{C_2}(\text{adj } A)^2 - 5_{C_3}(\text{adj } A)^3 + 5_{C_4}(\text{adj } A)^4 - 5_{C_5}(\text{adj } A)^5$

$= (I - (\text{adj } A))^5$

Cofactor of $A = \begin{bmatrix} (-1)^{1+1} \cdot 2 & (-1)^{1+2} \cdot 0 \\ (-1)^{2+1} \cdot (-1) & (-1)^{2+2} \cdot 2 \end{bmatrix}$

$= \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$

Transpose of cofactor of $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

$\therefore \text{adj } A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

Now, $I - \text{ad jA}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$

Now let,

$P = I - \text{adj } A = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$

$\therefore P^2 = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$$P^4 = P^2 \cdot P^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$P^5 = P^4 \cdot P = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -5 \\ 0 & -1 \end{bmatrix}$$

$$\therefore B = \begin{bmatrix} -1 & -5 \\ 0 & -1 \end{bmatrix}$$

Now sum of elements $= -1 - 5 - 1 + 0 = -7$

Question120

Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number an $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is____
[29-Jun-2022-Shift-2]

Answer: 1

Solution:

$$M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}; M^2 = \begin{bmatrix} -\alpha^2 & 0 \\ 0 & -\alpha^2 \end{bmatrix} = -\alpha^2 I$$

$N = M^2 + M^4 + \ldots\ldots + M^{98} = [-\alpha^2 + \alpha^4 - \alpha^6 + \ldots]I$

$= -\alpha^2 \dfrac{(1 - (-\alpha^2)^{49})}{1 + \alpha^2} \cdot I$

$I - M^2 = (1 + \alpha^2)I$

$(I - M^2)N = -\alpha^2(\alpha^{98} + 1) = -2$

$\alpha = 1$

Question121

Let S = { $\sqrt{n}$: 1 ≤ n ≤ 50 and n is odd }.

Let a ∈ S and A = $\begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$ **.**

If $\sum_{a \in S}$ **det(adj A) = 100λ, then λ is equal to : ion:**
[24-Jun-2022-Shift-1]

Options:

A. 218

B. 221

C. 663

D. 1717

Answer: B

Solution:

Given, $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}_{3\times3}$ $S = \{\sqrt{n} : 1 \le n \le 50 \text{ and } n \text{ is odd}\}$

$\therefore S = \{1, \sqrt{3}, \sqrt{5}, \sqrt{7}, \ldots, \sqrt{49}\}$

We know,

$|\text{adj}\, A| = |A|^{n-1}$

Here, n = order of matrix.

Here, $n = 3$

$\therefore |adj A| = |A|^{3-1} = |A|^2$

Now, $|A| = \begin{vmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{vmatrix}$

$= 1(1-0) - 0 + a(0-(-a))$

$= a^2 + 1$

$\therefore |adj A| = |A|^2 = (a^2+1)^2$

Now, $\sum_{a \in S} \det(adj A)$

$= \sum_{a \in S} (a^2+1)^2$

$= (1^2+1)^2 + ((\sqrt{3})^2+1)^2 + ((\sqrt{5})^2+1)^2 + \ldots + ((\sqrt{49})^2+1)^2$

$= (1^2+1)^2 + (3+1)^2 + (5+1)^2 + \ldots + (49+1)^2$

$= 2^2 + 4^2 + 6^2 + \ldots + 50^2$

$= 2^2(1^2 + 2^2 + 3^2 + \ldots + 25^2)$

$= 4.\ \dfrac{25.26.51}{6} = 100.221$

$\therefore 100K = 100.221$

$\Rightarrow K = 221$

Question122

Let A = $\begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ and B = A − I . If ω = $\frac{\sqrt{3}i-1}{2}$, then the number of elements in the set {n ∈ {1, 2, …, 100} : $A^n + (\omega B)^n = A + B$} is equal to [25-Jul-2022-Shift-1]

Answer: 17

Solution:

Here $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$

We get $A^2 = A$ and similarly for

$B = A - I = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$

We get $B^2 = -B \Rightarrow B^3 = B$

$\therefore A^n + (\omega B)^n = A + (\omega B)^n \quad \text{for } n \in N$

For ω^n to be unity n shall be multiple of 3 and for B^n to be B. n shell be $3, 5, 7, \ldots 99$

$\therefore n = \{3, 9, 15, \ldots\ldots 99\}$

Number of elements $= 17$

Question123

Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, a, b ∈ C. If for some $n \in \mathbb{N}$, $A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$

then $n + a + b$ is equal to
[25-Jul-2022-Shift-2]

Answer: 24

Solution:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} = I + B$$

$$B^2 = \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & ab \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$B^3 = 0$

$\therefore A^n = (1 + B)^n = {}^nC_0 I + {}^nC_1 B + {}^nC_2 B^2 + {}^nC_3 B^3 + \ldots$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & na & na \\ 0 & 0 & nb \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & \frac{n(n-1)ab}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & na & na + \frac{n(n-1)}{2}ab \\ 0 & 1 & nb \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 48 \\ 0 & 0 & 1 \end{bmatrix}$$

On comparing we get $na = 48$, $nb = 96$ and

$na + \frac{n(n-1)}{2}ab = 2160$

$\Rightarrow a = 4, n = 12$ and $b = 8$

$n + a + b = 24$

Question124

Let A be a 2 × 2 matrix with $\det(A) = -1$ and $\det((A + I)(\text{Adj}(A) + I)) = 4$. Then the sum of the diagonal elements of A can be :
[26-Jul-2022-Shift-1]

Options:

A. -1

B. 2

C. 1

D. $-\sqrt{2}$

Answer: B

Solution:

$$|(A+I)(adjA+I)|=4$$

$$\Rightarrow |A \operatorname{adj} A + A + \operatorname{adj} A + I| = 4$$

$$\Rightarrow |(A)I + A + \operatorname{adj} A + I| = 4$$

$$|A| = -1 \Rightarrow |A + \operatorname{adj} A| = 4$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} adjA = \begin{bmatrix} a & -b \\ -c & d \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} (a+d) & 0 \\ 0 & (a+d) \end{vmatrix} = 4$$

$$\Rightarrow a + d = \pm 2$$

Question125

Let A = $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ **and B =** $\begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$ **, then the value of** $A'BA$ **is:**
[26-Jul-2022-Shift-2]

Options:

A. 1224

B. 1042

C. 540

D. 539

Answer: D

Solution:

$$A'BA = [\,1 \quad 1 \quad 1\,] \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix} A$$

$$= \left[\; 9^2+12^2-15^2 \quad -10^2+13^2+16^2 \quad 11^2-14^2+17^2 \;\right] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$= [9^2+12^2-15^2-10^2+13^2+16^2+11^2-14^2+17^2]$
$= [(9^2-10^2)+(11^2+12^2)+(13^2-14^2)+(16^2-15^2)+17^2]$
$= [-19+265+(-27)+31+289]$
$= [585-46] = [539]$

Question126

The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where

a, b, c, d $\in \{-1, 0, 1, 2, 3, \ldots, ., 10\}$, such that $A = A^{-1}$, is _________. [26-Jul-2022-Shift-2]

Answer: 50

Solution:

$\because A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^2 = \begin{bmatrix} a^2+bc & b(a+d) \\ c(a+d) & bc+d^2 \end{bmatrix}$

For A^{-1} must exist $ad-bc \neq 0 \ldots\ldots$ (i)
and $A = A^{-1} \Rightarrow A^2 = I$
$\therefore a^2+bc = d^2+bc = 1$
and $b(a+d) = c(a+d) = 0$
Case I : When $a = d = 0$, then possible values of (b, c) are $(1, 1)$, $(-1, 1)$ and $(1, -1)$ and $(-1, 1)$.
Total four matrices are possible.
Case II: When $a = -d$ then (a, d) be $(1, -1)$ or $(-1, 1)$.
Then total possible values of (b, c) are $(12+11)\times 2 = 46$.
$\therefore$ Total possible matrices $= 46+4 = 50$.

Question127

Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to
[27-Jul-2022-Shift-1]

Options:

A. -10

B. -6

C. 6

D. 10

Answer: D

Solution:

$$A^2 = \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix} = \begin{bmatrix} -3 & -8 \\ 8 & 21 \end{bmatrix}$$

$$\alpha A^2 + \beta A = \begin{bmatrix} -3\alpha & -8\alpha \\ 8\alpha & 21\alpha \end{bmatrix} + \begin{bmatrix} \beta & 2\beta \\ -2\beta & -5\beta \end{bmatrix}$$

$$= \begin{bmatrix} -3\alpha + \beta & -8\alpha + 2\beta \\ 8\alpha - 2\beta & 21\alpha - 5\beta \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

On Comparing
$8\alpha = 2\beta, -3\alpha + \beta = 2, 21\alpha - 5\beta = 2$
$\Rightarrow \alpha = 2, \beta = 8$
So, $\alpha + \beta = 10$

Question128

Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices $A \in S$ such that the sum of all the diagonal elements of $A^T A$ is 6 is ________.
[27-Jul-2022-Shift-1]

Answer: 5376

Solution:

Sum of all diagonal elements is equal to sum of square of each element of the matrix.

i.e., $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ then $t_r(A \cdot A^T)$

$= a_1^2 + a_2^2 + a_3^2 + b_1^2 + b_2^2 + b_3^2 + c_1^2 + c_2^2 + c_3^2$

$\because a_i, b_i, c_i \in \{-1, 0, 1\}$ for $i = 1, 2, 3$

$\therefore$ Exactly three of them are zero and rest are 1 or -1.

Total number of possible matrices ${}^9C_3 \times 2^6$

$= \dfrac{9 \times 8 \times 7}{6} \times 64$

$= 5376$

Question129

Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$.

If $A^2 + \gamma A + 18I = O$, then det(A) is equal to ________.
[27-Jul-2022-Shift-2]

Options:

A. -18

B. 18

C. -50

D. 50

Answer: B

Solution:

Characteristic equation of A is given by

$|A - \lambda I| = 0$

$\begin{vmatrix} 4-\lambda & -2 \\ \alpha & \beta - \lambda \end{vmatrix} = 0$

$\Rightarrow \lambda^2 - (4 + \beta)\lambda + (4\beta + 2\alpha) = 0$

So, $A^2-(4+\beta)A+(4\beta+2\alpha)I=0$
$|A|=4\beta+2\alpha=18$

Question130

Consider a matrix $A=\begin{bmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{bmatrix}$, where α, β, γ are three distinct natural numbers. If $\frac{\det(\operatorname{adj}(\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A))))}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}}=2^{32}\times 3^{16}$, then the number of such 3 - tuples (α, β, γ) is _________.
[27-Jul-2022-Shift-2]

Answer: 42

Solution:

$$\det(A)=\begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ \beta+\gamma & \gamma+\alpha & \alpha+\beta\end{vmatrix}$$

$R_3\to R_3+R_1$

$$\Rightarrow(\alpha+\beta+\gamma)\begin{vmatrix}\alpha & \beta & \gamma\\ \alpha^2 & \beta^2 & \gamma^2\\ 1 & 1 & 1\end{vmatrix}$$

$\therefore\det(A)=(\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$
Also, $\det(adj(adj(adj(adj(A)))))$
$=(\det(A))^{2^4}=(\det(A)^{16}.$
$\therefore\ \frac{(\alpha+\beta+\gamma)^{16}(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}}=(4.13)^{16}$
$\Rightarrow\alpha+\beta+\gamma=12$
$\Rightarrow(\alpha,\beta,\gamma)$ distinct natural triplets
$={}^{11}C_2-1-{}^{3}C_2(4)=55-1-12=42$

Question131

Let the matrix A = $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and the matrix $B_0 = A^{49} + 2A^{98}$. If $B_n = Adj(B_{n-1})$ for all $n > 1$, then $\det(B_4)$ is equal to :
[28-Jul-2022-Shift-1]

Options:

A. 3^{28}

B. 3^{30}

C. 3^{32}

D. 3^{36}

Answer: C

Solution:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1$$

Now $B_0 = A^{49} + 2A^{98} = (A^3)^{16} \cdot A + 2(A^3)^{32} \cdot A^2$

$$B_0 = A + 2A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$|B_0| = 9$

Since, $B_n = Adj\,|B_{n-1}| \Rightarrow |B_n| = |B_{n-1}|^2$

Hence $|B_4| = |B_3|^2 = |B_2|^4 = |B_1|^8 = |B_0|^{16} = |3^2|^{16} = 3^{32}$

Question132

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}$, $\alpha, \beta \in R$. Let α_1 be the value of α which satisfies $(A+B)^2 - A^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ and α_2 be the value of α which satisfies $(A+B)^2 - B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to _______.
[28-Jul-2022-Shift-1]

Answer: 2

Solution:

$$(A+B)^2 = A^2 + B^2 + AB + BA$$

$$= A^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\therefore B^2 + AB + BA = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \ldots\ldots.(1)$$

$$AB = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}\begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \beta-1 & 1 \\ \alpha+2\beta & 2 \end{bmatrix}$$

$$BA = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix} = \begin{bmatrix} \beta+2 & \alpha-\beta \\ 1 & -1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \beta^2+1 & \beta \\ \beta & 1 \end{bmatrix}$$

By (1) we get

$$\begin{bmatrix} \beta^2+2\beta & \alpha+1 \\ \alpha+3\beta+1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\therefore \alpha = 1 \beta = 0 \Rightarrow \alpha_1 = 1$$

Similarly if $A^2 + AB + BA = 0$ then

$$\left(A^2 = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix} = \begin{bmatrix} -1 & -1-\alpha \\ 2+2\alpha & \alpha^2-2 \end{bmatrix}\right)$$

$$\begin{bmatrix} 2\beta & \alpha-\beta+1-1-\alpha \\ \alpha+2\beta+1+2+2\alpha & \alpha^2-2+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \beta = 0 \text{ and } \alpha = -1 \Rightarrow \alpha_2 = -1$$

$$\therefore |\alpha_1 - \alpha_2| = |2| = 2$$

Question133

Let A and B be any two 3 × 3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?
[28-Jul-2022-Shift-2]

Options:

A. $A^4 - B^4$ is a smmetric matrix

B. $AB - BA$ is a symmetric matrix

C. $B^5 - A^5$ is a skew-symmetric matrix

D. $AB + BA$ is a skew-symmetric matrix

Answer: C

Solution:

(A) $M = A^4 - B^4$
$M^T = (A^4 - B^4)^T = (A^T)^4 - (B^T)^4$
$= A^4 - (-B)^4 = A^4 - B^4 = M$
(B) $M = AB - BA$
$M^T = (AB - BA)^T = (AB)^T - (BA)^T$
$= B^T A^T - A^T B^T$
$= -BA - A(-B)$
$= AB - BA = M$
(C) $M = B^5 - A^5$
$M^T = (B^T)^5 - (A^T)^5 = -(B^5 + A^5) \neq -M$
(D) $M = AB + BA$
$M^T = (AB)^T + (BA)^T$
$= B^T A^T + A^T B^T = -BA - AB = -M$

Question134

Let A and B be two 3 × 3 non-zero real matrices such that AB is a zero matrix. Then
[29-Jul-2022-Shift-1]

Options:

A. the system of linear equations $AX = 0$ has a unique solution

B. the system of linear equations $AX = 0$ has infinitely many solutions

C. B is an invertible matrix

D. adj(A) is an invertible matrix

Answer: B

Solution:

AB is zero matrix
$\Rightarrow |A| = |B| = 0$
So neither A nor B is invertible
If $|A| = 0$
$\Rightarrow |\operatorname{adj} A| = 0$ so $\operatorname{adj} A$
$AX = 0$ is homogeneous system and $|A| = 0$
So, it is having infinitely many solutions

Question135

The number of matrices of order 3 × 3, whose entries are either 0 or 1 and the sum of all the entries is a prime number, is ________. [29-Jul-2022-Shift-1]

Answer: 282

Solution:

In a 3×3 order matrix there are 9 entries.
These nine entries are zero or one.
The sum of positive prime entries are $2, 3, 5$ or 7 .
Total possible matrices $= \frac{9!}{2!.7!} + \frac{9!}{3!.6!} + \frac{9!}{5!.4!} + \frac{9!}{7!.2!} = 36 + 84 + 126 + 36 = 282$

Question136

Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation?

[29-Jul-2022-Shift-2]

Options:

A. $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$

D. $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Answer: C

Solution:

$A = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

(1) $R_1 \to R_1 + R_2$; $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ possible

(2) $R_1 \leftrightarrow R_2$; $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ possible

(3) Option is not possible

(4) $R_2 \to R_2 + 2R_1$; $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$ possible

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Question137

Let X $= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and A $- \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $k \in N$, if $X'A^kX - 33$, then k is equal to ________.
[29-Jul-2022-Shift-2]

Answer: 10

Solution:

$$X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$$

$$X^TA^KX = 33$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$\text{As } A^2 = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^8 = \begin{bmatrix} 1 & 0 & 24 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{10} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 24 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{for } K \to \text{Even } A^K = \begin{bmatrix} 1 & 0 & 3K \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$X^T A^K X = 33$ (This is not correct)

$$[\,1 \quad 1 \quad 1\,]\begin{bmatrix} 1 & 0 & 3K \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= [\,1 \quad 1 \quad 3K+1\,]\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [3K+3]$$

$\therefore 3K+3 = 33 \therefore K = 10$

But it should be dropped as 33 is not matrix

If K is odd

$X^T A^K X = 33$

$X^T A A^{K-1} X = 33$

$$[\,1 \quad 1 \quad 1\,]\begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 3k-3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 33$$

$$[\,-1 \quad 3 \quad 8\,]\begin{bmatrix} 3k-2 \\ 1 \\ 1 \end{bmatrix} = [33]$$

$[-3k+13] = [33]$

$k = 20/3$ (not possible)

Question138

Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is

24 Feb 2021 Shift 1

Answer: 540

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2 = 7$

Case-I : Seven (1's) and two (0's)

$\frac{9!}{7!2!}=36$

Case-II: One (2) and three (1's) and five (0's)

$\frac{9!}{5!3!}=504$

$\therefore$ Total $=540$

Question139

If $A=\begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2+A)(I_2-A)^{-1}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2+b^2)$ is ....

[25 Feb 2021 Shift 1]

Answer: 13

Solution:

$A=\begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2+A)(I_2-A)^{-1}=\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$

$\Rightarrow |(I_2+A)(I_2-A)^{-1}|=a^2+b^2$

$\Rightarrow\ a^2+b^2=\frac{|I_2+A|}{|I_2-A|}$ . . . (i)

Now, $I_2+A=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}+\begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$

$=\begin{bmatrix} 1 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 1 \end{bmatrix}$

Similarly, $I_2 - A = \begin{bmatrix} 1 & \tan\left(\frac{\theta}{2}\right) \\ -\tan\left(\frac{\theta}{2}\right) & 1 \end{bmatrix}$

Here, $|I_2 + A| = \left|I_2 - A\right| = \left(1 + \tan^2\left(\frac{\theta}{2}\right)\right)$

$\Rightarrow \dfrac{|I_2 + A|}{|I_2 - A|} = 1 \;...\text{(ii)}$

From Eqs. (i) and (ii),

$a^2 + b^2 = 1$

Now, $13(a^2 + b^2) = 13 \times 1 = 13$

Question140

If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is [2021, 25 Feb. Shift-II]

Options:

A. 4

B. 1

C. 2

D. 3

Answer: B

Solution:

Given, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix}$

Given, $AA^T = I_2$ i.e.

$$\begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}\begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1+\alpha^2 & \alpha - \alpha\beta \\ \alpha - \alpha\beta & \alpha^2 + \beta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equating these matrices,
$c1 = \alpha^2 + 1$ gives, $\alpha = 0$
$\alpha(1 - \beta) = 0$
$\alpha^2 + \beta^2 = 1$

Put $\alpha = 0$ in $\alpha^2 + \beta^2 = 1$, we get $0 + \beta^2 = 1$, gives $\beta = \pm 1$
where we take $\beta^4 = 1$
$\therefore \ \alpha^4 + \beta^4 = 0^4 + 1 = 1$

Question141

Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1 , then the possible number of such matrices is
[2021, 26 Feb. Shift-I]

Options:

A. 4

B. 1

C. 6

D. 12

Answer: A

Solution:

Let A be the matrix as follows, $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, since A is symmetric matrix.

Now, $A^2 = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

$= \begin{bmatrix} a^2 + b^2 & ab + bc \\ ab + bc & b^2 + c^2 \end{bmatrix}$

Given that, diagonal entries of A^2 is 1 . i.e. $a^2 + b^2 + b^2 + c^2 = 1$
or $a^2 + 2b^2 + c^2 = 1$
Case $1 a = 0$
Then, $2b^2 + c^2 = 1$
(a) $c = 0$ gives, $b^2 = \frac{1}{2}$ or $b = \pm \frac{1}{\sqrt{2}}$ $\therefore a = 0, b = 1/\sqrt{2}, c = 0$ (2 matrices) $a = 0, b = -1/\sqrt{2}, c = 0$

(b) $b = 0$, gives $c^2 = 1$ or $c = \pm 1 \therefore a = 0, b = 0, c = 1$
and $a = 0, b = 0, c = -1$ (2 matrices)
Case $2 b = 0$, then $a^2 + c^2 = 1$
(a) $a = 0$, then $c = \pm 1$ $a = 0, b = 0, c = 1$ and $a = 0, b = 0, c = -1$
This is repeated case.
(b) $c = 0$, then $a = \pm 1$ $a = 1, b = 0, c = 0$ and $a = -1, b = 0, c = 0$ Again 2 matrices.

Thus, only acceptable matrices are as follows

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$$

Then possible number of such matrices are 4.

Question142

If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ **satisfies the equation**

$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **for some real numbers** α **and** β**, then**

$\beta - \alpha$ **is equal to**
[2021, 26 Feb. Shift-II]

Answer: 4

Solution:

$$A = \left. \begin{matrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{matrix} \right]$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

And so on,

$$A^{19} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{19} & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^{20} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{20} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c So, " $A^{20} + \alpha A^{19} + \beta A$

$$= \begin{bmatrix} 1+\alpha+\beta & 0 & 0 \\ 0 & 2^{20}+\alpha 2^{19}+2\beta & 0 \\ 3\alpha+2\beta & 0 & 1-\alpha-\beta \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

On comparing, we get

$1 - \alpha - \beta = 1$

$\Rightarrow \alpha + \beta = 0$

and $2^{20} + \alpha 2^{19} + 2\beta = 4$

$\Rightarrow 2^{20} + \alpha(2^{19} - 2) = 4$ [use, $\alpha + \beta = 0$]

$\Rightarrow \alpha = \dfrac{4 - 2^{20}}{2^{19} - 2} = -2$

$\Rightarrow \beta = 2$

$\therefore \beta - \alpha = 2 - (-2) = 2 + 2 = 4$

Question143

Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$.

If $Tr(A)$ denotes the sum of all diagonal elements of the matrix A, then $Tr(A) - Tr(B)$ has value equal to
[2021, 18 March Shift-1]

Options:

A. 1

B. 2

C. 0

D. 3

Answer: B

Solution:

Given, $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix} \dots \text{(i)}$

and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix} \dots \text{(ii)}$

Multiply with 2 in Eq. (ii), we get

$4A - 2B = \begin{bmatrix} 4 & -2 & 10 \\ 4 & -2 & 12 \\ 0 & 2 & 4 \end{bmatrix} \quad \cdots\cdots\cdot \text{(iii)}$

Adding Eqs. (i) and (iii),

$5A = \begin{bmatrix} 5 & 0 & 10 \\ 10 & -5 & 15 \\ -5 & 5 & 5 \end{bmatrix}$

$\therefore \ A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix}$

$\therefore Tr(A) = 1 - 1 + 1 = 1$
From Eq. (i),

$$B = \frac{1}{2}\left\{\begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix}\right\}$$

$$= \begin{bmatrix} 0 & 1 & -1 \\ 2 & -1 & 0 \\ -2 & 1 & 0 \end{bmatrix}$$

$\therefore Tr(B) = 0 - 1 + 0 = -1$
Hence, $Tr(A) - Tr(B) = 1 - (-1) = 2$

Question144

Let I be an identity matrix of order 2 × 2 and $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of $n \in N$ for which $P^n = 5I - 8P$ is equal to
[2021, 18 March Shift-II]

Answer: 6

Solution:

Method (1)

Given, $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$

Now, $P^n = 5I - 8P = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 16 & -8 \\ 40 & -24 \end{bmatrix}$

$\Rightarrow P^n = \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix} \quad \cdots\cdots\cdot (i)$

Now, $P^2 = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$

$= \begin{bmatrix} -1 & 1 \\ -5 & 4 \end{bmatrix}$

Again $P^3 = P \cdot P^2 = \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix}$ Now,

$P^6 = P^3 \cdot P^3 = \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 10 & -7 \end{bmatrix}$

$\Rightarrow \; P^6 = \begin{bmatrix} -11 & 8 \\ -40 & 29 \end{bmatrix} = P^n \quad [\text{from Eq. (i)} \;]$

$\therefore \; n = 6$

Method (2)

Given $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$

Characteristics equation is $|P - \lambda| = 0$

$\begin{vmatrix} 2-\lambda & -1 \\ 5 & -3-\lambda \end{vmatrix} = 0$

$\Rightarrow \; (2-\lambda)(-3-\lambda) + 5 = 0$

$\Rightarrow \; \lambda^2 + \lambda - 1 = 0$

By Cayley-Hamilton Theorem,

$P^2 + P - 1 = 0$

$\Rightarrow \; P^2 = 1 - P$

$\Rightarrow \; P^3 + P^2 - P = 0$

$\Rightarrow \; P^3 = P - P^2 = P - (1 - P)$ [from Eq. (i)]

$\Rightarrow \; P^3 = 2P - 1$

$\Rightarrow \; P^3 = 2P - 1$

Now, $P \cdot P^3 = P(2P - 1)$

$\Rightarrow P^4 = 2P^2 - P = 2(1 - P) - P$[from Eq. (i)]

$\Rightarrow \; P^4 = -3P + 21$

Again, $P \cdot (P^4) = P(-3P + 21)$

$\Rightarrow \; P^5 = -3P^2 + 2P$

$= -3(1 - P) + 2P$[from Eq. (i)]

$= 5P - 31$

and $P(P^5) = P(5P - 31)$

$11 \Rightarrow P^6 = 5P^2 - 3P = 5(1 - P) - 3P$

$\Rightarrow p^6 = 51 - 8P = P^n \quad \text{(given)}$

$\therefore n = 6$

Question145

Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$, $i = \sqrt{-1}$. Then, the system of linear equations

$A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has

[2021, 16 March Shift-I]

Options:

A. a unique solution.

B. infinitely many solutions.

C. no solution.

D. exactly two solutions.

Answer: C

Solution:

Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$

$A^2 = A \cdot A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}\begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$

$= \begin{bmatrix} i^2+i^2 & -i^2-i^2 \\ -i^2-i^2 & i^2+i^2 \end{bmatrix}$

$= \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$A^4 = A^2 \cdot A^2$

$= \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$= \begin{bmatrix} 4+4 & -4-4 \\ -4-4 & 4+4 \end{bmatrix} = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$

Similarly, $A^8 = A^4 \cdot A^4$

$= \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}\begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix}$

$= \begin{bmatrix} 64+64 & -64-64 \\ -64-64 & 64+64 \end{bmatrix}$

$= \begin{bmatrix} 128 & -128 \\ -128 & 128 \end{bmatrix}$ Now, $A^8\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$

$\begin{bmatrix} 128 & -128 \\ -128 & 128 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$

$128x - 128y = 8$

$\Rightarrow x - y = \frac{1}{16} \quad \cdots\cdots (i)$

and $-128x + 128y = 64$

$\Rightarrow x - y = -\frac{1}{2} \quad \cdots\cdots (ii)$

From Eqs. (i) and (ii), we get these two lines are parallel. So, there will be no solution.

Question146

The total number of 3 × 3 matrices A having entries from the set (0, 1, 2, 3), such that the sum of all the diagonal entries of AA^T is 9 , is equal to
[2021, 16 March Shift-I]

Answer: 766

Solution:

Set $S : \{0, 1, 2, 3\}$

$$\text{Let } A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a \cdot a & a \cdot b & a \cdot c \\ b \cdot a & b \cdot b & b \cdot c \\ c \cdot a & c \cdot b & c \cdot c \end{bmatrix}$$

where $a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$b = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$c = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

Now, as per question

$ca \cdot a + b \cdot b + c \cdot c = 9$

$\Rightarrow (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2)$

$+(c_1^2 + c_2^2 + c_3^2) = 9$

$[a_i, b_i, c_i \in S]$

$9 = (1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1)$

or $9 = (1 + 4 + 4 + 0 + 0 + 0 + 0 + 0 + 0)$

or $9 = (9 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0)$

or $9 = (4 + 1 + 1 + 1 + 1 + 1 + 0 + 0 + 0)$

Total permutations in case $1 = 1$

Total permutations in case 2 = $\frac{9!}{6!2!} = 252$

In case 3 = $\frac{9!}{8!} = 9$

In case 4 = $\frac{9!}{5!3!} = 504$

Total permutations $= 1 + 252 + 9 + 504 = 766$

Question147

Let A = $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and a = $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that A = X B, where X $= \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$ and $k \in R$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$, then the value of k is [2021, 16 March Shift-II]

Answer: 1

Solution:

$A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

$A = XB$

$X = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$

$XB = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

$XB = \frac{1}{\sqrt{3}}\begin{bmatrix} b_1 - b_2 \\ b_1 + kb_2 \end{bmatrix}$

As, $A = XB$

So, $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{\sqrt{3}}\begin{bmatrix} b_1 - b_2 \\ b_1 + kb_2 \end{bmatrix}$

$\sqrt{3}a_1 = b_1 - b_2 \quad \cdots\cdots \text{(i)}$

$\sqrt{3}a_2 = b_1 + kb_2 \quad \cdots\cdots \text{(ii)}$

And as given, $a_1^{\,2} + a_2^{\,2} = \frac{2}{3}(b_1^{\,2} + b_2^{\,2})$

Adding, Eq. (i) ^{2}+ Eq. (ii) ^{2},

$3a_1^{\,2} + 3a_2^{\,2} = (b_1 - b_2)^2 + (b_1 + kb_2)^2$

$\Rightarrow 2(b_1^{\,2} + b_2^{\,2})$

$= \; 2b_1^{\,2} + b_2^{\,2}(k^2 + 1) + 2b_1b_2(k - 1)$

$\Rightarrow b_2^{\,2}(k^2 + 1 - 2) + 2b_1b_2(k - 1) = 0$

$\Rightarrow (k - 1)[b_2^{\,2}(k + 1) + 2b_1b_2] = 0$

So, $\quad k = 1$

Question148

Let P = $\begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ **and A =** $\begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega + 1 \end{bmatrix}$

where, $\omega = \frac{-1 + i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3 . If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to.........
[16 Mar 2021 Shift 1]

Answer: 36

Solution:

Given, $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega + 1 \end{bmatrix}$

$|(P^{-1}AP - I_3)|^2 = \alpha\omega^2$

$\Rightarrow \quad |\,(P^{-1}AP - I_3)(P^{-1}AP - I_3)\,| \; = \alpha\omega^2$

$\Rightarrow P^{-1}APP^{-1}AP - P^{-1}API_3 - I_3P^{-1}AP + I_3 \cdot I_3 | = \alpha\omega^2$

$\Rightarrow \ P^{-1}A^2P - P^{-1}AP - P^{-1}AP + I_3 | \ = \alpha\omega^2$

$[\because PP^{-1} = I \text{ and } IA = A]$

$\Rightarrow \ | P^{-1}A^2P - 2P^{-1}AP + PP^{-1} | = \alpha\omega^2$

$\Rightarrow \ | P^{-1}(A^2 - 2A + I_3)P | = \alpha\omega\omega^2$

$\Rightarrow \ | P^{-1} | | A - I_3 |^2 | P | = \alpha\omega^2$

$\Rightarrow \ | P^{-1}P | | A - I_3 |^2 = \alpha\omega^2$

$\Rightarrow |A - I_3|^2 = \alpha\omega^2$

Consider,

$$A - I_3 = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega+1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A - I_3| = \begin{vmatrix} 1 & 7 & \omega^2 \\ -1 & -\omega-1 & 1 \\ 0 & -\omega & -\omega \end{vmatrix}$$

On applying $C_2 \to C_2 - C_3$, we get

$$|A - I_3| = \begin{vmatrix} 1 & 7-\omega^2 & \omega^2 \\ -1 & -\omega-2 & 1 \\ 0 & 0 & -\omega \end{vmatrix}$$

On applying $C_2 \to C_2 - C_3$, we get

$$|A - I_3| = \begin{vmatrix} 1 & 7-\omega^2 & \omega^2 \\ -1 & -\omega-2 & 1 \\ 0 & 0 & -\omega \end{vmatrix}$$

$= -\omega[(-\omega-2) - (-7+\omega^2)]$

$= -\omega(-\omega - 2 + 7 - \omega^2)$

$= -\omega(1 - 2 + 7)$

$= -6\omega$

$|A - I_3| = -6\omega$

$|A - I_3|^2 = 36\omega^2 = \alpha\omega^2$

$\therefore \ \alpha = 36$

Question149

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det[A^{10} - \text{Adj}(2A)^{10}]$ is equal to .....
[17 Mar 2021 Shift 1]

Answer: 16

Solution:

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$

$\det(A^4) + \det[A^{10} - [\operatorname{Adj}(2A)]^{10}]$

$$A \cdot A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}, A^2 = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 0 & -1 \end{bmatrix}$$

$$A^4 = A^3 \cdot A$$

$$= \begin{bmatrix} 8 & 9 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 16 & 15 \\ 0 & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 2^n & 2^n - (-1)^n \\ 0 & (-1)^n \end{bmatrix}$$

$$A^{10} = \begin{bmatrix} 2^{10} & 2^{10} - 1 \\ 0 & 1 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 4 & 6 \\ 0 & -2 \end{bmatrix}$$

$$\operatorname{adj}(2A) = \begin{bmatrix} -2 & -6 \\ 0 & 4 \end{bmatrix}$$

$$\operatorname{adj}(2A) = -2\begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} \quad \left[\begin{array}{c} \because x = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ \\ \operatorname{adj}(x) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \end{array}\right]$$

$$[\operatorname{adj}(2A)]^2 = 4\begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} = 4\begin{pmatrix} 1 & -3 \\ 0 & 4 \end{pmatrix}$$

$$[\operatorname{adj}(2A)]^3 = 4\begin{pmatrix} 1 & -3 \\ 0 & 4 \end{pmatrix} \times (-2)\begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} = -8\begin{pmatrix} 1 & 9 \\ 0 & -8 \end{pmatrix}$$

$$[\operatorname{adj}(2A)]^3 = 4\begin{pmatrix} 1 & -3 \\ 0 & 4 \end{pmatrix} \times (-2)\begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} = -8\begin{pmatrix} 1 & 9 \\ 0 & -8 \end{pmatrix}$$

$$[\operatorname{adj}(2A)]^n = (-2)^n \begin{bmatrix} 1 & (-1)^n[2^n-(-1)^n] \\ 0 & (-1)^n 2^n \end{bmatrix}$$

$$[\operatorname{adj}(2A)]^{10} = 2^{10}\begin{pmatrix} 1 & -(2^{10}-1) \\ 0 & 2^{10} \end{pmatrix}$$

Now, $A^{10} - [\operatorname{adj}(2A)]^{10}$

$$= \begin{bmatrix} 2^{10} & 2^{10}-1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2^{10} & -(2^{20}-2^{10}) \\ 0 & 2^{20} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2\cdot 2^{10}-2^{20}-1 \\ 0 & 1-2^{20} \end{bmatrix}$$

$\det|A^{10} - \operatorname{adj}(2A)^{10}| = 0$

$\therefore \det(A^4) + \det[A^{10} - \operatorname{adj}(2A)^{10}] = (16)^4 + 0 = 16$

Question150

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \ldots.. + A^{20}$, then the sum of all the elements of the matrix M is equal to
[2021, 27 July Shift-II]

Answer: 2020

Solution:

We have, $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 6 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

....
......

$$A^n = \begin{bmatrix} 1 & n & \sum n \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix}$$

Now, $M = A + A^2 + A^3 + \ldots + A^{20}$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} +$$

$$\ldots + \begin{bmatrix} 1 & n & \sum n \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & \sum n & \sum(\sum r) \\ 0 & 20 & \sum n \\ 0 & 0 & 20 \end{bmatrix}$$

Now, $\sum_{r=1}^{20} n = 1 + 2 + \ldots + 20 = \frac{20 \times 21}{2} = 210$

$\sum_{r=1}^{20} (\sum r) = \sum_{r=1}^{20} \frac{r(r+1)}{2} = \frac{1}{2} \sum_{r=1}^{20} (r^2 + r)$

$= \frac{1}{2}\left[\frac{20 \times 21 \times 41}{6} + \frac{20 \times 21}{2}\right]$

$= \frac{1}{2}[2870 + 210] = 1540$

Hence, $M = \begin{bmatrix} 20 & 210 & 1540 \\ 0 & 20 & 210 \\ 0 & 0 & 20 \end{bmatrix}$

Sum of all elements $= 2020$.

Question151

$$\mathbf{S} = \left\{ n \in N \;\middle|\; \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^2 \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \; \forall\, a, b, c, d \in R \right\}$$

, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is [2021, 25 July Shift-1]

Answer: 11

Solution:

Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $A = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n$

$\Rightarrow AX = 1X$

$A = 1$

$A^2 = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$

$\Rightarrow A = i\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow A^8 = i^4 1 = 1$

$\therefore n = 8$

$\Rightarrow n$ is a multiple of 8 .

$16, 24, \ldots\ldots .96$

Number of elements $= \dfrac{96-16}{8} + 1 = 11$

Question152

If P = $\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}$, then P^{50} is

[2021, 25 July Shift-II]

Options:

A. $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

Answer: A

Solution:

Given, $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$

$\Rightarrow P^2 = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$\Rightarrow P^3 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{bmatrix}$

$\Rightarrow P^4 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

$\vdots$

$\Rightarrow P^n = \begin{bmatrix} 1 & 0 \\ \frac{n}{2} & 1 \end{bmatrix}$

Hence, $P^{50} = \begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

Question153

Let $A = [a_{ij}]$ be a real matrix of order 3×3, such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1, 2, 3$. Then,
the sum of all the entries of the matrix A^3 is equal to
[2021, 22 July Shift-II]

Options:

A. 2

B. 1

C. 3

D. 9

Answer: C

Solution:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = A$$

Sum of elements of each row is 1 .

Let X be $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Then, $AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$AX = \begin{bmatrix} a_{11}+ & a_{12}+ & a_{13} \\ a_{21}+ & a_{22}+ & a_{23} \\ a_{31}+ & a_{32}+ & a_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$AX = X$

Replace X with AX.

$A \cdot AX = AX \Rightarrow A^2X = X$

Again, replace X with AX,

$A^2(AX) = AX$

$A^3X = X$

Let $A^3 = \begin{bmatrix} a_{11}' & a_{12}' & a_{13}' \\ a_{21}' & a_{22}' & a_{23}' \\ a_{31}' & a_{32}' & a_{33}' \end{bmatrix}$

Then,

$$A^3X = \begin{bmatrix} a_{11}' & a_{12}' & a_{13}' \\ a_{21}' & a_{22}' & a_{23}' \\ a_{31}' & a_{32}' & a_{33}' \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$A^3X = \begin{bmatrix} a_{11}' + & a_{12}' + & a_{13}' \\ a_{21}' + & a_{22}' + & a_{23}' \\ a_{31}' + & a_{32}' + & a_{33}' \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

So, $a_{11}' + a_{12}' + a_{13}' = 1$

$a_{21}' + a_{22}' + a_{23}' = 1$

$a_{31}' + a_{32}' + a_{33}' = 1$

$\therefore$ Sum $= 3$

Question154

Let A = $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then, the number of 3 × 3 matrices B with entries from the set {1, 2, 3, 4, 5} and satisfying AB = BA is
[2021, 22 July Shift-II]

Answer: 3125

Solution:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} B = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & a_1 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

AB = BA

$$\begin{pmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} a_2 & a_1 & a_3 \\ b_2 & b1 & b_3 \\ c_2 & c1 & c_3 \end{pmatrix}$$

$$\left\{ \begin{array}{c} b_1 = a_2 \\ b_2 = a_1 \\ b_3 = a_3. \end{array} \right. \quad \left\{ \begin{array}{c} a_1 = b_2 \\ a_2 = b_1 \\ a_3 = b_3. \end{array} \right. \quad \left\{ \begin{array}{c} C_1 = C_2 \\ C_2 = C_1 \\ C_3 = C_3. \end{array} \right.$$

$$B = \begin{pmatrix} a_1 & a_2 & a_3 \\ a_2 & a_1 & a_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

Number of distinct elements in B is 5 $\{a_1, a_2, a_3, c_1, c_3\}$ and according to question, $a_{ij} \in \{1, 2, 3, 4, 5\}$.
So, number of matrices $= 5 \times 5 \times 5 \times 5 \times 5 = 3125$

Question155

Let A = $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ and B = $7A^{20} - 20A^7 + 2l$, where l is an identity matrix of order 3 × 3. If B = $[b_{ij}]$, then b_{13} is equal to ...
[2021, 20 July Shift-1]

Answer: 910

Solution:

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$B = 7A^{20} - 20A^7 + 21$$

$$\therefore A^2 = A \cdot A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -3 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^4 = A^3A = \begin{pmatrix} 1 & -3 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow A^4 = \begin{pmatrix} 1 & -4 & 6 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow A^n = \begin{pmatrix} 1 & -n & \frac{n^2-n}{2} \\ 0 & 1 & -n \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore\ b_{13} = 7\left(\frac{20 \times 19}{2}\right) - 20\left(\frac{7 \times 6}{2}\right) + 2(0)$$

$$\Rightarrow b_{13} = 1330 - 420 = 910$$

Question156

If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$, and $Q = A^TBA$, then the inverse of the matrix $AQ^{2021}A^T$ is equal to
[26 Aug 2021 Shift 1]

Options:

A. $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

B. $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

C. $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

D. $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

Answer: B

Solution:

$$AA^T = \begin{vmatrix} \frac{1}{5} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{vmatrix} \begin{vmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{vmatrix}$$

$$AA^T \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Now, $Q^2 = A^TBAA^TBA$

$\Rightarrow Q^2 = A^TB^2A$

Again, $Q^3 = (A^TBA)(A^TB^2A) = A^TB^3A$

Similarly,

$Q^{2021} = A^TB^{2021}A$

$AQ^{2021}A^T = A(A^TB^{2021}A)A^T$

$= (AA^T)B^{2021}(AA^T) = B^{2021}$

$$B = \begin{vmatrix} 1 & 0 \\ i & 1 \end{vmatrix}$$

$$B^2 = \begin{vmatrix} 1 & 0 \\ 2i & 1 \end{vmatrix}, \text{similarly } B^{2021} = \begin{vmatrix} 1 & 0 \\ 2021i & 1 \end{vmatrix}$$

$$(B^{2021})^{-1} = \frac{\text{adj}(B^{2021})}{|B^{2021}|} = \begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$$

Question157

Let A be a 3 × 3 real matrix. If det(2 Adj(2 Adj(Adj(2A)))) = 2^{41} , then the value of det(A^2) equal
[26 Aug 2021 Shift 2]

Answer: 4

Solution:

We have, A is a square matrix of 3×3. Now,
$2\,\text{Adj}(2\,\text{Adj}(\text{Adj}(2A)))$
$= 2\,\text{Adj}(2\,\text{Adj}(2^{3-1}\,\text{adj}\,A))$ $[\because \text{adj}(KA) = K^{n-1}$ adj $A]$
$= 2\,\text{Adj}(2\,\text{Adj}(4\,\text{Adj}\,A))$
$= 2\,\text{Adj}(2 \times 4^{3-1}\,\text{Adj}\,\text{Adj}\,A)$
$= 2\,\text{Adj}(32\,\text{Adj}\,\text{Adj}\,A)$
$= 2 \times 32^{3-1}\,\text{Adj}\,\text{Adj}\,\text{Adj}\,A$
$= 2^{11}\,\text{Adj}\,\text{Adj}\,\text{Adj}\,A$
$= 2^{11}\,\text{Adj}(|A|^{3-2}A)$
$= 2^{11}\,\text{Adj}(|A|A)$
$= 2^{11} \times |A|^{3-1}\,\text{Adj}\,A$
$= 2^{11} \times |A|^{2}\,\text{Adj}\,A$
Now, $|2\,\text{Adj}(2\,\text{Adj}(\text{Adj}(2A)))| = 2^{41}$
$\Rightarrow |\,2^{11} \times |A|^{2}\,\text{Adj}\,A = 2^{41}$
$\Rightarrow (2^{11})^{3}(|A|^{2})^{3}|\text{Adj}\,A| = 2^{41}$
$\Rightarrow 2^{33}\,|A|^{6}\,|A|^{3-1} = 2^{41}$
$\Rightarrow |A|^{6} \times |A|^{2} = 2^{8}$
$\Rightarrow |A|^{8} = 2^{8}$
$\Rightarrow |A| = \pm 2$
Now, $|A^{2}| = |A|^{2} = (\pm 2)^{2} = 4$

Question158

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A$, $\alpha, \beta \in R$, I is a 2×2 identity matrix, then $4(\alpha - \beta)$ is equal to :
[27 Jul 2021 Shift 1]

Options:

A. 5

B. $\frac{8}{3}$

C. 2

D. 4

Answer: D

Solution:

$$A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}, |A|=6$$

$$A^{-1}=\frac{\text{ad jA}}{|A|}=\frac{1}{6}\begin{bmatrix}4 & -2\\ 1 & 1\end{bmatrix}=\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}=\begin{bmatrix}\alpha & 0\\ 0 & \alpha\end{bmatrix}+\begin{bmatrix}\beta & 2\beta\\ -\beta & 4\beta\end{bmatrix}$$

$$\left.\begin{aligned}\alpha+\beta=\frac{2}{3}\\ \beta=-\frac{1}{6}\end{aligned}\right\} \Rightarrow \alpha=\frac{2}{3}+\frac{1}{6}=\frac{5}{6}$$

$4(\alpha-\beta)=4(1)=4$

Question159

Let A and B be two 3 × 3 real matrices such that(A^2-B^2) is invertible matrix. If $A^5=B^5$ and$A^3B^2=A^2B^3$, then the value of the determinant ofthe matrix A^3+B^3 is equal to :
[27 Jul 2021 Shift 2]

Options:

A. 2

B. 4

C. 1

D. 0

Answer: D

Solution:

$C=A^2-B^2; |C|\neq 0$
$A^5=B^5$ and $A^3B^2=A^2B^2$
Now, $A^5-A^3B^2=B^5-A^2B^3$
$\Rightarrow A^3(A^2-B^2)+B^3(A^2-B^2)=0$
$\Rightarrow (A^3+B^3)(A^2-B^2)=0$
Post multiplying inverse of A^2-B^2 :
$A^3+B^3=0$

Question160

Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-1} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j, \end{cases}$ then $\det(3\,\mathrm{Adj}(2A^{-1}))$ is equal to ______.
[20 Jul 2021 Shift 2]

Answer: 108

Solution:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$|A| = 4$

$|3\,\mathrm{adj}(2A^{-1})| = |\,3.2^2\mathrm{adj}(A^{-1})|$

$= 12^3\,|\,\mathrm{adj}(A^{-1})\,| = 12^3\,|\,A^{-1}|^2 = \dfrac{12^3}{|A|^2} = \dfrac{12^3}{16} = 108$

Question161

If the matrix $A = \begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is
[27 Aug 2021 Shift 1]

Options:

A. $\frac{1}{2}$

B. $-\frac{1}{2}$

C. -1

D. 1

Answer: A

Solution:

Given matrix,

$A = \begin{bmatrix} 0 & 2 \\ K & -1 \end{bmatrix}$

Characteristic equation of A is

$|A - \lambda I| = 0$

$\Rightarrow \begin{vmatrix} -\lambda & 2 & \\ K & -1 & -\lambda \end{vmatrix} = 0$

$\Rightarrow \lambda(1 + \lambda) - 2k = 0$

$\Rightarrow \lambda^2 + \lambda - 2k = 0$

$\because$ Every square matrix satisfied its own characteristic equation.

$\therefore A^2 + A - 2KI = 0$

$\Rightarrow A^2 = 2KI - A$

$\Rightarrow A^4 = 4K^2 I + A^2 - 2(2KI)(A)$

$\Rightarrow A^4 = 4K^2 I + 2KI - A - 4KA$

$\Rightarrow A^4 = (4K^2 + 2K)I - (1 + 4K)A$ ...(i)

Now, $A(A^3 + 3I) = 2I$

$\Rightarrow A^4 = 2I - 3A$ ...(ii)

Comparing Eqs. (i) and (ii), we get

$1 + 4K = 3$

$\Rightarrow K = \frac{1}{2}$

Question162

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $A^{2025} - A^{2020}$ is equal to

[2021, 26 Aug. Shift-II]

Options:

A. $A^6 - A$

B. A^5

C. $A^5 - A$

D. A^6

Answer: A

Solution:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Now,

$$A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^4 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

......

......

$$A^n = \begin{pmatrix} 1 & 0 & 0 \\ n-2 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ n-1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^{2025} - A^{2020} = \begin{pmatrix} 1 & 0 & 0 \\ 2024 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 2019 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now,

$$A^6 - A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore A^{2025} - A^{2020} = A^6 - A$

Question163

Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$$ **, then the matrix A^{31} is equal to:**

[Jan. 7, 2020 (I)]

Options:

A. A

B. I_3

C. A^2

D. A^3

Answer: D

Solution:

Solution of $x^2 + x + 1 = 0$ is ω, ω^2
So, $\alpha = \omega$ and
$\omega^4 = \omega^3 . \omega = 1 . \omega = \omega$

$$A^2 = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$\Rightarrow A^4 = I$
$\Rightarrow A^{30} = A^{28} \times A^3 = A^3$

Question164

The number of all 3 × 3 matrices A, with enteries from the set {−1, 0, 1} such that the sum of the diagonal elements of AA^T is 3, is______.
[NA Jan. 8, 2020 (I)]

Answer: 672

Solution:

Let $A = [a_{ii}]_{3 \times 3}$
It is given that sum of diagonal elements of AA^T is 3 i.e., $tr(AA^T) = 3$
$a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + \ldots\ldots + a_{33}^2 = 3$
Possible cases are

$$\left.\begin{array}{cc} 0,0,0,0,0,0,1,1,1 & \rightarrow \; 1 \\ 0,0,0,0,0,0,-1,-1,-1, & \rightarrow \; 1 \\ 0,0,0,0,0,0,1,1,-1 & \rightarrow \; 3 \\ 0,0,0,0,0,0,-1,1,-1 & \rightarrow \; 3 \end{array}\right\} \; {}^9C_6 \times 8 = 84 \times 8 = 672$$

Question165

If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then $10A^{-1}$ is equal to:

[Jan. 8, 2020 (II)]

Options:

A. $A - 4I$

B. $6I - A$

C. $A - 6I$

D. $4I - A$

Answer: C

Solution:

Characteristics equation of matrix ' A ' is $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 2 \\ 9 & 4-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 6\lambda - 10 = 0$$

$\therefore A^2 - 6A - 10I = 0$
$\Rightarrow A^{-1}(A^2) - 6A^{-1} - 10I\,A^{-1} = 0$
$\Rightarrow 10A^{-1} = A - 6I$

Question166

If $A = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}$, $\left(\theta = \frac{\pi}{24}\right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $i = \sqrt{-1}$, then which one of the following is not true?
[Sep. 04, 2020 (I)]

Options:

A. $0 \le a^2 + b^2 \le 1$

B. $a^2 - d^2 = 0$

C. $a^2 - c^2 = 1$

D. $a^2 - b^2 = \frac{1}{2}$

Answer: D

Solution:

$\because A = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix}$

$\therefore A^n = \begin{bmatrix} \cos n\theta & i\sin n\theta \\ i\sin n\theta & \cos n\theta \end{bmatrix}, n \in N$

$\because A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\therefore A^5 = \begin{bmatrix} \cos\theta & i\sin\theta \\ i\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\therefore a = \cos 5\theta,\ b = i\sin 5\theta = c,\ d = \cos 5\theta$

$\therefore a^2 - b^2 = \cos^2 5\theta + \sin^2 5\theta = 1$

$a^2 - c^2 = \cos^2 5\theta + \sin^2 5\theta = 1$

$a^2 - d^2 = \cos^2 5\theta - \cos^2 5\theta = 1$

$a^2 + b^2 = \cos^2 5\theta - \sin^2 5\theta = \cos 10\theta = \cos\frac{10\pi}{24}$

and $0 < \cos\frac{5\pi}{12} < 1 \Rightarrow 0 \le a^2 + b^2 \le 1$

$\therefore a^2 - b^2 = \frac{1}{2}$ is wrong.

Question167

Let A = $\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, x ∈ R and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _______.

[NA Sep. 03, 2020 (I)]

Answer: 10

Solution:

$A^2 = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$

$x^4 = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$

$= \begin{bmatrix} (x^2+1)^2+x^2 & x(x^2+1)+x \\ x(x^2+1)+x & x^2+1 \end{bmatrix}$

Given that $(x^2+1)^2+x^2 = 109$

$x^4+3x^2-108 = 0$

$\Rightarrow (x^2+12)(x^2-9) = 0$

$\therefore x^2 = 9$

$a_{22} = x^2+1 = 9+1 = 10$

Question168

Let a, b, c ∈ R be all non-zero and satisfy$a^3+b^3+c^3 = 2$. If the matrix

A = $\begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ satisfies$A^T A = I$, then a value of abc can be:

[Sep. 02, 2020 (II)]

Options:

A. $-\frac{1}{3}$

B. $\frac{1}{3}$

C. 3

D. $\frac{2}{3}$

Answer: B

Solution:

Given : $A^T A = I$

$$\Rightarrow = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \sum a^2 & \sum ab & \sum ab \\ \sum ab & \sum a^2 & \sum ab \\ \sum ab & \sum ab & \sum a^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So, $\sum a^2 = 1$ and $\sum ab = 0$

Now, $a^3 + b^3 + c^3 - 3abc$

$= (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

$= (a+b+c)(1-0)$

$= \sqrt{(a+b+c)^2} = \sqrt{\sum a^2 + 2\sum ab} = \pm 1$

$\Rightarrow 2 - 3abc = 1 \Rightarrow abc = \frac{1}{3}$

or $2 - 3abc = -1 \Rightarrow abc = 1.$

Question169

Let A = $\begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$ **. If** $AA^T = I_3$**, then |p| is :**

[Jan. 11, 2019 (I)]

Options:

A. $\frac{1}{\sqrt{5}}$

B. $\frac{1}{\sqrt{3}}$

C. $\frac{1}{\sqrt{2}}$

D. $\frac{1}{\sqrt{6}}$

Answer: C

Solution:

$$A = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix}$$

$$\therefore A\,.\,A^T = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix} \times \begin{bmatrix} 0 & p & p \\ 2q & q & -q \\ r & -r & r \end{bmatrix}$$

$$= \begin{bmatrix} 4q^2+r^2 & 2q^2-r^2 & -2q^2+r^2 \\ 2q^2-r^2 & p^2+q^2+r^2 & p^2-q^2-r^2 \\ -2q^2+r^2 & p^2-q^2-r^2 & p^2+q^2+r^2 \end{bmatrix}$$

Given, $AA^T = I$

$\therefore 4q^2+r^2 = p^2+q^2+r^2 = 1$

$\Rightarrow p^2-3q^2 = 0$ and $r^2 = 1-4q^2$

and $2q^2-r^2 = 0 \Rightarrow r^2 = 2q^2$

$\therefore p^2 = \frac{1}{2}, q^2 = \frac{1}{6}$ and $r^2 = \frac{1}{3}$

$\therefore |p| = \frac{1}{\sqrt{2}}$

Question170

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21}+q_{31}}{q_{32}}$ is equal to:

[Jan. 12, 2019 (I)]

Options:

A. 10

B. 135

C. 15

D. 9

Answer: A

Solution:

$$P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix}$$

$$\Rightarrow P^4 = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 90 & 12 & 1 \end{bmatrix}$$

$$\Rightarrow P^5 = = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 90 & 12 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 15 & 1 & 0 \\ 135 & 15 & 1 \end{bmatrix}$$

$$\because Q - P^5 = I_3$$

$$\therefore Q = I_3 + P^6 = \begin{bmatrix} 2 & 0 & 0 \\ 15 & 2 & 0 \\ 135 & 15 & 2 \end{bmatrix}$$

$$\frac{q_{21} + q_{31}}{q_{32}} = \frac{15 + 135}{15} = 10$$

Question171

Let A = $\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, ($\alpha \in R$) such that $A^{32} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ Then a value of α is :
[April 8, 2019 (I)]

Options:

A. $\frac{\pi}{32}$

B. 0

C. $\frac{\pi}{64}$

D. $\frac{\pi}{16}$

Answer: C

Solution:

$A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

$A^2 = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

$= \begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$

Similarly, $A^4 = A^2 . A^2 = \begin{bmatrix} \cos 4\alpha & -\sin 4\alpha \\ \sin 4\alpha & \cos 4\alpha \end{bmatrix}$

and so on $A^{32} = \begin{bmatrix} \cos 32\alpha & -\sin 32\alpha \\ \sin 32\alpha & \cos 32\alpha \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Then $\sin 32\alpha = 1$ and $\cos 32\alpha = 0$

$\Rightarrow 32\alpha = n\pi + (-1)^n\frac{\pi}{2}$ and $32\alpha = 2n\pi + \frac{\pi}{2}$

$\Rightarrow \alpha = \frac{n\pi}{32} + (-1)^n\frac{\pi}{64}$ and $\alpha = \frac{n\pi}{16} + \frac{\pi}{64}$ where $n \in z$

Put $n = 0, \alpha = \frac{\pi}{64}$

Hence, required value of α is $\frac{\pi}{64}$

Question172

The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, $(x, y \in R, x \neq y)$ for which $A^T A = 3I_3$ is:
[April 09, 2019 (II)]

Options:

A. 2

B. 3

C. 6

D. 4

Answer: D

Solution:

Given, $A^T A = 3I_3$

$$\begin{bmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix} = 31$$

$$\Rightarrow \begin{bmatrix} 8x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$\Rightarrow 8x^2 = 3$ and $6y^2 = 3 \Rightarrow x = \pm\sqrt{\frac{3}{8}}$ and $y = \pm\sqrt{\frac{1}{2}}$

Number of combinations of $(x, y) = 2 \times 2 = 4$

Question173

If A is a symmetric matrix and B is a skew-symmetrix matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to:

[April 12, 2019 (I)]

Options:

A. $\begin{bmatrix} -4 & -1 \\ -1 & 4 \end{bmatrix}$

B. $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

C. $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

D. $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

Answer: B

Solution:

Let $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$ and $B = \begin{bmatrix} 0 & d \\ -d & 0 \end{bmatrix}$

Then, $A + B = \begin{bmatrix} a & c+d \\ c-d & b \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$

On comparing each term,

$a = 2, b = -1, c - d = 5, c + d = 3$

$\Rightarrow a = 2, b = -1, c = 4, d = -1$

Now, $AB = \begin{bmatrix} 2 & 4 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

Question174

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^{20}$. Then the sum of the elements of the first column of B is?
[Online April 16, 2018]

Options:

A. 211

B. 210

C. 231

D. 251

Answer: C

Solution:

Here $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$\therefore A^2 = A \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$

also $A^3 = A^2 \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 3 & 1 \end{bmatrix}$

and, $A^4 = A^3 \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 10 & 4 & 1 \end{bmatrix}$

On observing the pattern, we come to a conclusion that,

$A = \begin{bmatrix} 1 & 0 & 0 \\ n & 1 & 0 \\ \frac{n(n+1)}{2} & n & 1 \end{bmatrix}$

$\therefore A^{20} = [1, 0, 0; 20, 1, 0; 210, 20, 1]$

Therefore, sum of first column of $A^{20} = [1 + 20 + 210] = 231$

Question175

For two 3×3 matrices A and B, let $A + B = 2B^T$ and $3A + 2B = I_3$, where B^T is the transpose of B and I_3 is 3×3 identity matrix. Then : quad
[Online April 9, 2017]

Options:

A. $5A + 10B = 2I_3$

B. $10A + 5B = 3I_3$

C. $B + 2A = I_3$

D. $3A + 6B = 2I_3$

Answer: B

Solution:

$A^T + B^T = 2B$

$\because [(A+B)^T = (2B^T)^T]$

$\Rightarrow B = \dfrac{A^T + B^T}{2} = A + \left(\dfrac{B^T + A^T}{2}\right) = 2B^T$

$\Rightarrow 2A + A^T = 3B^T \Rightarrow A = \dfrac{3B^T - A^T}{2}$

Also, $3A + 2B = I_3$ ....(i)

$\Rightarrow 3(3B^T - A^T 2) + 2\left(\dfrac{A^T + B^T}{2}\right) = I_3$

$\Rightarrow 11B^T - A^T = 2I_3$ ......(ii)

Add (i) and (ii)

$35 \sim B = 7I_3$

$\Rightarrow B = \dfrac{I_3}{5} \Rightarrow 11\dfrac{I_3}{5} - A = 2I_3$

$\Rightarrow 11\dfrac{I_3}{5} - 2I_3 = A \Rightarrow A = \dfrac{I_3}{5}$

$\because 5A = 5B = I_3$

$\Rightarrow 10A + 5B = 3I_3$

Question176

If P = $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ **, A =** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ **and** $Q = PAP^T$**, then**$P^T Q^{2015} P$ **is ;**

[Online April 9, 2016]

Options:

A. $\begin{bmatrix} 0 & 2015 \\ 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 2015 & 0 \\ 1 & 2015 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2015 \\ 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 2015 & 1 \\ 0 & 2015 \end{bmatrix}$

Answer: C

Solution:

$P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ $P^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{-1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

$PP^T = P^T P = I$

$Q^{2015} = (PAP^T)(PAP^T)$ ———— (2015 terms)

$= PA^{2015}P^T$

$P^T Q^{2015} P = A^{2015}$

$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

$\therefore A^{2015} = \begin{bmatrix} 1 & 2015 \\ 0 & 1 \end{bmatrix}$

Question177

If A = $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then which one of the following statements is not correct?
[Online April 10, 2015]

Options:

A. $A^2 + I = A(A^2 - I)$

B. $A^4 - I = A^2 + I$

C. $A^3 + I = A(A^3 - I)$

D. $A^3 - I = A(A - I)$

Answer: A

Solution:

Given that

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow A^2 = -I$$

$$A^3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^2 + I = A^3 - A$$
$$-I + I = A^3 - A$$
$$A^3 \neq A$$

Question178

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to: [2015]

Options:

A. (2,1)

B. (-2,-1)

C. (2,-1)

D. (-2,1)

Answer: B

Solution:

Given that $AA^T = 9I$

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & 2 \\ 2 & -1 & b \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1+4+4 & 2+2-4 & a+4+2b \\ 2+2-4 & 4+1+4 & 2a+2-2b \\ a+4+2b & 2a+2-2b & a^2+4+b^2 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$\Rightarrow a+4+2b=0 \Rightarrow a+2b=-4$ ......(i)

$2a+2-2b=0 \Rightarrow 2a-2b=-2$ .....(ii)

$\Rightarrow a-b=-1$

Subtract (ii) from (i)

$a+2b=-4$

$a-b=-1$

$- \quad + \quad +$

$3b=-3$

$b=-1$

and $a=-2$

(a, b) = (–2, –1)

Question179

If $A = \begin{bmatrix} 1 & 2 & x \\ 3 & -1 & 2 \end{bmatrix}$ **and** $B = \begin{bmatrix} y \\ x \\ 1 \end{bmatrix}$ **be such that** $AB = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$**,then:**

[Online April 12, 2014]

Options:

A. $y = 2x$

B. $y = -2x$

C. $y = x$

D. $y=-x$

Answer: A

Solution:

Let $A=\begin{bmatrix}1 & 2 & x\\ 3 & -1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}y\\ x\\ 1\end{bmatrix}$

$AB=\begin{bmatrix}1 & 2 & x\\ 3 & -1 & 2\end{bmatrix}\begin{bmatrix}y\\ x\\ 1\end{bmatrix}$

$\Rightarrow\begin{bmatrix}6\\ 8\end{bmatrix}=\begin{bmatrix}y+2x+x\\ 3y-x+2\end{bmatrix}$

$\Rightarrow\begin{bmatrix}6\\ 8\end{bmatrix}=\begin{bmatrix}y+3x\\ 3y-x+2\end{bmatrix}$

$\Rightarrow y+3x=6$ and $3y-x=6$

On solving, we get

$x=\frac{6}{5}$ and $y=\frac{12}{5}$

$\Rightarrow y=2x$

Question180

Let A and B be any two 3 × 3 matrices. If A is symmetric and B is skewsymmetric, then the matrix AB – BA is:
[Online April 19, 2014]

Options:

A. skewsymmetric

B. symmetric

C. neither symmetric nor skewsymmetric

D. I or – I, where I is an identity matrix

Answer: B

Solution:

Let A be symmetric matrix and B be skew symmetric matrix.
$\therefore A^T = A$ and $B^T = -B$
Consider $(AB - BA)^T = (AB)^T - (BA)^T$
$= B^T A^T - A^T B^T = (-B)(A) - (A)(-B)$
$= -BA + AB = AB - BA$
This shows $AB - BA$ is symmetric matrix.

Question181

If p, q, r are 3 real numbers satisfying the matrix equation,

$$\begin{bmatrix} p & q & r \end{bmatrix}\begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$$ **then $2p + q - r$ equals :**

[Online April 22, 2013]

Options:

A. – 3

B. – 1

C. 4

D. 2

Answer: A

Solution:

Given

$$\begin{bmatrix} p & q & r \end{bmatrix}\begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3p + 3q + 2r & 4p + 2q & p + 3q + 2r \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$$

$\Rightarrow 3p + 3q + 2r = 3$ ......(i)
$4p + 2q = 0 \Rightarrow q = -2p$ ......(ii)
$p + 3q + 2r = 1$ ......(iii)
On solving (i), (ii) and (iii), we get
$p = 1, q = -2, r = 3$
$\therefore 2p + q - r = 2(1) + (-2) - (3) = -3.$

Question182

The matrix $A^2 + 4A - 5I$, where I is identity matrix and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, equals
[Online April 9, 2013]

Options:

A. $4\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$

B. $4\begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$

C. $32\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$

D. $32\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Answer: A

Solution:

$A^2 + 4A - 5I = A \times A + 4A - 5I$

$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} + 4\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} + \begin{bmatrix} 4 & 8 \\ 16 & -12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

$= \begin{bmatrix} 9+4-5 & -4+8-0 \\ -8+16-0 & 17-12-5 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 8 & 0 \end{bmatrix} = 4\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$

Question183

If A = $\begin{pmatrix} \alpha-1 \\ 0 \\ 0 \end{pmatrix}$, B = $\begin{pmatrix} \alpha+1 \\ 0 \\ 0 \end{pmatrix}$ be two matrices, then AB^T is anon-zero matrix for |a| not equal to
[Online May 7, 2012]

Options:

A. 2

B. 0

C. 1

D. 3

Answer: C

Solution:

Let $A = \begin{pmatrix} \alpha-1 \\ 0 \\ 0 \end{pmatrix}, B = \begin{pmatrix} \alpha+1 \\ 0 \\ 0 \end{pmatrix}$
be two matrices.

$AB^T = \begin{pmatrix} \alpha-1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} \alpha+1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \alpha^2-1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Thus, AB^T is non-zero matrix for $|\alpha| \neq 1$

Question184

If A = $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 2 & 1 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -2 & 1 \end{bmatrix}$ then AB equals

[Online May 26, 2012]

Options:

A. I

B. A

C. B

D. 0

Answer: A

Solution:

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -2 & 1 \end{bmatrix}$

$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

Question185

If $\omega \neq 1$ is the complex cube root of unity and matrix H $= \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$,

then H^{70} is equal to
[2011RS]

Options:

A. 0

B. $-H$

C. H^2

D. H

Answer: D

Solution:

$$H^2 = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = \begin{bmatrix} \omega^2 & 0 \\ 0 & \omega^2 \end{bmatrix}$$

We observed that $H^k = \begin{bmatrix} \omega^k & 0 \\ 0 & \omega^k \end{bmatrix}$

$$\therefore H^{70} = \begin{bmatrix} \omega^{70} & 0 \\ 0 & \omega^{70} \end{bmatrix} = \begin{bmatrix} \omega^{69}\omega & 0 \\ 0 & \omega^{69}\omega \end{bmatrix} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = H \ [\because \omega^{3n} = 1]$$

Question186

Let A and B be two symmetric matrices of order 3.
Statement-1: A(BA) and (AB)A are symmetric matrices.
Statement-2: AB is symmetric matrix if matrix multiplication of A with B is commutative.
[2011]

Options:

A. Statement-1 is true, Statement-2 is true; Statement-2 isnot a correct explanation for Statement-1.

B. Statement-1 is true, Statement-2 is false.

C. Statement-1 is false, Statement-2 is true.

D. Statement-1 is true, Statement-2 is true; Statement-2 isa correct explanation for Statement-1.

Answer: A

Solution:

Given that A and B are symmetric matrix
A' = A
B' = B
Now (A(BA))' = (BA)'A' = (A'B')A' = (AB)A = A(BA)(∵ product of matrices are associative)
Similarly, ((AB)A)' = A'(B'A') = A (BA) = (AB)A
So, A(BA) and (AB)A are symmetric matrices.
Again (AB)' = B'A' = BA
Now if BA = AB, then AB is symmetric matrix

Question187

The number of 3 × 3 non-singular matrices, with four entries as 1 and all other entries as 0, is
[2010]

Options:

A. 5

B. 6

C. at least 7

D. less than 4

Answer: C

Solution:

$\begin{bmatrix} 1 & \ldots & \ldots \\ \ldots & 1 & \ldots \\ \ldots & \ldots & 1 \end{bmatrix}$ are 6 non-singular matrices because 6 blanks will be filled by 5 zeros and 1 one.

$\begin{bmatrix} 1 & \ldots & \ldots \\ \ldots & 1 & \ldots \\ \ldots & \ldots & 1 \end{bmatrix}$ are 6 non-singular matrices.

Total = 6 + 6 = 12
So, required cases are more than 7, non-singular 3 × 3 matrices.

Question188

Let A = $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and B = $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, a, b ∈ N . Then

[2006]

Options:

A. there cannot exist any B such that AB = BA

B. there exist more than one but finite number of B's such that AB = BA

C. there exists exactly one B such that AB = BA

D. there exist infinitely many B¢s such that AB = BA

Answer: D

Solution:

Given that $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

$AB = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}$

$BA = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}$

Hence, AB = BA only when a = b

∴ There can be infinitely many B's for which AB = BA

Question189

If A and B are square matrices of size n × n such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true? [2006]

Options:

A. $A = B$

B. $AB = BA$

C. either of A or B is a zero matrix

D. either of A or B is identity matrix

Answer: B

Solution:

Given that $A^2 - B^2 = (A - B)(A + B)$

$A^2 - B^2 = A^2 + AB - BA - B^2$

$\Rightarrow AB = BA$

Question190

If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction
[2005]

Options:

A. $A^n = nA - (n-1)I$

B. $A^n = 2^{n-1}A - (n-1)I$

C. $A^n = nA + (n-1)I$

D. $A^n = 2^{n-1}A + (n-1)I$

Answer: A

Solution:

Given that $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $A^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$

Therefore we observed that $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

Now $nA - (n-1)I = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} = A^n$

$\therefore nA - (n-1)I = A^n$

Question191

If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

[2003]

Options:

A. $\alpha = 2ab, \beta = a^2 + b^2$

B. $\alpha = a^2 + b^2, \beta = ab$

C. $\alpha = a^2 + b^2, \beta = 2ab$

D. $\alpha = a^2 + b^2, \beta = a^2 - b^2.$

Answer: C

Solution:

$A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} \Rightarrow A . A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

$= \begin{bmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{bmatrix}$

$\alpha = a^2 + b^2; \beta = 2ab$
